

TMF630 REST API Design Guidelines Part 1: Practical guidelines for RESTful APIs naming, CRUD, filtering, tasks, history, notifications

TMF630 Part 1

Maturity Level: General Availability (GA)	Team Approved Date: 04-Sep-2025
Release Status: Pre-production	Approval status: Team Approved
Version: 5.0.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2025. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the [TM FORUM IPR Policy](#), must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an “AS IS” basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648

TM Forum Web Page: www.tmforum.org

Table of Contents

Notice	2
Executive Summary	6
Conventions	7
Keywords to Indicate Requirement Levels	7
Pattern Notation	7
Formatting Conventions for Examples	8
1. General	9
1.1. API Resource Archetypes	9
1.2. REST Levels	9
1.3. REST API Specification Information	10
1.4. API Implementation Technology	10
1.5. HTTP Headers	11
1.6. Authentication and Authorization	15
2. Managed Entity Model	20
2.1. Resource Schemas	20
2.2. Resource State and Uniform Contract Verbs	20
2.3. URI Naming Standard	20
2.4. Resource id, href and @type	21
2.5. Resource Naming Convention	22
2.6. Identifier Syntax and Uniform Contract Verbs	23
2.7. Resource Collection Naming	23
2.8. DateTime Format	24
3. Uniform Contract	26
3.1. Uniform Contract Operations	26
3.2. API Media Types	27
3.3. API Response Status and Exception Codes	28
3.4. User, Application and Extended Error Codes	34
3.5. Representations	35
3.6. Common Information Model	35
4. Query Resources Patterns	37
4.1. Query single Resource all Attributes	39
4.2. Querying multiple Resources	42
4.3. Query partial Resource Representation by Attribute Selection	50

4.4. Query Resources with Attribute Filtering	57
4.5. Query Resources with Pagination	73
4.6. Query Sub-Resources / Sub-Resource Pattern	95
4.7. Query Resources with Attribute Filtering and Attribute Selection	99
4.8. Sorting	99
5. Modify Resources Patterns	101
5.1. Uniform Contract Operations for modifying Resources	101
5.2. PATCH Syntax Alternatives	101
5.3. Modify single-valued Attributes of a Resource	102
5.4. Modify multi-valued Attributes of a Resource	108
5.5. JSON-Patch Extension to manage Arrays	112
6. Create Resource Patterns	117
6.1. Creating a single Resource	117
6.2. Create a single Resource with Attribute Selection	123
7. Delete Resource Pattern	126
8. Operations on Collection Resources	128
8.1. Adding a non-existing element to a collection	128
8.2. Updating an existing element of a collection	128
8.3. Removing an existing element from a collection	129
8.4. Mixed operations	129
8.5. Response Codes	129
9. Task Resource Pattern	138
9.1. Modeling complex Operations with Task Resources	138
9.2. Short-running Tasks	139
9.3. Long-running Tasks	146
9.4. Task Lifecycle	146
9.5. Task Cancellation	148
9.6. Type-selection by Server	151
9.7. Using Task with Sub-Resource based Response	153
10. Asynchronous Operation Pattern	163
10.1. Asynchronous Operation & Monitor	165
10.2. Asynchronous Operations & Notifications	166
10.3. Expect Header	169
10.4. Monitor Pattern	171
11. Notification Patterns	184

11.1. Create Subscription & register Listener	184
11.2. Retrieve all Subscriptions	190
11.3. Retrieve Subscription	191
11.4. Delete Subscription	193
11.5. Publishing Events	194
11.6. Event Types	197
12. History Pattern	206
12.1. Revisioning of Resources	207
12.2. History Resource	207
12.3. History Operations	209
12.4. History Links via the Link Header	237
13. Miscellaneous	239
13.1. API Versioning	239
13.2. National Language Support	241
13.3. Resource Scheme Customisation – removing non-mandatory Attributes	245
14. Event Management	246
14.1. Decoupling	246
14.2. Event API Relationship to Notification Events by a generalized Event Model	246
14.3. Event Resource Model	247
14.4. Topic/Event Resource Graph	251
14.5. Event Management as API for Consumers	252
14.6. Event Management in a multi-cloud Scenario	252
References	254
Administrative Appendix	259
Document History	259
Acknowledgments	260

Executive Summary

The “REST API Design Guidelines” document provides guidelines and design patterns used in designing TM Forum REST APIs. The document is organized in multiple parts as follows:

- **Part One:** Practical guidelines for RESTful APIs; naming, CRUD, filtering, tasks, history, notifications
- **Part Two:** Advanced guidelines for RESTful API polymorphism, extension patterns, depth and expand directive, entity RefOrValue
- **Part Three:** Guidelines for extending TMF Open APIs with hypermedia support
- **Part Four:** Advanced guidelines for RESTful APIs; lifecycle management, common tasks
- **Part Five:** JSON Patch extension to manage arrays
- **Part Six:** JSON Path extension
- **Part Seven:** JSON Schema patterns
- **Part Eight:** API Specialization

This part (Part One) of the document provides recommendations and guidelines for the design and implementation of Entity CRUD operations and Task operations.

It also provides information on filtering and attribute selection. Finally, it provides information on supporting notification management in REST based systems.

The uniform contract establishes a set of methods that are expected to be reused by services within a given collection or inventory.

Conventions

Keywords to Indicate Requirement Levels

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in [\[RFC8174\]](#).

Pattern Notation

To formally describe patterns, the notation defined in [\[RFC5234\]](#) and extended in [\[RFC9110\]](#) (section 5.6.1) is applied:

- `[ ]` Square brackets enclose an optional element.
- `( )` Elements enclosed in parentheses are treated as a single element whose contents are strictly ordered.
- `/` (forward slash) Elements separated by a forward slash are alternatives.
- `*` (star) preceding an element means repetition without separator between different values; a number preceding the star symbol defines the minimal number of values, defaulting to 0, and a number following the star symbol defines the maximal number of values, defaulting to unbounded; thus, `1*` indicates an unbounded repetition having at least one value.
- `#` (hash) preceding an element means repetition with a `,` (comma) as separator between different values; a number preceding the hash symbol defines the minimal number of values, defaulting to 0, and a number following the hash symbol defines the maximal number of values, defaulting to unbounded; thus, `1#` indicates an unbounded comma-separated repetition having at least one value.
- `" "` One or more symbols enclosed in quotation marks represent the enclosed symbols as literals.

Deviating from the RFC5234 notation:

- `{ }` A string enclosed in curly braces represents a value of an element named by the enclosed string.

For better readability, spaces may be inserted between elements, which do not become part of the pattern's production. Only if enclosed in quotation marks spaces are meant to be part of the pattern production.

Formatting Conventions for Examples

In most examples, attributes occur that normally would be long UUID values; as simplification of the layout (avoidance of unnecessary line breaks), these values are shortened to a length of 4 characters only, followed by two dots:

`bd69..` instead of `bd694f23-627f-4553-a1df-7e79d62001ac`

For the same reason, in examples we abbreviate the leading part of URLs by a placeholder:

`{apiRoot}` instead of `https://mycsp.com/tmf-api`

To increase readability, we do not show the encoded form of URLs, but use the unencoded form. Of course, in a real implementation the URLs must be encoded as per [\[RFC3986\]](#).

Chapter 1. General

1.1. API Resource Archetypes

The following section describes the resource archetypes supported by the TMF REST APIs.

A REST API is composed of 3 distinct resource archetypes and should align each resource to just one of these:

- **Managed Resource** – e.g. a database record or an object instance – its representation includes: fields with values, links to related resources and child resources of different resource types. Is unambiguously identified by **id**. Client can create, query, update and delete resources.

Examples: `/troubleTicket/{id}`, `/productOrder/{id}`.

- **Task Resource** – resources that are executable functions – with associated input and output parameters. Necessary where the required action cannot be mapped to standard CRUD methods.

Examples: `/checkProductOfferingQualification/{id}`, `/cancelProductOrder/{id}`

- **Collection Resource** – server managed *collection* of either Managed or Task resources. In the TMF REST API Design collections are anonymous. A resource with no identifier represents the resource collection (matching the resource type).

Examples: `/troubleTicket`, `/productOrder`. Note that when a query filter is applied to a collection, the result is also a collection resource, which even may be empty.

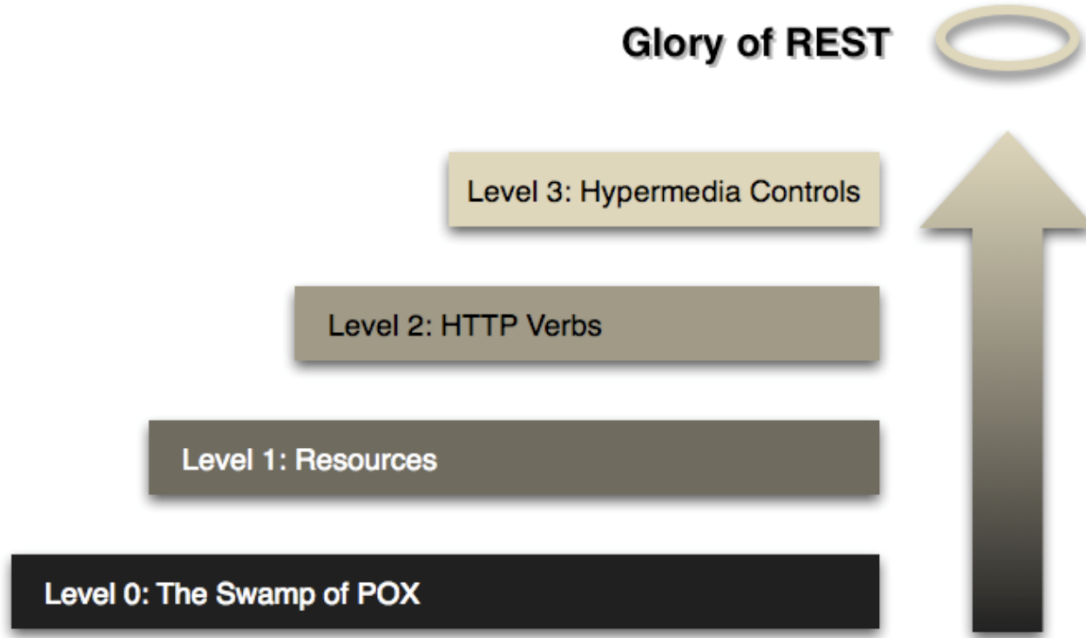
- **Child-Resource** – object or collection existentially dependent on some other resource ("the parent"), i.e. cannot exist without its parent. Can only be created or accessed with prior knowledge of the parent.

Examples: `/geographicAddress/{id}/geographicSubAddress` (a collection of GeographicSubAddress objects), `/productOrder/{id}/productOrderItem/{itemId}` (an object).

1.2. REST Levels

All APIs must implement Level 2 of the *Richardson Maturity Model* [RMM]

Level 3 is not mandatory and not specified in this part of the Design Pattern Guideline. Level 3 issues are addressed in the REST Guidelines Part 3 sections as they relate to workflow oriented patterns.



The levels of the Richardson Maturity Model for RESTful APIs

1.3. REST API Specification Information

For each REST API specification, the following information **MUST** be included:

- Purpose of the API.
- URL of resources and API including version number.
- HTTP verbs supported.
- Representations supported: JSON
- Response schema (*and where POST, PATCH are supported – request schema*).
- Links supported (Optional in L2 APIs)
- Response status codes supported.

The API **MUST** be described using *Open API Specification 3.x* [\[OAS3\]](#)

1.4. API Implementation Technology

APIs are independent of the dependent technology implementation.

REST APIs embrace all aspects of the Hypertext Transfer Protocol, HTTP/1.1 or HTTP/2, including its request methods, response codes, and message headers.

1.5. HTTP Headers

The following describes the header protocol elements that SHOULD be used by the TMF REST APIs.

If not stated otherwise, headers are not explicitly included in the TMF Open API specification files.

1.5.1. Caching HTTP Headers

Cache-Control, **Expires**, and **Date** in response headers SHOULD be used when caching is required as specified in [\[RFC9111\]](#).

```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-control: max-age=60, must-revalidate
```

- **Cache-Control** and **Expires** response headers MAY be used to discourage caching. If a response must not be cached, Cache-Control can be added with the value **no-cache** and **no-store**.
- Caching SHOULD be encouraged. The **no-cache** directive will prevent any cache from serving cached responses. This SHOULD not be used unless absolutely necessary.
- Expiration caching headers SHOULD be used with **200 OK** responses to **GET** and **HEAD** requests.
- Expiration caching headers MAY be used with 3xx and 4xx responses – this helps reduce the amount of redirecting and error-triggering load on a REST API.



Date/Time values of the **Date** and **Expires** headers are defined in [\[RFC9110\]](#) resp. [\[RFC9111\]](#) to be formatted according to [\[RFC5322\]](#); for example: **Sun, 06 Nov 1994 08:49:37 GMT**

1.5.2. Negotiation HTTP Headers

Client Side

The negotiation of the media type required by the client and provided by the server involves a number of headers in both the request and the response.

- The client MAY include any HTTP Header as specified in [\[RFC9110\]](#), [\[RFC9111\]](#),[\[RFC9112\]](#).
- The client MUST expect any HTTP header as specified in [\[RFC9110\]](#), [\[RFC9111\]](#),[\[RFC9112\]](#).

The media type is the form of the response payload that the client would like to receive; this is specified using the standard HTTP header: **Accept**

- The client MUST use the **Accept** HTTP header to specify the media type.
- The **Accept** HTTP header MUST have a value matching [RFC9110].
- The client MUST support **application/json** by default.

As per [RFC8259 \(JSON\)](#) (section 8.1), UTF-8 MUST be used (if not in a closed ecosystem) as character set. Thus, stating a character set in the **Content-Type** or **Accept** header is in general not necessary. Also, using the **Accept-Charset** header SHOULD NOT be used, and even is deprecated as defined in [\[RFC9110\]](#) (section 12.5.2).

If a media type is specified that is not supported by the server then the response must be returned using the default, **application/json**, i.e. as if no media type was specified.

- The client MAY specify the locale of the response in the request header.
- The client SHOULD use the **Accept-Language** HTTP header to specify the locale.
- The **Accept-Language** HTTP header MUST have a value matching the pattern:
"Locale: " {ISO639} ["_" {ISO3166-Alpha2}]
with {ISO-639} denoting a language code defined in [\[ISO639\]](#)
and {ISO3166-Alpha2} denoting a 2-letter country code defined in [\[ISO3166\]](#)
(see [the pattern notation](#) described above).
- The client MUST expect **en_GB** by default.

The **Accept-Language** header uses the [\[ISO639\]](#) language code and [\[ISO3166\]](#) country code standards separated by an underscore ("_"), with the second half of the format being optional. If a language is specified that is not supported by the server then the response payload must be returned as **en_GB** (for additional information see [National Language Support](#)).

Server Side

As part of the content negotiation, just as the client can request particular behavior from the server, the server needs to respond back saying; "this is what you're getting".

- The server MAY include any HTTP header as specified in [\[RFC9110\]](#), [\[RFC9111\]](#), [\[RFC9112\]](#).
- The server MUST expect any HTTP header as specified in [\[RFC9110\]](#), [\[RFC9111\]](#), [\[RFC9112\]](#).
- The server SHOULD specify the media type of the response in the response header
- The server MUST use the **Content-Type** HTTP header to specify media type
- The **Content-Type** HTTP header MUST have a value matching in [\[RFC9110\]](#), [\[RFC9111\]](#),

[RFC9112].

- The server MUST support `application/json` by default.

The above rules work in the same way as the `Accept: application/json` is supported as the default media type of the server. This is returned either when requested or when no `Accept` header is present. The server SHOULD provide this header back to the client for the sake of clarity and so that the client does not have to “detect” the actual media type of the response. However, it is not required and the client SHOULD assume that either the requested media type or the default is being returned.

- The server SHOULD specify the `Content-Length` HTTP header. Client can thus know whether it has read the correct number of bytes from the connection and can make a HEAD request to find out how large the entity-body is, without downloading it.

As per [RFC8259] (section 8.1), UTF-8 must be used (if not in a closed ecosystem) as character set. Thus, stating a character set in the `Content-Type` or `Accept` header is in general not necessary. Also, using the `Accept-Charset` header should not be used, and even is deprecated as defined in [RFC9110].

```
`HTTP/1.1 200 OK
Content-Type: application/json
Content-Language: en_GB
Content-Length: 1024
ETag: "x234dff"
```

- The server SHOULD use `Last-Modified` HTTP header in the responses to specify the time at which the resources were created.
- The server SHOULD use `ETag` in responses. The entity tag may be any string value, so long as it changes along with the resource’s representation. Note that if the response is a list of entities (e.g. as result of a GET request to a collection resource, with or without query parameters), the `ETag` value corresponds to the complete response.
- The server MUST use `Location` HTTP header to specify the URI of a newly created resource, which may be used for example part of the hypermedia support.

The following specifies the locale within the response; this matches the `Accept-Language` HTTP header in the request message.

- The server MAY specify the locale of the response in the response header.
- The server SHOULD use the `Content-Language` HTTP header to specify the locale.
- The `Content-Language` HTTP header MUST have a value matching the template:

Locale: ISO-639, ("_", ISO3166-Alpha2)?

- The server MUST support **en_GB** by default.
- The supported locales MUST be documented (see [National Language Support](#))

1.5.3. Lost Update Headers

To avoid the "Lost Update Problem", the implementation SHOULD use the **ETag** and **If-Match** HTTP headers to realize optimistic locking:

- When a representation of the resource's state is returned to the client, the server MUST include an entity tag in the **ETag** header of the response. Its value MUST uniquely identify the resource's state at the time of the response. It MUST NOT be changed until the resource is modified.
- When a client requests any change to the request (e.g. by a POST, PATCH or DELETE), it MUST return the entity tag in the **If-Match** header of the request.
- When the client does not provide the **If-Match** header, the server MUST reject the request with a **428 Precondition Required** status code.
- The server MUST check the value of the **If-Match** header against the current entity tag of the resource. If the values do not match, it MUST reject the request with a **412 Precondition Failed** status code, and return the current entity tag in the **ETag** header of the response.
- If the values match, the server MUST accept the request as if the mechanism was not applied at all, update the resource, and change the entity tag to a new value. The new value MUST be returned in the **ETag** header of the response.
- The client can then decide if it wants to retry the request with the new entity tag, or if it wants to fetch the new state of the resource and adapt its request correspondingly.

1.5.4. Rate Limit Headers

- The server SHOULD specify a rate limit. Rate limiting of an API can be defined on a per-user basis and per-application basis. If an operation allows X requests per rate limit window, then it allows the client to make X requests per window.
- The server SHOULD return a **429 Too Many Requests** HTTP response when this limit is exceeded.
- The server SHOULD return the following HTTP headers:
 - **X-Rate-Limit-Limit**: The number of requests allowed in the current period
 - **X-Rate-Limit-Remaining**: The number of requests remaining in the current period
 - **X-Rate-Limit-Reset**: The number of seconds left in the current period.

1.5.5. Custom Header

- The server SHOULD return the HTTP header **X-Total-Count** in a response to a **GET** List resource request.

The HTTP header must specify the total number of matching resources to the **GET**.

Example:

REQUEST

```
GET {apiRoot}/troubleTicket/troubleTicket?offset=0&limit=20
```

RESPONSE

```
Content-Type: application/json
X-Total-Count: 200
```



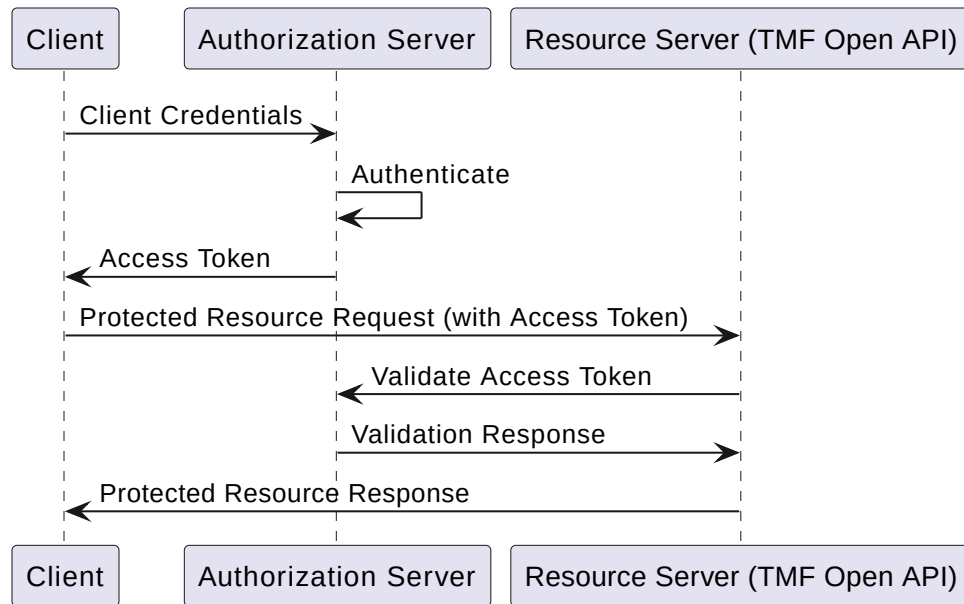
In the above example, the body of the payload was omitted for focus on headers.

1.6. Authentication and Authorization

Many TMF Open APIs expose access to sensitive information and provide access to both customer and business operation details. Care should be taken to secure these Open API interfaces to prevent unauthorized access. Each API should be secured following the [principle of least privilege](#).

OAS provides many different mechanisms to secure APIs and one implementation method will not satisfy every provider's implementation. Additionally, one API may have different authentication strategies depending on where the service is deployed or which client is consuming the service. The TM Forum recommends all Open API interfaces are secured and provides general guidelines and methods to implement authentication and authorization of these interfaces.

The most common method of securing APIs is using OAuth 2.0. OAuth separates the authentication provider from the resource. The authorization server issues access tokens which are used to access the protected resource. This allows consumption of the API without sharing credentials with the API provider. OAuth 2.0 is implemented in various ways using *flows*. The most common flow for machine-to-machine interfaces is the *Client Credentials Flow*. Below is a sequence diagram illustrating the *Client Credentials Flow*.



This security method can be applied to TMF Open APIs by decorating the specifications using standard OAS security definitions. Below is an example of a Client Credential Flow security definition as specified in [\[OAS3\]](#):

```

components:
  securitySchemes:
    OAuth2:
      type: oauth2
      flows:
        authorizationCode:
          authorizationUrl: https://example.com/oauth/authorize
          tokenUrl: https://example.com/oauth/token
          scopes:
            read: Grants read access
            write: Grants write access
            admin: Grants access to admin operations
  
```

Once defined this security definition can be applied to all methods in the API or applied to individual methods.

Example: Security applied to ALL methods:

```

openapi: 3.0.0
info:
  title: Security
  version: 0.0.1
  
```

```

security:
- OAuth2:
  - read
  - write
paths:
  "/resource":
    get:
      summary: Health check
      description: Health check
      operationId: GetResource
      responses:
        '200':
          description: Success
          content:
            application/json:
              schema:
                type: object
                description: Base schema
                properties:
                  status:
                    type: string
                    description: Healthcheck - status
                    example: OK
components:
  securitySchemes:
    OAuth2:
      type: oauth2
      flows:
        authorizationCode:
          authorizationUrl: https://example.com/oauth/authorize
          tokenUrl: https://example.com/oauth/token
          scopes:
            read: Grants read access
            write: Grants write access
            admin: Grants access to admin operations

```

Example: Security applied to individual methods:

```

openapi: 3.0.0
info:
  title: Security
  version: 0.0.1
security:
- OAuth2:

```

```
- read
- write
paths:
  /resource:
    get:
      summary: Health check
      description: Health check
      operationId: GetResource
      security:
        - OAuth2: [read] # Use OAuth with a different scope
      responses:
        '200':
          description: Success
          content:
            application/json:
              schema:
                type: object
                description: Base schema
                properties:
                  status:
                    type: string
                    description: Healthcheck - status
                    example: OK

  /users:
    post:
      summary: Add a user
      security:
        - OAuth2: [write]
      requestBody:
        description: Optional description in *Markdown*
        required: true
        content:
          application/json:
            schema:
              type: object
              properties:
                name:
                  type: string
      responses:
        '201':
          description: Created

  /ping:
    get:
      summary: Checks if the server is running
      security: [] # No security
```

```
responses:
  '200':
    description: Server is up and running
  default:
    description: Something is wrong
components:
  securitySchemes:
    OAuth2:
      type: oauth2
      flows:
        authorizationCode:
          authorizationUrl: https://example.com/oauth/authorize
          tokenUrl: https://example.com/oauth/token
      scopes:
        read: Grants read access
        write: Grants write access
        admin: Grants access to admin operations
```

Chapter 2. Managed Entity Model

Managed Resources represent entities that are acted upon by the REST API.

Examples: a Trouble Ticket, a Logical Port, an Order, a Work, etc.

2.1. Resource Schemas

There is a direct mapping between the managed entities and the corresponding Resources in the REST model.

The mapping between the managed entity types and the corresponding Resource model **MUST** be included as schema definitions in the API specification.

Individual managed resources **MUST** have a unique identifier field called **id**.

2.2. Resource State and Uniform Contract Verbs

The specific information or data that characterizes a resource at a particular moment is referred to as the resource's state. It includes attributes or properties of the resource and can change over time.

A REST API acts upon its resources by operations defined by Uniform Contract Verbs (**POST**, **GET**, **DELETE**, **PATCH**). Operations on Resources affect the state of the corresponding managed entities.

2.3. URI Naming Standard

Resource identifiers determine the actual resources that a service exposes. A resource identifier is like a unique ID assigned to one or more service resources. To standardize the syntax used to represent them the Web's Uniform Resource Identifier (URI) syntax is **RECOMMENDED**, being the most common syntax used.

The URI path may convey the REST API's resource model, with each forward slash separated path segment corresponding to a unique resource within the model's hierarchy.

Characters that require encoding **MUST NOT** be used in the URI.

The structure of a Resource Path URI is given by the following expression:

```
{resourcePath} = *[ {resourceName} "/" {resourceID} "/" ] ( {resourceName} [ "/"  
{resourceID} ] / {taskName} [ "/" {taskId} ] )
```



The above follows the notation defined in section [Pattern Notation](#)

For example in our sample Trouble Ticket Management application:

```
{resourcePath} = troubleTicket/{id}
```

Resource URIs SHOULD be formed according to the following base pattern:

```
{resourceURI} = {apiRoot} "/" {resourcePath}
```

apiRoot points to the server-URL of the API, defined in the servers.url attribute of the API specification.

All APIs SHOULD use HTTP/TLS, "HTTPS" as the scheme.

Example:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/b890..
```

2.4. Resource id, href and @type

Every Resource has mandatory attributes:

- **id**: A unique identifier in the context of the application collection

Example: `id=f043..`

- **href**: The URI of the resource as per **Location** header

Example: `href=troubleTicket/f043..`

(may be a full URL – it is up to each implementation to decide if it is using a full or relative URIs)

- **@type**: The class-type of a REST resource.

Example: `"@type": "TroubleTicket"`

The **id** of a new resource is expected to be provided by the implementation and as such is

not provided as input to a create operation (**POST**).

The **id** must be immutable, and within the collection its value must be unique and must never be reused for any other resource of this collection.

2.4.1. id implementations

The id values of the resource SHOULD be taken randomly from a large range of possible values, so that from a few known id values no other existing value can be computed or "guessed" (large entropy) with sufficient probability.

Possible variants are:

- The standard UUID format: 36 char String format (including the 4 separators "-") defined in [\[RFC4122\]](#).
 - **Example:** `bd694f23-627f-4553-a1df-7e79d62001ac`
 - **advantage:** supported by many programming languages
 - **disadvantages:** slow to create new values; possible (but very unlikely) to double-use the same value
- Large integers, where to create a new value the last one is incremented by a random number
 - **Example:** int64, with random increment from the range 1 to 10,000
 - **advantages:** much faster; surely no double-usage
 - **disadvantage:** requires custom implementation

2.5. Resource Naming Convention

Managed resource names and collection resource names MUST be nouns, while task resource names SHOULD be verbs or start with a verb.

Names of managed resources MUST be upper camel case.

e.g. `ServiceProblem`, `TroubleTicket`, `ServiceOrder` – as a resource

Names of task resources MUST be lower camel case.

e.g. `close`, `cancel`, `group`, `send`, `check`, `queryProductRecommendation`, `cancelProductOrder` – as a task

Names in URI (individual resources, collections, child resources, task resources) MUST be lower camel case.

e.g. `/account/billingAccount` not `/Account/Billingaccount`

Names of resource properties MUST be lower camel case.

e.g. `"billingAccount": { ... }` not `"Billingaccount": { ... }`

The resource name MUST NOT expose technical or implementation details.

Abbreviations MUST NOT be used as the full name of a resource. They SHOULD only be used as part of the name if the name otherwise becomes too long. If using an abbreviation, it MUST be used in a uniform way (i.e. use always the same abbreviation).

e.g. `ResourceSpecification` not `ResourceSpec`

Acronyms MUST NOT be used as the full name of a resource, but resource names may contain an acronym if generally well known in the industry. If used, it MUST be in upper camel case, i.e. with leading capital and the rest lower case.

e.g. `CdrTransaction` not `CDR`

The names of the API resources MUST be derived from the TMF Information Framework (SID), but in cases where the SID uses abbreviations, the API resource name SHOULD use the full word not the abbreviation.

The resource URI SHOULD NOT use meaningless words, like

`Management, Maintenance, Provision`

e.g. instead of `ResourcePoolManagement`, it SHOULD be `ResourcePool`.

2.6. Identifier Syntax and Uniform Contract Verbs

URI names MUST NOT contain the names of HTTP verbs (task resources should be used if needed for clarity).

2.7. Resource Collection Naming

Collection names MUST not use a Collection postfix. Collections are implicit when the resource name is used.

Examples:

```
{apiRoot}/troubleTicket
```

represents the Trouble Ticket Collection, and

```
GET {apiRoot}/troubleTicket
```

retrieves all the items from the Trouble Ticket Collection
(i.e. all the resources of type `troubleTicket` contained within the application).

2.8. DateTime Format

All datetime attributes in every data model MUST be described into a format aligned with [\[RFC3339\]](#) (section 5.6).

Assuming that the time zones of the dates and times are the same (e.g., all in UTC), expressed using the same string (e.g., all "Z" or all "+00:00"), the datetime format looks as follows:

```
YYYY-MM-DDThh:mm:ssZ
```

```
YYYY = 4 digits for the year (e.g. 2020,2021,...)
MM   = 2 digit for the Month (e.g. 01,02,03,...,12)
DD   = 2 digits for the day (e.g. 01,..31)
T    = symbol for standard time
hh   = 2 digits for hours (e.g. 00,..23)
mm   = 2 digits for minutes (e.g. 01,..60)
ss   = 2 digits for seconds (e.g. 00,01,..60)
Z    = symbol for the UTC standard time without deviation
+    = sign for a positive deviation (e.g. 2021-01-01T17:30:10+01.00)
-    = sign for a negative deviation (e.g. 2021-01-01T17:30:10-01.00)
```

Datetime examples from [\[RFC3339\]](#):

```
1985-04-12T23:20:50.52Z
```

This represents 20 minutes and 50.52 seconds after the 23rd hour of April 12th, 1985 in UTC.

```
1996-12-19T16:39:57-08:00
```

This represents 39 minutes and 57 seconds after the 16th hour of December 19th, 1996 with an offset of -08:00 from UTC (Pacific Standard Time).

Note that this is equivalent to 1996-12-20T00:39:57Z in UTC.

```
1990-12-31T23:59:60Z
```

This represents the leap second inserted at the end of 1990.

```
1990-12-31T15:59:60-08:00
```

This represents the same leap second in Pacific Standard Time, 8 hours behind UTC.

```
1937-01-01T12:00:27.87+00:20
```

This represents the same instant of time as noon, January 1, 1937, Netherlands time. Standard time in the Netherlands was exactly 19 minutes and 32.13 seconds ahead of UTC by law from 1909-05-01 through 1937-06-30. This time zone cannot be represented exactly using the hh:mm format, and this timestamp uses the closest representable UTC offset.



It is recommended that all dates provide the offset as per [\[RFC3339\]](#), however in the case that the offset (timezone) is missing the date is considered as local time as per [\[ISO8601\]](#).

Chapter 3. Uniform Contract

HTTP provides a set of generic methods, such as **GET**, **PUT**, **PATCH**, **POST**, **DELETE**, **HEAD** and **OPTIONS**, that are pre-defined in the HTTP specification. The complete protocol interactions include a set of response codes, plus syntax for expressing various parameters that can be encoded in HTTP messages. As described below, not all HTTP verbs are used in TMF Open APIs.

The REST uniform contract is based on three fundamental elements:

- resource identifier syntax – How can we express where the data is being transferred to or from?
- methods – What are the protocol mechanisms used to transfer the data?
- media types – What type of data is being transferred?

The following section describes the guidelines for modeling operations and for specifying what media types to use.

3.1. Uniform Contract Operations

All API operations are based on the REST Uniform Contract operations.

- **GET** and **POST** MUST NOT be used to tunnel other request methods in the TMF Open API specifications, although it is recognized as a practice in an implementation. Exceptions are TTask Resources as defined in this Guideline.
- **HEAD** MAY be used, but it MUST NOT be used for other purposes than to retrieve response metadata as defined in [\[RFC9110\]](#).

The following describe the relationships between the elements of the API and the Uniform Contract.

Only legitimate resources can be acted upon using Uniform operation and not any dependent entities.

Operations on Entities are mapped to operations on the corresponding resources:

Operation on Entities	Uniform API Operation	Description
Retrieve an individual Entity	GET Managed Resource	GET with {id} as last path element MUST be used to retrieve a representation of an individual managed resource.

Operation on Entities	Uniform API Operation	Description
Query Entities from a Collection	GET Collection Resource	GET with query parameters MUST be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST on a collection resource MUST be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH on a managed resource or a collection resource MUST be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE on a managed resource MUST be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST on a collection of task resources must be used to execute Task Resources
Replacement of an Entity	PUT	The PUT operation is not defined across the TM Forum Open APIs and SHOULD NOT be used.
Other Operation Semantics		GET and POST MUST NOT be used to tunnel other operations.

3.2. API Media Types

When defining methods for REST services, we can further specify the types of data a given method can process. For example, a **GET** method may be able to retrieve a Trouble Ticket representation in JSON as well as other formats. Each is represented by its own media type.

All TM Forum Open APIs MUST support JSON (**application/json**) as the default resource representation.

Mandatory Media Type Support

- REST APIs MUST support the **application/json** media type by default.
- The default resource representation MUST be JSON.
- An API MUST use only the **Accept** and **Content-Type** headers to control media type representation. Other mechanisms SHOULD NOT be supported.

Media Type Support for PATCH Operations

- For partial updates, the following media types MUST be used:
 - **JSON Merge Patch:** see [\[RFC7396\]](#) → "application/merge-patch+json"
 - **JSON Patch:** see [\[RFC6902\]](#) → "application/json-patch+json"
 - **JSON PATCH Query:** TMF specific extension, see [\[TMF630-5\]](#) → "application/json-patch-query+json"

Deprecated in V5, to be removed in V6:

If [application/json](#) is used in PATCH, JSON Merge rules MUST be applied as per [RFC7396 \(JSON Merge Patch\)](#).



Support for PATCH requests using [application/json](#) as the MIME type is deprecated in TMF Open API version V5 and will be completely removed in V6.

In V6, PATCH with [application/json](#) MUST NOT be supported.

3.3. API Response Status and Exception Codes

REST APIs MUST use the exception and response codes [\[IANA-STATUS\]](#).

3.3.1. Success Responses

In particular, the following response codes MUST be used to indicate that the client's request was accepted successfully.

They MUST NOT be used to communicate errors in the response

Status Code	Rule
200	OK SHOULD be used to indicate success of synchronously executing the requested action when the action's result data is returned as payload. GET: always (even if no data matches the query conditions; then returns the empty result set) PATCH: always POST: in case of a task resource, if the task has completed synchronously and the task was not persisted. otherwise, never DELETE: never
201	Created MUST be used to indicate successful resource creation. Return message SHOULD contain a resource representation and a Location header with the created resource's URI. In case of a task resource, indicates the creation of the task, not necessarily the start of processing the task itself. GET, PATCH, DELETE: never POST: always, with exception of a task resource that is not persisted.

Status Code	Rule
202	Accepted MUST be used to indicate that the request has been accepted, but does NOT signal the start or completion of processing the requested action. The request might eventually be acted upon or not. See [RFC9110] (section 15.3.3).
204	No Content SHOULD be used when the response body is intentionally empty. GET, POST, PATCH: never DELETE: always
206	Partial Content MUST be used for partial resource returned in response or for Offset Pagination – Range Based Style

3.3.2. Redirection Responses

The following response codes MUST be used to indicate that the client must take some additional action in order to complete their request:

Status Code	Rule
301	Moved Permanently SHOULD be used to relocate resources
303	See Other SHOULD be used to refer the client to a different URI – can be used with a Location header containing the URI of a resource that shows the outcome of an asynchronous task.
304	Not Modified SHOULD be used to preserve bandwidth

Status Code	Rule
307	Temporary Redirect SHOULD be used to tell clients to resubmit the request to another URI

3.3.3. Client Error Responses

The following status codes indicate an error on the client's side:

Status Code	Rule
400	Bad Request MAY be used to indicate nonspecific failure. The request could not be understood by the server. The client SHOULD NOT repeat the request without modifications
401	Unauthorized MUST be used when there is a problem with the client's credentials
403	Forbidden SHOULD be used to forbid access regardless of authorization state. <i>For example, a client may be authorized to interact with some, but not all of a REST API's resources. If the client attempts a resource interaction that is outside its permitted scope, the REST API SHOULD respond with 403.</i>
404	Not Found MUST be used when a client's provided URI cannot be mapped to a resource URI
405	Method Not Allowed MUST be used when the HTTP method is not supported
406	Not Acceptable SHOULD be used to indicate that the requested resource is capable of generating only content not acceptable according to the Accept headers sent in the request

Status Code	Rule
409	Conflict SHOULD be used to indicate that the request could not be completed due to a conflict with the current state of the target resource.
410	Gone SHOULD be used to indicate that the requested resource is no longer available at the server and no forwarding address is known.
411	Length required MUST be used to indicate that the server refuses to accept the request without a defined Content-Length.
412	Precondition Failed MUST be used to indicate that the precondition given in one or more of the request-header fields evaluated to false when it was tested on the server – SHOULD in particular be used to indicate that the entity tag value provided by the client in the `If-Match` header did not match the current value of the resource's entity tag (i.e. that the state of the resource has changed in the meantime).
413	Request Entity Too Large MUST be used to indicate that the server is refusing to process a request because the request entity is larger than the server is willing or able to process
414	Request-URI Too Long SHOULD be used to indicate that the server is refusing to service the request because the Request-URI is longer than the server is willing to interpret
415	Unsupported Media Type SHOULD be used to indicate that the request entity has a media type which the server or resource does not support. <i>For example, the client uploads an image as image/svg+xml, but the server requires that images use a different format.</i>

Status Code	Rule
422	Unprocessable Entity SHOULD be used to indicate that the request was well-formed but was unable to be followed due to semantic errors. In particular, SHOULD be used by the server to indicate that a business rule prevents the correct execution of the requested function.
428	Precondition Required SHOULD be used to indicate that the client did not send a previously received entity tag value in the If-Match header.
429	Too Many Requests SHOULD be used to indicate that the client has sent too many requests in a given amount of time ("rate limiting").

3.3.4. Server Fault Responses

The following status codes indicate a server fault:

Status Code	Rule
500	Internal Server Error MUST be used to indicate that a generic server-side error message, given when an unexpected condition was encountered and no more specific message is suitable.
501	Not Implemented SHOULD be used to indicate that the server either does not recognize the request method, or it lacks the ability to fulfil the request. Usually this implies future availability (e.g., a new feature of a web-service API).
503	Service Unavailable MUST be used to indicate that the server is currently unavailable (because it is overloaded or down for maintenance). Generally, this is a temporary state.



The TM Forum Open APIs won't present in the OAS specifications all the

return codes listed in the Design Guidelines.

3.4. User, Application and Extended Error Codes

The HTTP 4xx or 5xx standard error codes should always be used in the response header.

An API MAY support user and application specific error codes.

User and Application specific Error Codes MUST be represented in the Error Representation (Body) of the response.

Sub codes are possible (400-2) however they MUST be in the Error Representation (Body)

Error Representation in body of the response MUST have the following structure:

Property	Description	Mandatory
code	Application related code (as defined in the API or from a common list)	Mandatory
reason	Text that explains the reason for error. This can be shown to a client user.	Mandatory
message	Text that provide more details and corrective actions related to the error. This can be shown to a client user.	Optional
status	HTTP error code extension like 400-2	Optional
referenceError	url pointing to documentation describing the error	Optional
@type	The class-type of a REST resource.	Mandatory
@schemaLocation	it provides a link to the schema describing a REST resource.	Optional

Example:

```
400 Bad Request
Content-type:application/json
```

```
{
  "code": "ERR001",
  "reason": "Missing mandatory field",
  "message": "Please provide an Authorisation header",
  "@type": "Error"
```

```
}
```

Example:

when HTTP return code is extended (400-2):

```
400 Bad Request
Content-type:application/json
```

```
{
  "code": "ERR001",
  "reason": "Missing mandatory field",
  "message": "Please provide an Authorisation header",
  "status": "400-2",
  "@type": "Error"
}
```

The error codes if supported SHOULD be defined in the API specification.

@type and **@schemaLocation** SHOULD be used for runtime extension of the error.

3.5. Representations

- (Well-formed) JSON MUST be the default for resource representation
- XML and other formats may optionally be supported via content negotiation with the client
- Media type selection other than the default MUST be signaled with the **Content-Type** resp. **Accept** header.
- Documents (individual resources) MUST have a unique identifier field called **id**.
- All documentation SHOULD link to a schema (JSON schema or W3C XML Schema)
- Additional envelopes must not be created. A REST API must leverage the message “envelope” provided by HTTP.

3.6. Common Information Model

When applicable, the data types used to define the information model in API parameters SHOULD follow the [TMF Information Framework](#) reference model. In particular:

- Where there are fields defined in the TMF Information Framework for an API resource

then the same field names SHOULD be used in the API.

- The mandatory fields from the TMF Information Framework SHOULD be represented in the API resource model.
- The optional fields from the TMF Information Framework MAY be represented in the API resource model.
- If mandatory fields for the API resource are not found in the TMF Information Framework then the fields MAY be added to the TMF Information Framework.
- If optional fields for the API resource are not found in the TMF Information Framework then the fields MAY be added to the TMF Information Framework.

Chapter 4. Query Resources Patterns

The following section describes the structure and constraints of query operations.

Example:

All examples are relative to the management of TroubleTicket entities having the following JSON representation.



To keep examples shorter, we do not use the troubleTicketRelationship element of the overall structure.

```
{
  "id": "b890..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/b890..",
  "creationDate": "2022-05-31T07:34:45.968Z",
  "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2022-05-31T07:34:45.968Z",
  "name": "complaint over last bill",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Bill Dispute",
  "attachment": [
    {
      "description": "Scanned disputed bill",
      "attachmentType": "billCopy",
      "mimeType": "image/png",
      "name": "March Bill",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  }
},
```



```

    },
    {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "statusChangeHistory": [
    {
      "statusChangeDate": "2019-05-31T07:34:45.968Z",
      "statusChangeReason": "Need more information from the customer",
      "status": "pending",
      "@type": "StatusChange"
    }
  ],
  "@schemaLocation": "{apiRoot}/troubleTicket/v5/schema/troubleTicket.yaml",
  "@type": "TroubleTicket"
}

```

4.1. Query single Resource all Attributes

The following pattern **MUST** be used to retrieve the representation of a resource identified by the resource identifier **{resourceID}**:

```
GET {apiRoot} "/" {resourceName} "/" {resourceID}
```

If the resource exists, the complete resource representation (with all the attributes) **MUST** be returned.



When the resource does not have a value for an attribute, the attribute **MUST NOT** be included in the returned representation. Returning an empty value or the JSON value **null** for an attribute is not allowed.

The returned representation **MUST** contain a field called **id** and that field **MUST** be populated with the resource identifier value.

If the request is successful then the returned code MUST be **200 OK**.

If there is no matching resource then **404 Not Found** MUST be returned.

HTTP Response Codes – Success
200 OK
HTTP Response Codes – Failure
400 Bad Request
404 Not Found
405 Method Not Allowed
500 Internal Server Error
...

The exceptions must use the exception codes [\[IANA_STATUS\]](#) as explained in section [API Response Status and Exception Codes](#).

Example: Retrieving a single TroubleTicket with an **id** of **b890..**:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/b890..
```

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "id": "b890..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/b890..",
  "creationDate": "2022-05-31T07:34:45.968Z",
  "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle ",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2022-05-31T07:34:45.968Z",
  "name": "complaint over last bill",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
```

```
"statusCode": "2019-05-31T07:34:45.968Z",
"statusChangeReason": "Need more information from the customer",
"ticketType": "Bill Dispute",
"attachment": [
  {
    "description": "Scanned disputed bill",
    "attachmentType": "billCopy",
    "mimeType": "image/png",
    "name": "March Bill",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2019-05-31T07:34:45.968Z",
    "text": "This is quite important, please approach me as soon as possible",
    "@type": "Note"
  }
],
"relatedEntity": [
  {
    "role": "disputedBill",
    "@type": "RelatedEntity",
    "entity": {
      "id": "6936..",
      "href": "{apiRoot}/customerBillManagement/v5/customerBill/6936..",
      "name": "March 2019 Bill",
      "@type": "EntityRef",
      "@referredType": "CustomerBill"
    }
  }
],
"relatedParty": [
  {
    "role": "member",
    "partyOrPartyRole": {
      "id": "eb10..",
```

```

        "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
    "role": "reporter",
    "partyOrPartyRole": {
        "id": "387d..",
        "name": "Jacob Jac Miller",
        "@type": "PartyRef",
        "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
    "role": "customer",
    "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
}
],
"statusChangeHistory": [
    {
        "statusChangeDate": "2019-05-31T07:34:45.968Z",
        "statusChangeReason": "Need more information from the customer",
        "status": "pending",
        "@type": "StatusChange"
    }
],
"@schemaLocation": "{apiRoot}/troubleTicket/v5/schema/troubleTicket.yaml",
"@type": "TroubleTicket"
}

```

4.2. Querying multiple Resources

The following section describes how to use the **GET** operation to retrieve multiple resources

without specifying id's.

The following pattern must be used to retrieve the representation of all resources for a resource type corresponding to {resourceName}:

```
GET {apiRoot} "/" {resourceName}
```

The complete resource representations (with all the attributes) of all the matching entities must be returned.

The returned representation of each entity must contain a field called **id**.

If the request is successful then the returned code MUST be **200 OK**; otherwise, the appropriate error code SHOULD be returned.



If the collection of resource (possibly after applying any query parameters) is an empty list, the return code will still be **200 OK**.

HTTP Response Codes – Success
200 OK
HTTP Response Codes – Failure
400 Bad Request
405 Method Not Allowed
500 Internal Server Error
...

The exceptions must use the exception codes [\[IANA-STATUS\]](#) as explained in section [API Response Status and Exception Codes](#).

Example: Retrieving all TroubleTickets:



A unbounded request to a resource collection will be paginated as described in the [Query Resources with pagination](#)

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket
```

RESPONSE

200 OK

Content-Type: application/json

```
[
  {
    "id": "87b9..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/87b9..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "description": "I ordered a new mobile line 2 week ago, but the mobile device
did not arrived",
    "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
    "lastUpdate": "2023-05-31T07:34:45.968Z",
    "name": "Commerce problem",
    "priority": "High",
    "requestedResolutionDate": "2022-05-31T07:34:45.968Z",
    "resolutionDate": "2023-05-31T07:34:45.968Z",
    "severity": "Minor",
    "status": "resolved",
    "statusChangeDate": "2023-05-31T07:34:45.968Z",
    "statusChangeReason": "Apply full credit",
    "ticketType": "Commerce Issue",
    "troubleTicketCharacteristic": [
      {
        "id": "1",
        "name": "orderReferenceNumber",
        "valueType": "string",
        "value": "Order_888",
        "@type": "StringCharacteristic"
      },
      {
        "id": "2",
        "name": "orderCreationDate",
        "valueType": "string",
        "value": "2023-05-31",
        "@type": "StringCharacteristic"
      }
    ],
    "attachment": [
      {
        "description": "Scanned Order",
        "attachmentType": "orderCopy",
        "mimeType": "image/png",
        "name": "New Mobile Order",
```

```

        "url": "https://mycsp.com/docloader?docnum=3534888",
        "@type": "AttachmentRef"
    },
],
"channel": {
    "id": "f4b4..",
    "name": "self service",
    "@type": "ChannelRef"
},
"note": [
    {
        "id": "8eb1..",
        "author": "Jack Smith",
        "date": "2023-05-31T07:34:45.968Z",
        "text": "I did not recived the mobile device",
        "@type": "Note"
    }
],
"relatedEntity": [
    {
        "role": "disputedOrder",
        "@type": "RelatedEntity",
        "entity": {
            "id": "3e10..",
            "href": "{apiRoot}/productOrdering/v5/productOrder/3e10..",
            "name": "New Mobile Order",
            "@type": "EntityRef",
            "@referredType": "ProductOrder"
        }
    }
],
"relatedParty": [
    {
        "role": "member",
        "partyOrPartyRole": {
            "id": "bdfa..",
            "href": "{apiRoot}/partyManagement/v5/individual/bdfa..",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
        "role": "reporter",

```

```

    "partyOrPartyRole": {
      "id": "16d3..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "0747..",
      "href": "{apiRoot}/partyManagement/v5/customer/0747..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"statusChangeHistory": [
  {
    "statusChangeDate": "2023-05-28T07:34:45.968Z",
    "statusChangeReason": "trouble ticket created ",
    "status": "acknowledged",
    "@type": "StatusChange"
  },
  {
    "statusChangeDate": "2023-05-28T08:34:45.968Z",
    "statusChangeReason": "start process",
    "status": "inProgress",
    "@type": "StatusChange"
  },
  {
    "statusChangeDate": "2023-05-31T07:34:45.968Z",
    "statusChangeReason": "Resend new device to the customer",
    "status": "resolved",
    "@type": "StatusChange"
  }
],
"@type": "TroubleTicket"
},
{
  "id": "e457..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/e457..",

```

```

"creationDate": "2022-05-31T07:34:45.968Z",
"description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle ",
"expectedResolutionDate": "2019-06-10T07:34:45.968Z",
"lastUpdate": "2022-05-31T07:34:45.968Z",
"name": "complaint over last bill",
"priority": "High",
"requestedResolutionDate": "2022-05-31T07:34:45.968Z",
"resolutionDate": "2022-05-31T07:34:45.968Z",
"severity": "Minor",
"status": "resolved",
"statusChangeDate": "2022-05-31T07:34:45.968Z",
"statusChangeReason": "Apply full credit",
"ticketType": "Bill Dispute",
"troubleTicketCharacteristic": [
  {
    "id": "1",
    "name": "disputeAmount",
    "valueType": "number",
    "value": "40",
    "@type": "NumberCharacteristic"
  },
  {
    "id": "2",
    "name": "creditAmount",
    "valueType": "number",
    "value": "40",
    "characteristicRelationship": [
      {
        "id": "4",
        "relationshipType": "dependency",
        "@type": "CharacteristicRelationship"
      }
    ],
    "@type": "NubmerCharacteristic"
  },
  {
    "id": "3",
    "name": "currency",
    "valueType": "string",
    "value": "USD",
    "@type": "StringCharacteristic"
  },
  {
    "id": "4",

```

```

        "name": "creditReason",
        "valueType": "string",
        "value": "Network connectivity issues",
        "@type": "StringCharacteristic"
    },
],
"attachment": [
    {
        "description": "Scanned disputed bill",
        "attachmentType": "billCopy",
        "mimeType": "image/png",
        "name": "March Bill",
        "url": "https://mycsp.com/docloader?docnum=3534555",
        "@type": "AttachmentRef"
    }
],
"channel": {
    "id": "17d1..",
    "name": "self service",
    "@type": "ChannelRef"
},
"note": [
    {
        "id": "8f47..",
        "author": "Jack Smith",
        "date": "2019-05-31T07:34:45.968Z",
        "text": "This is quite important, please approach me as soon as
possible",
        "@type": "Note"
    }
],
"relatedEntity": [
    {
        "role": "disputedBill",
        "@type": "RelatedEntity",
        "entity": {
            "id": "e9e1..",
            "href": "{apiRoot}/customerBillManagement/v5/customerBill/e9e1..",
            "name": "March 2019 Bill",
            "@type": "EntityRef",
            "@referredType": "CustomerBill"
        }
    }
],
"relatedParty": [

```

```

{
  "role": "member",
  "partyOrPartyRole": {
    "id": "e271..",
    "href": "{apiRoot}/partyManagement/v5/individual/e271..",
    "name": "Sandy Smith",
    "@type": "PartyRef",
    "@referredType": "Individual"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
  "role": "reporter",
  "partyOrPartyRole": {
    "id": "a5cc..",
    "name": "Jacob Jac Miller",
    "@type": "PartyRef",
    "@referredType": "Individual"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
  "role": "customer",
  "partyOrPartyRole": {
    "id": "3fd5..",
    "href": "{apiRoot}/partyManagement/v5/customer/3fd5..",
    "name": "Jack Smith",
    "@type": "PartyRoleRef",
    "@referredType": "Customer"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
}
],
"statusChangeHistory": [
  {
    "statusChangeDate": "2022-05-28T07:34:45.968Z",
    "statusChangeReason": "trouble ticket created ",
    "status": "acknowledged",
    "@type": "StatusChange"
  },
  {
    "statusChangeDate": "2022-05-28T08:34:45.968Z",
    "statusChangeReason": "start process",
    "status": "inProgress",
    "@type": "StatusChange"
  }
]

```

```

    },
    {
      "statusChangeDate": "2022-05-29T07:34:45.968Z",
      "statusChangeReason": "Need more information from the customer",
      "status": "pending",
      "@type": "StatusChange"
    },
    {
      "statusChangeDate": "2022-05-31T07:34:45.968Z",
      "statusChangeReason": "Apply full credit",
      "status": "resolved",
      "@type": "StatusChange"
    }
  ],
  "externalIdentifier": [
    {
      "id": "bf3d..",
      "owner": "BMC Remedy",
      "externalIdentifierType": "Incident",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "TroubleTicket"
}
]

```

4.3. Query partial Resource Representation by Attribute Selection

The following section describes how to select a subset of the attributes of an entity to be present in a returned representation.

Attribute selection **MUST** be enabled on all first level attributes.

Attribute selection **MAY** optionally be enabled on all attributes and inner classes. Selection of the attributes on inner classes can be achieved using the `.(dot)` notation; for more details see [Complex Attribute filtering using the `.\(dot\)` notation](#).

An attribute selector directive called **fields** **MUST** be used to specify the attributes to be returned as part of a partial representation of a resource.

4.3.1. Attribute Inclusion

The following pattern **MUST** be used to retrieve the partial representation with the listed attributes of a resource identified by the resourceID:

```
GET {apiRoot} "/" {resourceName} [ "/" {resourceID} ] "?fields="
1#{attributeName}
```



The above follows the notation defined in section [Pattern Notation](#)

Example: `?fields=status,attachment.name`

If no attribute selector directive is provided, then the complete resource representation (with all the attributes) **MUST** be returned.

In order to indicate that no resource properties should be returned the following directive can be used:

`fields=None`



In all cases, and even if the directive `fields=None` is being used, the following rules apply:

- `id` and `href` **MUST** allways be returned in the resource body representation (where available).
- `@type` **MUST** be returned in the resource body representation.

The attribute selector `fields` **MAY** be used with other Uniform Operations (see [Lang](#) for an example where `fields` can be used with `POST` operation)

Example:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/529c..?fields=None
Content-Type: application/json
```

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "id": "529c..",
  "href": "/tmf-api/troubleTicket/v5/troubleTicket/529c..",
  ...
}
```

```
"@type": "TroubleTicket"
}
```

If the request is successful, then the returned code MUST be **200 OK**.

Exceptions must use the exception [\[IANA-STATUS\]](#) as explained in section [API Response Status and Exception Codes](#).

Example:

Retrieve the troubleTicket with an **id** of **072f..** and populate the representations with the description and status attributes.



The **id**, **href** and **@type** attributes are always present.

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/072f..?fields=description,status
```

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "id": "072f..",
  "href": " /troubleTicket/troubleTicket/072f..",
  "description": "Customer complaint over last invoice.",
  "status": "In Progress",
  "@type": "TroubleTicket"
}
```

4.3.2. Exclusion Marker

The attribute selector directive **fields** provides the ability to filter the list of attributes to be returned in a partial response, which means it will select only the attributes provided and implicitly discards all the others.

There is also the option to revert this behaviour by using the exclusion marker **!** (not) to *exclude* optional attributes from the partial response.

The pattern for the request is extended as follows:

```
GET {apiRoot} "/" {resourceName} "/" {resourceID} "?fields=" 1#( [!]  
{attributeName} )
```



The above follows the notation defined in section [Pattern Notation](#)

Example:

In the following example we want to retrieve the trouble ticket with `id=f6ac..` and we want a partial response representation with all the attributes except `relatedEntity` and `channel`. In this case the producer of the API will return all attributes but discard the ones mentioned above:

REQUEST

```
GET  
{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..?fields=!relatedEntity,!channel
```

RESPONSE

```
200 OK  
Content-Type: application/json
```

```
{  
  "id": "f6ac..",  
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",  
  "creationDate": "2023-09-01T06:55:33.798Z",  
  "description": "Experiencing a network outage in my area. No internet or phone  
service available.",  
  "expectedResolutionDate": "2023-09-10T07:34:45.968Z",  
  "lastUpdate": "2023-08-31T07:34:45.968Z",  
  "name": "Network Outage Report",  
  "priority": "High",  
  "requestedResolutionDate": "2023-09-05T07:34:45.968Z",  
  "severity": "Critical",  
  "status": "pending",  
  "statusChangeDate": "2023-08-31T07:34:45.968Z",  
  "statusChangeReason": "Investigating the outage",  
  "ticketType": "Network Outage",  
}
```

```
"attachment": [
  {
    "description": "Screenshot of network status",
    "attachmentType": "Screenshot",
    "mimeType": "image/png",
    "name": "Network Status Screenshot",
    "url": "https://mycsp.com/docloader?docnum=2c2b..",
    "@type": "AttachmentRef",
    "id": "2c2b.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "22db..",
    "author": "Jack Smith",
    "date": "2023-08-31T07:34:45.968Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedParty": [
  {
    "role": "Member",
    "partyOrPartyRole": {
      "id": "1f6f..",
      "href": "{apiRoot}/partyManagement/v5/individual/1f6f..",
      "name": "Sandy Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "Reporter",
    "partyOrPartyRole": {
      "id": "3bfb..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "Customer",
    "partyOrPartyRole": {
```

```

      "id": "a32d..",
      "href": "{apiRoot}/partyManagement/v5/customer/a32d..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  ],
  "statusChangeHistory": [
    {
      "statusChangeDate": "2023-08-31T07:34:45.968Z",
      "statusChangeReason": "Investigating the outage",
      "status": "pending",
      "@type": "StatusChange"
    }
  ],
  "@schemaLocation": "{apiRoot}/troubleTicket/v5/schema/troubleTicket.yaml",
  "@type": "TroubleTicket"
}

```

4.3.3. Mixing Attribute Selection and Exclusion

If both markers are present (the explicit inclusion one and the exclusion one) in a single query the explicit inclusion one takes precedence and the exclusion marker will be evaluated after.



It is not recommended to use both markers in the same query as they can be mutually exclusive.

Example:

If we consider the following query:

REQUEST

```

GET
{apiRoot}/troubleTicket/v5/troubleTicket/1057..?fields=creationDate,lastUpdate,status,!relatedEntity,!channel

```

it is expected that the API producer will fulfill first the explicit one

```
fields=creationDate,lastUpdate,status
```

and then on the outcome, to apply the second marker

```
fields=!relatedEntity,!channel
```

In the above example, after the first part the partial result is:

```
{
  "id": "1057..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1057..",
  "creationDate": "2019-05-31T07:34:45.968Z",
  "lastUpdate": "2019-05-31T07:34:45.968Z",
  "status": "pending",
  "@type": "TroubleTicket"
}
```

and the exclusion marker part **!** doesn't have any effect (since the exclusion marked attributes have already been removed).



If both markers are referring to the same attribute the server **MUST** return a **409 - Conflict** return code.

4.3.4. Resolving name conflicts

If the resource has an attribute with the **fields**, the attribute selector directive **fields** **MUST** be used with the **.(dot)** notation to resolve the name conflict.

Example:

Assume an API with resource **SomeResource** having an attribute **fields** of type **string**. A request to retrieve attributes "attribute1" and "attribute2" of the resource **SomeResource** is done as defined for the **fields** directive:

```
GET {apiRoot}/{apiName}/v5/someResource?fields=attribute1,attribute2
```

And to select entities by a using the **fields** attribute as query parameter, the following form can be used:

```
GET {apiRoot}/{apiName}/v5/someResource?.fields={value}
```

This may also be combined, using the **fields** attribute as query parameter and the **fields** directive to select attributes:

```
GET
{apiRoot}/{apiName}/v5/someResource?fields=attribute1,attribute2&.fields={value}
```

4.4. Query Resources with Attribute Filtering

The following section describes how to retrieve resources using an attribute filtering mechanism. The filtering is based on using name value query parameters on entity attributes.

The basic expression is a sequence of attribute assertions being combined by **AND** to formulate a filtering expression:

```
GET {apiRoot} "/" {resourceName} "?" {attributeName} "=" {attributeValue} *( "&"
{attributeName} "=" {attributeValue} )
```

For examples:

```
GET
{apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged&creationDate=2017-
04-20
```



The above expressions match only for attribute value equality.



In query parameters, shortened date/time values, e.g. date-only format, are allowed for filters.

Attribute value filtering using the logical **OR** operator is supported by including multiple instances of the same attribute in the filtering expression, each with a different value:

```
{attributeName} "=" {attributeValue 1} *( "&" {attributeName} "=" {attributeValue
2} )
```

(multiple occurrences of same attributeName, with different attributeValues).

The “OR” behavior can also be achieved by providing a filtering expression containing a comma- or semicolon-separated list of the alternative attributeValues:

```
{attributeName} "=" 1#( {attributeValue} )
```

```
{attributeName} "=" {attributeValue} *( ";" {attributeValue} )
```

For example:

- GET {apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged&status=rejected
- GET {apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged,rejected
- GET {apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged;rejected

and in combination with **AND**-combined other attribute selection:

- GET
{apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged&status=rejected&type=BillDispute
- GET
{apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged,rejected&type=BillDispute
- GET
{apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged;rejected&type=BillDispute

The following operators MAY be used.



The operator is inserted as postfix to the attributeName, either using the .(dot) form or the URI-encoded form; it does not replace the = symbol:

Operator	.(dot) form	URL encoded form	Description
=	.eq	%3D	equal to Returns results where the search criteria fields is "equal to", which is equivalent as if no operator is provided
>	.gt	%3E	greater than Return results where the search criteria field is strictly greater than. Example: GET {apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate.gt=2017-04-20 or GET {apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate%3E=2017-04-20
>=	.gte	%3E%3D	greater than or equal to Return results where the search criteria field is equal to or greater than. Example: GET {apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate.gte=2017-04-20 or GET {apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate%3E%3D=2017-04-20

Operator	.(dot) form	URL encoded form	Description
<	.lt	%3C	Less than Return results where the search criteria field is strictly less than. Example: GET <code>{apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate.lt=2017-04-20</code> or GET <code>{apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate%3C=2017-04-20</code>
<=	.lte	%3C%3D	less than or equal to Return results where the search criteria field is equal to or less than Example: GET <code>{apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate.lte=2017-04-20</code> or GET <code>{apiRoot}/troubleTicket/v5/troubleTicket?expectedResolutionDate%3C%3D=2017-04-20</code>

Operator	.(dot) form	URL encoded form	Description
	.regex		Regex expression (case insensitive) NOTE: all regex special characters must be encoded. NOTE: the regex expression MUST be separated from the = operator by ~ (tilde) as delimiter. Example: Consider a regex expression to search for the words Compliant and Bill: <pre>~^.*?\bCompliant\b.*?\bbill\b.*?\$</pre> which URI encoded is equivalent with: <pre>~%5E.%2A%3F%5CbCompliant%5Cb.%2A%3F%5Cbbill%5Cb.%2A%3F%24</pre> [source,http] ---- GET {apiRoot}/troubleTicket/v5/troubleTicket?description.regex=~%5E.%2A%3F%5CbCompliant%5Cb.%2A%3F%5Cbbill%5Cb.%2A%3F%24 ---- NOTE: using an unconstrained regular expression can be a potential vulnerability risk and thus SHOULD NOT be used. See [OWASP-ReDoS].

Examples:

Using ANDed to describe "2017-04-20>dateTime>2013-04-20" :

- GET `{apiRoot}/troubleTicket/v5/troubleTicket?dateTime%3E=2013-04-20&dateTime%3C=2017-04-20`

Using logical OR operator to describe creationDate<2013-04-20 or creationDate<2017-04-20 :

- `GET {apiRoot}/troubleTicket/v5/troubleTicket?dateTime%3C=2013-04-20;2017-04-20`



The returned representation of each entity must contain a field called `id` and that field be populated with the resourceID.



The returned representation of each entity must contain a field called `@type` and that field be populated with the entity's type.

If the request is successful then the returned code MUST be `200 OK`.

If the request is successfully completed but results in an empty list (no resource matched the supplied search criteria) the return code MUST be `200 OK`.

The exceptions must use the exception codes [\[IANA-STATUS\]](#) as explained in section [API Response Status and Exception Codes](#).

Example:

Retrieve all Trouble Tickets with `creationDate` greater than 2023-09-04 and status acknowledged:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?creationDate.gt=2023-09-04&status=acknowledged
```

or

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?creationDate%3E2023-09-04&status=acknowledged
```

RESPONSE

```
200 OK
Content-Type: application/json
```

```
[
  {
    "@type": "TroubleTicket",
    "id": "f6ac..",
```

```

    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "resolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "4aaf..",
        "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",
        "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
      }
    ],
    "channel": {
      "@type": "ChannelRef",
      "id": "8be1..",
      "href": "https://mycsp.com/channel/8be1..",
      "name": "Web"
    },
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "owner": "DocumentManagementSystem",
        "externalIdentifierType": "Screenshot",
        "id": "4aaf..",
        "href": "https://mycsp.com/dms/screenshot/4aaf.."
      }
    ],
    "note": [
      {
        "@type": "Note",
        "id": "3153..",
        "author": "Mr. N. Bene",

```

```
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "620d..",
      "name": "Mr. N. Bene",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "status": "acknowledged"
  }
],
"troubleTicketSpecification": {
  "@type": "TroubleTicketSpecificationRef",
  "id": "4aaf..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicketSpecification/4aaf..",
  "name": "NetworkOutageTroubleTicket",
  "version": "1.2"
},
"troubleTicketCharacteristic": [
```

```

    {
      "id": "4aaf..",
      "name": "OutageAddress",
      "valueType": "string",
      "value": "912143 London Road, BOURNEMOUTH, BH83 2EN",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "4aaf..",
          "relationshipType": "dependency"
        }
      ],
      "@type": "StringCharacteristic"
    }
  ],
  {
    "@type": "TroubleTicket",
    "id": "db3b..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/db3b..",
    "name": "Network Exchange Issue Report",
    "description": "Experiencing issues with the network exchange in the
Bournemouth area. Interruption in internet and phone services.",
    "severity": "Critical",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:30:22.143Z",
    "requestedResolutionDate": "2023-09-04T08:30:22.143Z",
    "expectedResolutionDate": "2023-09-04T08:30:22.143Z",
    "resolutionDate": "2023-09-04T08:30:22.143Z",
    "lastUpdate": "2023-09-04T08:30:22.143Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:30:22.143Z",
    "statusChangeReason": "Urgent attention required due to widespread impact",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "58d9..",
        "href": "{apiRoot}/documentManagement/v5/document/58d9..",
        "name": "Network Exchange Status Screenshot",
        "description": "Screenshot of network exchange status",
        "url": "https://mycsp.com/documents/docloader?docnum=58d9.."
      }
    ]
  },
  "channel": {

```

```

    "@type": "ChannelRef",
    "id": "b291..",
    "href": "https://mycsp.com/channel/b291..",
    "name": "BackOffice"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "58d9..",
      "href": "https://mycsp.com/dms/screenshot/58d9.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "d0ad..",
      "author": "Ms. A.Smith",
      "date": "2023-09-04T08:30:22.143Z",
      "text": "This is critical, all the addresses from this exchange are
affected."
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntity",
      "role": "Exchange",
      "entity": {
        "@type": "EntityRef",
        "id": "3f8d..",
        "href": "{apiRoot}/resourceInventoryManagement/v5/resource/3f8d..",
        "name": "EXC-MK-2023",
        "@referredType": "Resource"
      }
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Ms. A.Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ]

```

```

    }
  }
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:30:22.143Z",
    "statusChangeReason": "Urgent attention required due to widespread
impact",
    "status": "acknowledged"
  }
],
"troubleTicketSpecification": {
  "@type": "TroubleTicketSpecificationRef",
  "id": "3f8d..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicketSpecification/3f8d..",
  "name": "NetworkExchangeTroubleTicket",
  "version": "1.2"
},
"troubleTicketCharacteristic": [
  {
    "id": "3f8d..",
    "name": "OutageAddress",
    "valueType": "string",
    "value": "912143 London Road, BOURNEMOUTH, BH83 2EN",
    "characteristicRelationship": [
      {
        "@type": "CharacteristicRelationship",
        "id": "3f8d..",
        "relationshipType": "dependency"
      }
    ]
  }
]
}
]

```

4.4.1. Complex attribute filtering using the .(dot) notation

Complex attribute value type may be filtered using a "." notation. This allows the consumer to specify the path to the attribute using the simple .(dot) notation.

```
"?" {attributeName.attributeName} "=" {attributeValue} *["&"
```

```
{attributeName.attributeName} "=" {attributeValue} ]
```



Although the `.(dot)` notation provides a simple filtering mechanism for advanced filtering capabilities, it is recommended that JSON Path, described in [\[TMF630-6\]](#), should be used.

The complete resource representations (with all the attributes) of all the matching entities must be returned (unless otherwise restricted via the `fields` operator).



If the attribute is an array then all the elements of the array must be evaluated for filter matching.

Example: Array attribute filtering using `.(dot)` notation:

Let's consider the following data definitions:

```
[
  {
    "@type": "TroubleTicket",
    "id": "f6ac..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "resolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "4aaf..",
        "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",
        "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
      }
    ]
  }
]
```

```
}
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "620d..",
      "name": "N. Bene",
```

```

        "@type": "PartyRef",
        "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
}
],
"statusChangeHistory": [
{
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "status": "acknowledged"
}
]
},
{
    "@type": "TroubleTicket",
    "id": "db3b..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/db3b..",
    "name": "Network Exchange Issue Report",
    "description": "Experiencing issues with the network exchange in the
Bourmouth area. Interruption in internet and phone services.",
    "severity": "Critical",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:30:22.143Z",
    "requestedResolutionDate": "2023-09-04T08:30:22.143Z",
    "expectedResolutionDate": "2023-09-04T08:30:22.143Z",
    "resolutionDate": "2023-09-04T08:30:22.143Z",
    "lastUpdate": "2023-09-04T08:30:22.143Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:30:22.143Z",
    "statusChangeReason": "Urgent attention required due to widespread impact",
    "attachment": [
        {
            "@type": "AttachmentRef",
            "id": "58d9..",
            "href": "{apiRoot}/documentManagement/v5/document/58d9..",
            "name": "Network Exchange Status Screenshot",
            "description": "Screenshot of network exchange status",
            "url": "https://mycsp.com/documents/docloader?docnum=58d9.."
        }
    ]
},
"channel": {
    "@type": "ChannelRef",

```

```
    "id": "b291..",
    "href": "https://mycsp.com/channel/b291..",
    "name": "BackOffice"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "58d9..",
      "href": "https://mycsp.com/dms/screenshot/58d9.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "d0ad..",
      "author": "A.Smith",
      "date": "2023-09-04T08:30:22.143Z",
      "text": "This is critical, all the addresses from this exchange are
affected."
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntity",
      "role": "Exchange",
      "entity": {
        "@type": "EntityRef",
        "id": "3f8d..",
        "href": "{apiRoot}/resourceInventoryManagement/v5/resource/3f8d..",
        "name": "EXC-MK-2023",
        "@referredType": "Resource"
      }
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "A.Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ]
}
```

```
    }
  ],
  "statusChangeHistory": [
    {
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T08:30:22.143Z",
      "statusChangeReason": "Urgent attention required due to widespread
impact",
      "status": "acknowledged"
    }
  ]
}
```

If the consumer wants to retrieve the trouble ticket where the note author is 'Smith' using the **.(dot)** notation:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?note.author=A.Smith
```

Using the above query, the producer implementation will search through all the TroubleTicket resources and against each resource it will search in all **note** arrays for the attribute called **author**.

If the filtering condition is placed against an array where the items type is **object** without specifying the subsequent attribute, the API producer can either silently ignore the filter or return an appropriate error code.

For example, the following query will result in an error return code:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?note=Smith
```

In the above example, the **note** attribute is an array of objects as such the above query is invalid as the API Producer doesn't know in which array item attribute to look in order to match the filter.

4.5. Query Resources with Pagination

The majority of the APIs from the TM Forum Open APIs catalog support extensive datasets of resources (e.g., party, account, order, etc.).

The APIs could potentially return millions, billions of results fetched from the databases.

When the clients call the APIs to retrieve these datasets, they are partitioned and presented back in subsets (pages).

The process of splitting the result datasets into individual pages is called pagination.

Any implementation **MUST** apply by default one or another pagination technique, to prevent the drain of resources (server, network, etc.) (see [OWASP API4:2023 Lack of Resources & Rate Limiting](#)).

To support the return of extensive collections of resources, the GET HTTP method from the APIs **MUST** support pagination.



Pagination requires a succession of requests to retrieve the entire dataset. As usual for RESTful APIs, the requests **MUST** be stateless, so that the individual page requests will be completely independent of each other, as they may be served by different server instances, or even different data centers.

Thus, the server cannot access information on the previous requests, and the client **MUST** provide all the necessary information in each request to retrieve the next page of results.

In particular, when combined with filtering, the client **MUST** provide all the filtering criteria in each request to retrieve the correct next page of results.

The following methods of performing pagination are supported:

- [Offset pagination](#)
 - [Query Based Style](#)
 - [Range Header Style](#)
- [Cursor-based pagination](#)

Each pagination method offers different advantages.

It is up for the API implementation to decide which method they want to use depending on

the requirements.

However, at least one of the pagination methods **MUST** be supported by each API implementation.



Pagination requires that the ordering of the result set does not change between the successive requests for the pages.

4.5.1. Offset Pagination – Query Based Style

The *Offset Pagination* is one of the most used methods of paginating through the datasets.

It consists of returning the results of a particular "page" to the client, where the page is computed by splitting them using the total record count by the number of results per page.

Using this method, the clients provide parameters indicating how many results they want per page (*limit*) and the index (*offset*) from where the server should start to return the entries.

The server implementation will use those parameters to calculate the result set, starting from the *offset* and counting *limit* entries.

There are a set of advantages for using the *Offset Pagination*:

- The consumer can see the total number of pages and their position in the total number of pages
- The consumer can easily select and navigate to a particular page
- The offset pagination can be very easily implemented

There are also a set of disadvantages associated with the *Offset Pagination*:

- The *Offset Pagination* doesn't scale for large databases. For an extensive database, the database will still have to read up to the value of the *offset* for every page requested and count the number of entries specified by the *limit* parameter.
- If there is a dataset with a high frequency (write operations), the page window could be skipping or returning duplicate results.

In order to support the *Offset query-based pagination*, the following query parameters **MUST** be supported:

Parameter	Type	Description
<i>offset</i>	integer	Requested index for the start of resources to be provided in response requested by the client
<i>limit</i>	integer	Requested number of resources to be provided in response requested by the client

Further clarifications on the *offset* and *limit* query parameters:

- *offset*
 - If the *offset* parameter is not provided it MUST be defaulted to zero
 - If the *offset* parameter is provided, it MUST be a non-negative integer.
 - If the *offset* parameter is provided as a negative value, the server MUST reply with **400 Bad Request**.
 - If the *offset* parameter is a positive integer but out of range for the collection, the server SHOULD respond with an empty collection with the status code **200 OK**.
- *limit*
 - If the *limit* parameter is provided, it MUST be a positive integer.
 - If the *limit* parameter is provided as a non-positive value, the server SHOULD reply with **400 Bad Request**.
 - If the *limit* parameter is not supplied, the server's default value will be used. If the server doesn't put a default value for the number of results (page size), all the results SHOULD be returned.
 - If the server has defined a max limit and the client requests a higher limit, the server MAY return **400 Bad Request** or MAY enforce the server defined limit.

If we consider the following dataset: [0,1,2,3,4,5,6,7,8,9,10] of items, an example of pagination will look like this:

Query	Dataset Returned	Page
offset=0&limit=4	[0,1,2,3]	Get the first 4 matching resources, equivalent to <i>Page 1</i>
offset=4&limit=4	[4,5,6,7]	Get the next 4 resources starting from the 4th position, equivalent to <i>Page 2</i>

Query	Dataset Returned	Page
offset=8&limit=4	[8,9,10]	Get the next 4 resources starting from the 8th position. In this case because there are no more four entries the server will return all the remaining three resources, equivalent to <i>Page 3</i>

Every time the server implementation receives the *offset* parameter, that is an indication that it needs to skip the entries up to the *offset* value and return from that position a number of entries up to the *limit* value.

A typical request using the *offset query-based pagination* looks like this:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10
```

If the *offset* query parameter is missing, then it MUST default to zero.

If the *limit* query parameter is missing, it MUST be defaulted to the server implementation limit.

?limit=20	Get the first twenty matching resources.
?offset=0&limit=20	Get the first twenty matching resources.
?offset=10&limit=20	Get the twenty resources starting after the tenth



It is always advisable that the server have a default and a top value for the *limit* parameter. The top value for the limit is preventing a possible resource attack on the server (see [\[OWASP-API4:2023-RL\]](#)).

For example, considering a server *limit* default value set to 20 and a top limit value set to 40, we have the following example:

?offset=0	Get the first 20 matching resources. (The server default <i>limit</i> value is set to 20)
?offset=10&limit=20	Get the 20 resources starting after the 10th
?offset=20&limit=30	Get the 30 resources starting after the 20th

?offset=50&limit=100	Get the 40 resources starting after the 50th. In this case the server top limit value is overriding the client request for the limit.
----------------------	--

For the query-based pagination, the server **MUST** return

- the HTTP header **X-Result-Count** in response, with the number of matching resources provided with this response, so that the client can calculate the next page.
- the HTTP header **X-Total-Count** in response, with the total number of matching resources so that the client can calculate the number of pages.

The HTTP Status Code returned by server in the success scenario **MUST** be **200 OK**, complemented of course for all the other HTTP Status Codes relevant for the error scenarios.

The server **MUST** return navigation links as [Web Linking HTTP header](#), to ease the navigation and avoid that client has to construct the related links (self, first, next, previous, last). The server can compute the navigation links' values based on the values of the *offset*, *limit* and the total number of results.

Assuming we are paginating using the *TroubleTicket* API with a dataset of 50 items in the database, below is an example of the navigation links returned to the consumer that made the following request:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10
```

RESPONSE

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10>; rel="self",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=0&limit=10>; rel="first",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=30&limit=10>; rel="next",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=10&limit=10>; rel="prev",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=40&limit=10>; rel="last"
```

The links can follow an *absolute path* or a *relative path*, it is up to the implementation to decide which version they want to use.

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10>; rel="self"
```

or

```
Link: </tmf-api/troubleTicket/v5/troubleTicket?offset=20&limit=10>; rel="self"
```

Assuming we are paginating using the *TroubleTicket* API and we have in the dataset a collection of **50** TroubleTicket resources.

A possible scenario will be a consumer making the following request:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10
```

RESPONSE

200 OK

X-Result-Count:10

X-Total-Count:50

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?offset=20&limit=10>;rel="self",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=0&limit=10>; rel="first",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=30&limit=10>;rel="next",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=10&limit=10>; rel="prev",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=40&limit=10>; rel="last"
```

```
[  
  {  
    "id": "8850..",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/8850..",  
    "status": "acknowledged",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "bfba..",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/bfba..",  
    "status": "acknowledged",  
    "@type": "TroubleTicket"  
  },  
  ...  
]
```

```
{
  "id": "d17a..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/d17a..",
  "status": "acknowledged",
  "@type": "TroubleTicket"
}
```

As seen in the above response sample, the server has returned 10 entries (*limit=10*) starting after the 20th (*offset=20*).

If the client will navigate to the last page, following the hypermedia link:

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?offset=40&limit=10>; rel="last"
```

the following API Request call will be made:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?offset=40&limit=10
```

Since this is the last page, the following response will be returned:

RESPONSE

```
200 OK
```

```
X-Result-Count:10
X-Total-Count:50
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?offset=40&limit=10>;rel="self",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=0&limit=10>;
rel="first",
      <{apiRoot}/troubleTicket/v5/troubleTicket?offset=30&limit=10>; rel="prev"
```

```
[
  {
    "id": "447e6..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/447e6..",
    "status": "acknowledged",
```

```
    "@type": "TroubleTicket"
  },
  {
    "id": "5f96..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/5f96..",
    "status": "acknowledged",
    "@type": "TroubleTicket"
  },
  ...
  {
    "id": "b580..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/b580..",
    "status": "acknowledged",
    "@type": "TroubleTicket"
  }
]
```

Attribute filtering may be combined in a single request as per the following example.

For example, if the customer will request all the TroubleTicket entries with the status "rejected" and the server has only five entries, the *X-Result-Count* header's value will be 5, the *X-Total-Count* header will change its value to 5, and the server will return only those entries matching the filter specified by the client:



If the *limit* query parameter is missing, the server will default the implementation limit. In the example below it was omitted on purpose.

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?status=rejected
```

RESPONSE

```
200 OK
```

```
X-Result-Count:5
X-Total-Count:5
Link:
<{apiRoot}/troubleTicket/v5/troubleTicket?status=rejected&offset=0&limit=10>;rel=
"self"
```

```
[
  {
    "id": "a3fc..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/a3fc..",
    "status": "rejected",
    "@type": "TroubleTicket"
  },
  {
    "id": "3a41..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/3a41..",
    "status": "rejected",
    "@type": "TroubleTicket"
  },
  {
    "id": "5449..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/5449..",
    "status": "rejected"
  },
  {
    "id": "4bc2..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/4bc2..",
    "status": "rejected",
    "@type": "TroubleTicket"
  },
  {
    "id": "6fc3..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/6fc3..",
    "status": "rejected",
    "@type": "TroubleTicket"
  }
]
```

4.5.2. Offset Pagination – Range Based Style

The [Offset Pagination](#) can also be implemented by using the [Range](#) header style defined in [\[RFC9110\]](#) (section 14.2).

The [Content-Range](#) header defined in [\[RFC9110\]](#) (section 14.4) MAY also be used to control the amount of data returned.

This header is present in the request and controls the minimum, and maximum values returned. Filtered queries return collections, and the [Content-Range](#) header is relative to the entities in the returned collection.

For example:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged
```

RESPONSE

200 OK

Content-Type: application/json
Accept-Ranges: items
Content-Range: items 1-10/30

```
[
  {
    "id": "1",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1",
    "status": "acknowledged",
    "@type": "TroubleTicket"
  },
  {
    "id": "2",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/2",
    "status": "acknowledged",
    "@type": "TroubleTicket"
  },
  ...
  {
    "id": "10",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/10",
    "status": "acknowledged",
    "@type": "TroubleTicket"
  }
]
```

A set of items to be noted in the above response:

- The **Accept-Ranges** header defined in [\[RFC9110\]](#) (section 14.3) is present and is indicating that the server can accept a range of units based on *items*

- The **Content-Range** header defined in [\[RFC9110\]](#) (section 14.4) provides the information regarding the *current page* (e.g., **1-10**) and the *total number of results* found (e.g., **30**), both values expressed in number of *items*.



The **Range** Header style's response code is **200 OK** (rather than **206 Partial Content**) because the request did not include the **Range** header (see [\[RFC9110\]](#) (section-15.3.7)).

The client must parse the **Content-Range** header to discover that not all items, but only a subset have been returned, and further requests are needed to retrieve the remaining ones.

If the client wants to navigate to a particular page, it will pass the range information in the **Range** header.

The following example shows how to use the **Range** header to iterate a collection of trouble tickets:

The example assumes that 15 trouble tickets match the filtering criteria and that the maximum amount of trouble tickets being retrieved by calls is set by default to 10.

For example:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?status=acknowledged
```

```
Range: items=0-9
```

RESPONSE

```
206 Partial Content
```

```
Content-Type: application/json
```

```
Accept-Ranges: items
```

```
Content-Range: items 0-9/15
```

```
[
  {
    "id": "0",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/0",
```

```
"status": "acknowledged",
"@type": "TroubleTicket"
},
{
  "id": "1",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1",
  "status": "acknowledged",
  "@type": "TroubleTicket"
},
...
{
  "id": "9",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/9",
  "status": "acknowledged",
  "@type": "TroubleTicket"
}
]
```

Since the request included the **Range** header, the server is allowed to respond with the **206 Partial Content** status code defined in RFC9110 (section-15.3.7), indicating the presence of the **Content-Range** response header.

If the server is not capable of returning the information on the total number of entries, it will return (*):

RESPONSE

206 Partial Content

Content-Type: application/json
Accept-Ranges: items
Content-Range: items 0-9/*

4.5.3. Cursor-based Pagination

The *cursor* represents a pointer to a particular item in the dataset.

The client, instead of sending the *offset* (index) to allow the server to compute the page entries, in the cursor pagination method, it will send the *cursor*. The server will return the results after or before the given pointer/cursor.

As part of the request, the client would also have to send, along the *cursor*, a parameter to

indicate the number of results per page, *limit*.

Although the cursor-based pagination addresses some of the drawbacks of the *offset-based pagination*, it has several tradeoffs:

- The cursor must be based on a unique column (or columns) having with unique values.
- There is no concept of the total number of pages of results in the set.
- The client can't jump to a specific page unless it has a cursor to that page or a navigation link.

Cursor

A *cursor* is an opaque string created by the server or available in the dataset that divides the results dataset into those that fall *before*, *after*, or *on* the cursor.

The *cursor* MUST be a unique value, based on sequential column or computed.

For example, consider the following simplified dataset:

```
[
  {
    "id": "1000",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1000",
    "@type": "TroubleTicket"
  },
  {
    "id": "1010",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1010",
    "@type": "TroubleTicket"
  },
  {
    "id": "1020",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1020",
    "@type": "TroubleTicket"
  },
  {
    "id": "1030",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1030",
    "@type": "TroubleTicket"
  },
  {
    "id": "1040",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1040"
```

```
},
{
  "id": "1050",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1050"
},
{
  "id": "1060",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1060",
  "@type": "TroubleTicket"
},
{
  "id": "1070",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1070",
  "@type": "TroubleTicket"
},
{
  "id": "1080",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1080",
  "@type": "TroubleTicket"
},
{
  "id": "1090",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1090",
  "@type": "TroubleTicket"
},
{
  "id": "1100",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1100",
  "@type": "TroubleTicket"
}
]
```

For the following entry:

```
{
  "id": "1010",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1010",
  "@type": "TroubleTicket"
}
```

The cursor value can be the entry id: "1010".

As such, the entry for "id=1000" will fall *before* the cursor, and all the other entries will fall

after.

```
[
  {
    "id": "1000",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1000",    --Note: "before"
the cursor
    "@type": "TroubleTicket"
  },
  {
    "id": "1010",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1010",    --Note: "on" the
cursor
    "@type": "TroubleTicket"
  },
  {
    "id": "1020",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1020",    --Note: "after"
the cursor
    "@type": "TroubleTicket"
  }
]
```

The dataset might change, but as long as the server can identify the *cursor*, the position of the element in the list, it can produce the *before* and *after* results.

Another example for a potential cursor can be the document ID generated by [MongoDB](#), because it is already defined by MongoDB as unique and based on timestamp (which allows ordering the results by natural order).

The id field here is 12 bytes in length and consist of

- a 4-byte timestamp value, representing the ObjectId's creation, measured in seconds since the Unix epoch
- a 5-byte random value
- a 3-byte incrementing counter, initialized to a random value

The cursor could also be an integer, a timestamp, etc and there should not have any limitation on what a service could use for the cursor. It is easier to either specify the cursors as String so that the content can be cast to what the service is using for the cursor, or simply leave it open-ended for the implementation service to decide.

Query Parameters

In order to support the cursor-based pagination, the following query parameters **MUST** be supported:

Parameter	Type	Description
<i>limit</i>	integer	Requested number of resources to be provided in response requested by the client
<i>before</i>	string	An opaque string value representing the page results before the cursor value
<i>after</i>	string	An opaque string value representing the page results after the cursor value

The value of *before* and *after* query parameters is the cursor value.

The client usually sends the *after* parameter to get to the next page, where the *before* is sent to get the previous page.

Further clarifications on the *limit* query parameter:

- *limit*
 - If the *limit* parameter is provided, it **MUST** be a positive integer.
 - If the *limit* parameter is provided as a non-positive value, the server **MUST** reply with **400 Bad Request**.
 - If the *limit* parameter is not supplied, the server's default value will be used. If the server doesn't put a default value for the number of results (page size), all the results should be returned.
 - If the server has defined a max limit and the client requests a higher limit, the server **MAY** return **400 Bad Request** or **MAY** enforce the server defined limit.

For example, to retrieve the previous five results before the cursor *1050*, paginating backward:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?limit=5&before=1050
```

RESPONSE

200 OK

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?limit=5&before=1050>;  
rel="self"  
      <{apiRoot}/troubleTicket/v5/troubleTicket?limit=5&after=1040>; rel="next"
```

```
[  
  {  
    "id": "1000",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1000",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1010",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1010",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1020",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1020",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1030",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1030",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1040",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1040",  
    "@type": "TroubleTicket"  
  }  
]
```



In the above example, the navigation link for *prev* is not present due to the fact that there are no other resources based on our example dataset.

If instead, we want to retrieve the next five results, paginating forward, the request will be:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?limit=5&after=1040
```

RESPONSE

200 OK

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?limit=5&after=1040>;  
rel="self",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?limit=5&after=1090>;  
rel="next",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?limit=5&before=1050>;  
rel="prev",
```

```
[  
  {  
    "id": "1050",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1050",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1060",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1060",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1070",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1070",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1080",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1080",  
    "@type": "TroubleTicket"  
  },  
  {  
    "id": "1090",  
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1090",  
    "@type": "TroubleTicket"  
  }  
]
```

```
]
```

Further combinations between the selectors *before* and *after* are possible.

For example, to retrieve all the results between the cursors *1010* and *1045*:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?after=1010&before=1050
```

RESPONSE

200 OK

Link: <{apiRoot}/troubleTicket/v5/troubleTicket?after=1010&before=1050>;
rel="self"

```
[
  {
    "id": "1020",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1020",
    "@type": "TroubleTicket"
  },
  {
    "id": "1030",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1030",
    "@type": "TroubleTicket"
  },
  {
    "id": "1040",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1040",
    "@type": "TroubleTicket"
  }
]
```

If the server doesn't recognize the cursor in the *before* or *after* query parameters, then it MUST respond with **400 Bad Request**.

If the limit value is set to one (1), then the *before* and *after* cursors would be pointing to the

same cursor.

Navigation Links

The server MUST return [Navigation Links](#) as Web Linking HTTP header, to ease the navigation and avoid that client has to construct the related links ([self](#), [next](#), [previous](#)).

```
Link: <{apiRoot}/troubleTicket/v5/troubleTicket?limit=2&after=1060>; rel="next",  
      <{apiRoot}/troubleTicket/v5/troubleTicket?limit=2&before=1030>; rel="prev",
```

The *rel=next* points to the next data page.



If omitted from the server response it means this is the last page of data.

The *rel=prev* points to the previous data page.



If omitted from the server response it means this is the first page of data.

4.5.4. "before" and "after" parameters

When the *after* parameter is being used, the returned results MUST have as its first item, the item immediately after the cursor in the dataset.



An exception is that if there are no items in the result list that fall after the cursor, the server MUST return an empty array.

When the *before* parameter is being used, the last item returned in the paginated data MUST be the item that is before the cursor.



An exception is that if there are no items that fall before the cursor, the server MUST return an empty array.

For example, considering the same dataset as before (described [here](#)), assuming the client will send the following request:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?limit=2&before=1040
```

RESPONSE

200 OK

```
[
  {
    "id": "1020",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1020",
    "@type": "TroubleTicket"
  },
  {
    "id": "1030",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1030",
    "@type": "TroubleTicket"
  }
]
```

As well for the *after* parameter, if the client will send the following request:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?limit=2&after=1040
```

RESPONSE

200 OK

```
[
  {
    "id": "1050",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1050",
    "@type": "TroubleTicket"
  },
  {
    "id": "1060",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1060",
    "@type": "TroubleTicket"
  }
]
```

If both parameters are omitted, depending on the server pagination technique, the pagination results MUST start with the first item from the result dataset.

4.5.5. Combining "after" and "before"

Clients MAY combine the *after* and *before* parameters together on the same request.

By doing so, it will create a range for the pagination request as the client is asking for all the results starting from immediately after the *after* cursor and continuing up until the *before* cursor. If the *after* cursor is not before the *before* cursor, the created range is empty.

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?after=1020&before=1060
```

RESPONSE

200 OK

```
[
  {
    "id": "1030",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1030",
    "@type": "TroubleTicket"
  },
  {
    "id": "1040",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1040",
    "@type": "TroubleTicket"
  },
  {
    "id": "1050",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/1050",
    "@type": "TroubleTicket"
  }
]
```



If the range exceeds the limit (either provided by the client or the server max limit), the server can limit the number of results returned – as if the *before* was not supplied.

4.5.6. Filtering with Cursor Pagination

Collection filtering based on different criteria can still be used in conjunction with the cursor technique, similar to the offset technique. The server will filter the result dataset based on the client's criteria, which will be the result dataset source. The cursor will be a value within the dataset. If the combination of the filtering criteria and cursor value doesn't produce any result, an empty array **MUST** be returned.

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?creationDate.gt=2013-04-20&status=acknowledged&after=1030
```

4.6. Query Sub-Resources / Sub-Resource Pattern

Resource definitions usually contain sub-objects or collections of sub-objects. Some objects/collections are **existentially dependent** on the resource/parent. *Existentially dependent* means that a child object cannot exist without its parent. An example of such a relationship is a line-item collection from an invoice.

This relationship is modelled through the **sub-resource pattern** (iterator).

Consider the *TroubleTicket* example, where the *troubleTicket* resource contains a collection of *notes*. There are cases where this collection of *notes* can reach hundreds of elements without the ability to paginate through them or without the ability to retrieve only one particular note.

These collections can be exposed as *sub-resources*, allowing a consumer to retrieve only the *note* it is interested in.

The *sub-resources* are exposed using a nested resource URL approach. The nested URL can convey the relationship that one resource belongs to another one.

In order for the **sub-resource pattern** to be applied the following conditions **MUST** be met:

- Sub-resource collection/array list **MUST** include ID as a property
- The value of the ID property **MUST** be unique within the sub-resource collection/array list.
- The value of the ID property **MUST** be immutable and never reused (i.e. unique for all times) within the sub-resource collection/array list.



Immutability as well as uniqueness over time exclude the use of the sub-

resource collection/array index as sub-resource ID.



Exposing sub-resources is an optional pattern that can be applied by every implementation when there is a need to do so.

4.6.1. Operations supported

The only operation allowed for *sub-resources* access SHOULD be **GET**.



All the other operations *create,update,delete* SHOULD be performed through the parent resource.



The here described operations are only supported where the API Specification mentions them explicitly.

The following pattern is used to model the *sub-resources*:

```
GET "/" {resource} "/" {resource-id} "/" {sub-resource}
```



If the resource identified by the resource-id does not exist, the **404 Not Found** response code MUST be returned. This does not contradict the general rule that the **404 Not Found** response code must not be in context of a GET on a collection (with or without query parameters), because this error refers to the resource that is queried for the sub-resource, not to the sub-resource itself.

The *sub-resources* could potentially be chained, allowing the exposure of a *sub-resource* that belongs to another *sub-resource*.

```
GET "/" {resource} "/" {resource-id} "/" {sub-resource} "/" {sub-resource-id}
  "/" {sub-resource2}
GET "/" {resource} "/" {resource-id} "/" {sub-resource} "/" {sub-resource-id}
  "/" {sub-resource2} "/" {sub-resource2-id}
```

For example:

```
GET
{apiRoot}/productOrderingManagement/v5/8fc1../productOrderItem/6346../qualificati
```

on/f832..



Although chaining mechanism MAY be used, chaining too many levels can result in a rather long and complicated URL. As such, it is recommended that the depth limit of the *sub-resources* URIs SHOULD be kept to 1.

For example:

```
{apiRoot}/troubleTicket/v5/troubleTicket/{troubleTicketID}/note
```

or

```
{apiRoot}/geographicAddressManagement/v5/geographicAddress/{geographicAddressId}/  
geographicSubAddress/{geographicSubAddressId}
```



It is important to note here the resource *context* or the relativity of the sub-resources *identifiers*. The sub-resource is only in the context of the parent resource.

Example:

The client's intention is to retrieve the note collection from the ticket with ID **fad8..**

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/fad8../note
```

RESPONSE

200 OK

```
[  
  {  
    "id": "3d5f..",  
    "author": "Jack Smith",  
    "date": "2019-05-31T07:34:45.968Z",  
    "text": "This is quite important, please approach me as soon as  
possible",  
    "@type": "Note"  
  },  
  ...  
]
```

```
{
  "id": "db39d6b-ad3c-4b89-84b4-210265d7ced2",
  "author": "Joe Maxim",
  "date": "2019-05-31T07:34:45.968Z",
  "text": "The service seems to be affected",
  "@type": "Note"
}
```



The sub-resources query operation follow the same rules as the query on the resources, described across [Query Resource Patterns](#).

4.6.2. Hypermedia Support

As explained in more detail in [\[TMF630-3\]](#), through hypermedia, a resource should expose one or multiple links via the Link header, specified by [\[RFC8288\]](#). If the *sub-resource* pattern is being used, then the parent resource MUST include the relevant links to the *sub-resources* exposed.

For example:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/ee61..
```

RESPONSE

200 OK

Link: <{apiRoot}/troubleTicket/troubleTicket/ee61..>; rel="self"
<{apiRoot}/troubleTicket/troubleTicket/ee61../note>; rel="collection"

The hypermedia relationship for "*collection*" and "*item*" is defined as per [RFC6573](#) and [IANA Link Relationships](#).

The *collection* link is used to indicate that the resource is a collection of sub-resources, that can be iterated to retrieve the sub-resource items, including application of pagination.

4.7. Query Resources with Attribute Filtering and Attribute Selection

Attribute filtering and attribute selection may be combined in a single request as per the following example:

REQUEST

```
GET
{apiRoot}/troubleTicket/v5/troubleTicket?fields=description,status&creationDate.g
t=2013-04-20&status=acknowledged
```

4.8. Sorting

To support sorting the **sort** directive is used in HTTP query parameters.

If sorting is used then following request parameter **MUST** be supported:

Sort-Query-Parameter

"sort" "=" ([{Sort-Direction}] {Sort-Field})

Sort-Direction

"-" / "+"

Sort-Field

A field to sort on.

The default direction is ascending order, the use of the modifier in front of the sort field name, "-", changes the sort order direction.

Sort the result set on the **name** field:

```
?sort=name
```

Sort the result set on the name in descending order:

```
?sort=-name
```

Nested resource attributes can be specified, e.g.:

```
?sort=parentResource.childResource.attributeName
```

Comma separated list of resource attributes can be provided in the request:

```
?sort=attributeName,...,attributeName
```

Sorting precedence is specified by the left to right order of the fields listed.

REQUEST

```
GET  
{apiRoot}/troubleTicket/v5/troubleTicket?sort=creationDate,statusChangeDate,note.  
date
```



The **.sort** format can be used (as for any directive) if there is an ambiguity regarding the semantic of sort within the query parameters.

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket?.sort=creationDate
```

Chapter 5. Modify Resources Patterns

The following section describes the patterns used to modify resources.

5.1. Uniform Contract Operations for modifying Resources

TMF REST APIs MUST only use **PATCH** to modify the attributes of an entity.

PUT MUST NOT be used.

5.2. PATCH Syntax Alternatives

PATCH MUST be used to update a limited subset of the attributes of an entity as described in [\[RFC5789\]](#), using the various **PATCH** syntax alternatives:

- **JSON Merge Patch** as per [\[RFC7396\]](#) with **application/merge-patch+json**
- **JSON PATCH** as per [\[RFC6902\]](#) with **application/json-patch+json**
- **JSON PATCH Query** as per [\[TMF630-5\]](#) with **application/json-patch-query+json**



In case of **PATCH** if **application/json** is provided then the default rule will be to apply the same rules as for JSON Merge Patch.



Using **application/json** for the **PATCH** operation is **deprecated** for Version 5 of the TMF Open APIs and **will be removed completely** in Version 6.

The following applies to all three alternatives:

The HTTP **PATCH** method is atomic, as per [RFC5789](#), i.e. it either succeeds completely or not at all.

If the request is processed successfully then the returned code MUST be **200 OK**.

If the request is processed asynchronously then the returned code MUST be **202 Accepted**.



The following attributes MUST NOT be modified via a PATCH request:

- **id**
- **href**
- **@type**
- **@baseType**
- **@schemaLocation**

Exceptions must use the exception codes [\[IANA-STATUS\]](#).

HTTP Response Codes – Success
200 OK 202 Accepted
HTTP Response Codes – Failure
400 Bad Request 404 Not Found 405 Method Not Allowed 409 Conflict 422 Unprocessable Content 500 Internal Server Error ...

Full representation of the modified resource **MUST** be included in the response if the request is successful, error code otherwise with [error representation](#) in body.

The **fields** selector (see section [Query partial Resource Representation by Attribute Selection](#)) can be used as part of the **PATCH** operation to achieve a partial response in order to reduce the size of the response payload (where applicable).

If the **PATCH** operation fails, then the **fields** selector will be ignored.

5.3. Modify single-valued Attributes of a Resource

Although all three alternative methods can be used to modify a single-valued attribute of a resource, the JSON Merge Patch method with the **application/merge-patch+json** content type is the most straightforward and simple way to do so.

The request payload contains the resource reduced to the attributes to be modified, and the respective values to be set. The server will then apply the change to the resource identified by the URL, retaining the values of all not provided attributes.

Example using *JSON Merge Patch*:

Change the status of the Trouble Ticket with **ID=e36b..** to "resolved"

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/e36b..  
Content-Type: application/merge-patch+json
```

```
{
  "@type": "TroubleTicket",
  "status": "resolved",
  "statusChangeReason": "Issue found and fixed"
}
```



Although the **@type** attribute must not be modified, it has to be included in the request to ensure that the server can correctly interpret the request as a **TroubleTicket** resource. Therefore, this attribute is mandatory also in the PATCH request payload.

RESPONSE

200 OK
Content-Type: Application/json

```
{
  "@type": "TroubleTicket",
  "id": "e36b..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/e36b..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T10:02:23.121Z",
  "lastUpdate": "2023-09-04T10:02:23.121Z",
  "priority": "High",
  "status": "resolved",
  "statusChangeDate": "2023-09-04T10:02:23.121Z",
  "statusChangeReason": "Issue found and fixed",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "f65b..",
      "href": "{apiRoot}/documentManagement/v5/document/f65b..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
    }
  ]
}
```

```
        "url": "https://mycsp.com/documents/docloader?docnum=f65b.."
      },
    ],
    "channel": {
      "@type": "ChannelRef",
      "id": "1d1d..",
      "href": "https://mycsp.com/channel/1d1d..",
      "name": "Web"
    },
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "f65b..",
      "href": "https://mycsp.com/dms/screenshot/f65b.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "873e..",
      "author": "Mr. A. Johnson",
      "date": "2023-09-04T08:59:27.255Z",
      "text": "This is critical, please address it urgently."
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntity",
      "role": "Exchange",
      "entity": {
        "@type": "EntityRef",
        "id": "8a2a..",
        "href": "{apiRoot}/resourceInventoryManagement/v5/resource/8a2a..",
        "name": "EXC-MK-1872",
        "@referredType": "Resource"
      }
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {

```

```

        "id": "e2f5..",
        "name": "Mr. S. Clark",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "statusChangeHistory": [
    {
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T08:59:27.255Z",
      "statusChangeReason": "Need more information from the customer",
      "status": "acknowledged"
    },
    {
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T10:02:23.121Z",
      "statusChangeReason": "Issue found and fixed",
      "status": "resolved"
    }
  ]
}

```

When JSON Merge Patch with content-type `application/merge-patch+json` is used to modify attributes of a resource, passing the `null` as attribute value will result in that particular attribute to be removed (as per [RFC5789](#)).

Example:

REQUEST

```

PATCH tmf-api/troubleTicket/v5/troubleTicket/0bfb..
Content-Type: application/merge-patch+json

```

```

{
  "@type": "TroubleTicket",
  "description": null
}

```

RESPONSE

200 OK

Content-Type: Application/json

```
{
  "@type": "TroubleTicket",
  "id": "0bfb..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/0bfb..",
  "name": "Network Outage Report",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T10:02:23.121Z",
  "lastUpdate": "2023-09-04T10:02:23.121Z",
  "priority": "High",
  "status": "resolved",
  "statusChangeDate": "2023-09-04T10:02:23.121Z",
  "statusChangeReason": "Issue found and fixed",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "f65b..",
      "href": "{apiRoot}/documentManagement/v5/document/f65b..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
      "url": "https://mycsp.com/documents/docloader?docnum=f65b.."
    }
  ],
  "channel": {
    "@type": "ChannelRef",
    "id": "1d1d..",
    "href": "https://mycsp.com/channel/1d1d..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "f65b..",
      "href": "https://mycsp.com/dms/screenshot/f65b.."
    }
  ]
}
```

```
],
"note": [
  {
    "@type": "Note",
    "id": "873e..",
    "author": "Mr. A. Johnson",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "8a2a..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/8a2a..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "e2f5..",
      "name": "Mr. S. Clark",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "status": "acknowledged"
  },
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T10:02:23.121Z",
```

```
    "statusChangeReason": "Issue found and fixed",  
    "status": "resolved"  
  }  
]  
}
```

5.4. Modify multi-valued Attributes of a Resource

When using the JSON Merge Patch method to modify a multi-valued attribute of a resource (e.g. an array), the value of the attribute (i.e. the complete array) is replaced with the new value provided in the request.

To modify only a specific element in a multi-valued attribute using JSON Merge Patch, the attribute must be provided entirely in the request, including all existing values and the new value to be added or modified. This can be cumbersome and inefficient, especially for large arrays.

Therefore, the JSON PATCH method can be used for modifying only parts of multivalued attributes, as it allows for more granular control over the changes made to the array.

To use JSON PATCH, the **Content-Type** MUST be set to

```
Content-Type: application/json-patch+json
```

and the payload consists of an array of directives for changing, adding, removing array elements as defined in the JSON PATCH specification.

Each directive specifies the operation to be performed (**add**, **replace**, **copy**, **move**, **test**, **remove**), the path to the to be changed element (a JSON Pointer), and (depending on the operation) the value to be added or modified.



The HTTP **PATCH** method is atomic, as per [\[RFC5789\]](#), i.e. either *all* operations succeed completely, or *none at all*.

Example:

Add a Note entry to the **note** array of an individual Trouble Ticket.

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/0bfb..  
Content-type: application/json-patch+json
```

```
[
  {
    "op": "add",
    "path": "/note",
    "value": [
      {
        "@type": "Note",
        "id": "389b..",
        "author": "George Yohani",
        "date": "2023-09-05T12:43:02.255Z",
        "text": "The issue has been identified."
      }
    ]
  }
]
```

where:

- **op** is the action to take. The following operations are supported:
add, remove, replace, copy, move, test
- **path** is a [JSON Pointer](#) to the field to change, relative to the API root.
- **value** is the value to use, optional depending on the value of **op**

For details, see the JSON PATCH specification [\[RFC6902\]](#).

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "@type": "TroubleTicket",
  "id": "0bfb..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/0bfb..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
}
```

```

"expectedResolutionDate": "2023-09-04T08:59:27.255Z",
"resolutionDate": "2023-09-04T10:02:23.121Z",
"lastUpdate": "2023-09-04T10:02:23.121Z",
"priority": "High",
"status": "resolved",
"statusChangeDate": "2023-09-04T10:02:23.121Z",
"statusChangeReason": "Need more information from the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "f65b..",
    "href": "{apiRoot}/documentManagement/v5/document/f65b..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=f65b.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "1d1d..",
  "href": "https://mycsp.com/channel/1d1d..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "f65b..",
    "href": "https://mycsp.com/dms/screenshot/f65b.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "873e..",
    "author": "Mr. A. Johnson",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  },
  {
    "@type": "Note",
    "id": "389b..",
    "author": "George Yohani",
    "date": "2023-09-05T12:43:02.255Z",

```

```
    "text": "The issue has been identified."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "8a2a..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/8a2a..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "e2f5..",
      "name": "Mr. S. Clark",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "status": "acknowledged"
  },
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T10:02:23.121Z",
    "statusChangeReason": "Issue found and fixed",
    "status": "resolved"
  }
]
}
```

The JSON Patch method relies on a JSON Pointer to identify the to be changed element. To address a specific element of an array, it needs the element's ID (or if that does not exist, the element's array index; but see the [discouragement of using the array index as ID](#)).

5.5. JSON-Patch Extension to manage Arrays

JSON Patch Query defined in part 5 of the Guideline extends the JSON Patch document by extending the JSON Pointer used as **path** by a query expression, as used for a query operation, of the form **?{attributeName}={attributeValue}**.

This enables value-based identification of the element to which the operation applies. Neither the element's ID nor its index within the array needs to be known in advance.

This can be further extended by using the **filter** query parameter defined in part 6 of the Guideline, which has a JSON Path as value, providing even more granular control of the target element's identification.

The **application/json-patch-query+json** media type is used to identify JSON Patch Query documents.

Care must be taken as both a JSON Pointer with additional query parameters and a JSON Path might match multiple elements within an array, and thus the operation will be applied to all of them.

The following is an example JSON Patch Query document, transferred in an HTTP **PATCH** request:

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/4f32..  
Content-type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "/attachment?id=f65b..e"  
  },  
  {  
    "op": "replace",  
    "path": "/note?author=Mr. A. Johnson",  
    "value": {  
      "@type": "Note",  
      "id": "873e..",  
    }  
  }  
]
```

```

    "author": "Mr. A. Johnson",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is VERY critical, please address it IMMEDIATELY."
  }
}
]
```

The URL of the request addresses a particular TroubleTicket, and then the JSON Pointer parts of the two elements of the payload's array identify which parts of that TroubleTicket will be affected by the respective operation.

The first operation, **remove**, will be performed on the attachment with the specified **id** value and remove it from the **attachment** collection of the TroubleTicket.

The second operation, **replace**, will replace the entire elements of the **note** collection whose "author" attribute has the value "Mr. A. Johnson". The operation will replace each of the matching elements (this could be many!) entirely with the provided value.



This must be applied very carefully, because as can be seen from the second operation of the example, many sub-resources might match the query criteria and thus will be replaced with the provided value, which may not be the intended outcome.

RESPONSE

```

200 OK
Content-Type: application/json-patch+query
```

```

{
  "@type": "TroubleTicket",
  "id": "4f32..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/4f32..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone
service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T10:02:23.121Z",
  "lastUpdate": "2023-09-04T10:02:23.121Z",
```

```
"priority": "High",
"status": "resolved",
"statusChangeDate": "2023-09-04T10:02:23.121Z",
"statusChangeReason": "Need more information from the customer",
"channel": {
  "@type": "ChannelRef",
  "id": "1d1d..",
  "href": "https://mycsp.com/channel/1d1d..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "f65b..",
    "href": "https://mycsp.com/dms/screenshot/f65b.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "873e..",
    "author": "Mr. A. Johnson",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is VERY critical, please address it IMMEDIATELY."
  },
  {
    "@type": "Note",
    "id": "389b..",
    "author": "George Yohani",
    "date": "2023-09-05T12:43:02.255Z",
    "text": "The issue has been identified."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "8a2a..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/8a2a..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
]
```

```

    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "e2f5..",
      "name": "Mr. S. Clark",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "status": "acknowledged"
  },
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T10:02:23.121Z",
    "statusChangeReason": "Issue found and fixed",
    "status": "resolved"
  }
]
}

```

To avoid having to provide the complete Note element where only the value **text** attribute is to be changed, the JSON Path variant can be applied, writing the directive for the second operation as:

```

[
  ...,
  {
    "op": "replace",
    "path": "/note?filter=$.[?(@.author=='Mr. A. Johnson')].text",
    "value": "This is VERY critical, please address it IMMEDIATELY."
  }
]

```

```
]
```

or by merging the JSON Pointer part into the JSON Path and using that directly as **path**:

```
[
  ...,
  {
    "op": "replace",
    "path": "$.note[?(@.author=='Mr. A. Johnson')].text",
    "value": "This is VERY critical, please address it IMMEDIATELY."
  }
]
```

Refer to [\[TMF630-5\]](#) for a complete list of examples of the use of JSON PATCH Query document for partial update of a resource.

Chapter 6. Create Resource Patterns

The following section describes the patterns used to create resources.

6.1. Creating a single Resource

POST MUST be used to create single resources.

The POST request body should include values for each attribute that is specified as mandatory (either in the schema or in the conformance profile).

The id (and href) of a new resource is expected to be provided by the implementation and as such is not provided as input to a create operation.

The implementation of the POST operation should ensure that attributes of the created entity are populated according to the request body, or by default values if the attribute is not present in the request body.

A successful synchronously processed creation MUST return a **201 Created** HTTP code on success.

The return message of a successful synchronously processed creation MUST contain a resource representation (including the implementation-provided **id** and **href**) and a **Location** header with the created resource's URI.

The **fields** parameter may be used in request to control the resource representation in response according to [Query partial Resource representation or attribute selection](#).

See [Create a single resource with attribute selection](#) for an example.

A creation operation MUST return a **202 Accepted** HTTP response code when the resource creation is asynchronous.

Response Code List – Success
201 Created 202 Accepted
Response Code List – Failure
400 Bad Request 405 Method Not Allowed 409 Conflict 422 Unprocessable Content 500 Internal Server Error ...

If operation is asynchronous, then please see the [asynchronous operation pattern](#) for more details.

Example:

Create a trouble ticket

REQUEST

```
POST {apiRoot}/troubleTicket/v5/troubleTicket
Content-Type: application/json
```

```
{
  "@type": "TroubleTicket",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone
service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T10:02:23.121Z",
  "priority": "High",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "8e4d..",
      "href": "{apiRoot}/documentManagement/v5/document/8e4d..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
      "url": "https://mycsp.com/documents/docloader?docnum=8e4d.."
    }
  ],
  "channel": {
    "@type": "ChannelRef",
    "id": "3df4..",
    "href": "https://mycsp.com/channel/3df4..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",

```

```
    "externalIdentifierType": "Screenshot",
    "id": "8e4d..",
    "href": "https://mycsp.com/dms/screenshot/8e4d.."
  },
],
"note": [
  {
    "@type": "Note",
    "id": "28da..",
    "author": "Mr. B. Smith",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "c43e..",
      "name": "Ms. L. Davis",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
]
}
```

RESPONSE

201 Created
Location: {apiRoot}/troubleTicket/v5/troubleTicket/3b31..
Content-Type: application/json

```
{
  "@type": "TroubleTicket",
  "id": "3b31..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/3b31..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone service available.",
  "severity": "High",
}
```

```
"ticketType": "NetworkIssues",
"creationDate": "2023-09-04T08:59:27.255Z",
"requestedResolutionDate": "2023-09-04T08:59:27.255Z",
"expectedResolutionDate": "2023-09-04T08:59:27.255Z",
"lastUpdate": "2023-09-04T08:59:27.255Z",
"priority": "High",
"status": "acknowledged",
"statusChangeDate": "2023-09-04T08:59:27.255Z",
"statusChangeReason": "Trouble Ticket raised by the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "8e4d..",
    "href": "{apiRoot}/documentManagement/v5/document/8e4d..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=8e4d.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "3df4..",
  "href": "https://mycsp.com/channel/3df4..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "8e4d..",
    "href": "https://mycsp.com/dms/screenshot/8e4d.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "28da..",
    "author": "Mr. B. Smith",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedParty": [
  {
```

```
{
  "role": "reporter",
  "partyOrPartyRole": {
    "id": "c43e..",
    "name": "Ms. L. Davis",
    "@type": "PartyRef",
    "@referredType": "Individual"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
},
],
"statusChangeHistory": [
  {
    "@type": "StatusChange",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Trouble Ticket raised by the customer",
    "status": "acknowledged"
  }
]
}
```

The resource created is identified by a service-generated URI. The **id** in the resource representation and the URI are related.



The **id**, **href** and **@type** attribute MUST be present in the response (see [\[Resource ID and href\]](#))



Empty string attributes (e.g., **date**: "") MUST NOT be passed as part of the resource payload when creating the resource.

The **fields** selector (see [\[fieldsOperator\]](#)) can be used as part of the POST operation to reduce the size of the response payload on success return codes.



If the create operation fails, then the **fields** selector will be ignored.

Example:

Create a trouble ticket

REQUEST

```
POST {apiRoot}/troubleTicket/v5/troubleTicket?fields=name,description,severity
Content-Type: application/json
```

```
{
  "@type": "TroubleTicket",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone
service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T10:02:23.121Z",
  "priority": "High",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "8e4d..",
      "href": "{apiRoot}/documentManagement/v5/document/8e4d..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
      "url": "https://mycsp.com/documents/docloader?docnum=8e4d.."
    }
  ],
  "channel": {
    "@type": "ChannelRef",
    "id": "3df4..",
    "href": "https://mycsp.com/channel/3df4..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "8e4d..",
      "href": "https://mycsp.com/dms/screenshot/8e4d.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "28da..",
      "author": "Mr. B. Smith",
      "date": "2023-09-04T08:59:27.255Z",
      "text": "This is critical, please address it urgently."
    }
  ]
}
```

```
],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "c43e..",
        "name": "Ms. L. Davis",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ]
}
```

RESPONSE

201 Created
Location: {apiRoot}/troubleTicket/v5/troubleTicket/fe6f..
Content-Type: application/json

```
{
  "@type": "TroubleTicket",
  "id": "fe6f..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/fe6f..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone service available.",
  "severity": "High"
}
```

6.2. Create a single Resource with Attribute Selection

Attribute selection MAY be used with **POST** operation.

Example:

REQUEST

```
POST {apiRoot}/troubleTicket/v5/troubleTicket?fields=none
```

Content-Type: application/json

```
{
  "@type": "TroubleTicket",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone
service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "priority": "High",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "e18a..",
      "href": "{apiRoot}/documentManagement/v5/document/e18a..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
      "url": "https://mycsp.com/documents/docloader?docnum=e18a.."
    }
  ],
  "channel": {
    "@type": "ChannelRef",
    "id": "6a86..",
    "href": "https://mycsp.com/channel/6a86..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "e18a..",
      "href": "https://mycsp.com/dms/screenshot/e18a.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "f0f0..",
      "author": "Ms. M. Johnson",
      "date": "2023-09-04T08:59:27.255Z",
```

```
    "text": "This is critical, please address it urgently."
  },
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "e18a..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/e18a..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "afca..",
      "name": "Ms. M. Johnson",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
]
}
```

RESPONSE

201 Created

Location: http:server:port/troubleTicketManagement/troubleTicket/f825..

Content-Type: application/json

```
{
  "id": "f825..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f825..",
  "@type": "TroubleTicket"
}
```

Chapter 7. Delete Resource Pattern

A Delete operation on a resource is used to remove an instance of a resource.



It is an implementation specific decision if the removal of the resource instance is either logical or physical.

A Delete uses the HTTP method DELETE on a Resource URI, for example:

REQUEST

```
DELETE /troubleTicket/troubleTicket/{id}
```

RESPONSE

```
204 No Content
```

or

```
202 Accepted
```

A Delete operation MUST use an HTTP DELETE on a resource.

A Delete operation MUST NOT accept a request payload.

If a DELETE method is successfully applied, the origin server SHOULD send

- a **204 No Content** status code if the action has been enacted and no further information is to be supplied,
- a **202 Accepted** status code if the action will likely succeed but has not yet been enacted

Response Code List – Success
204 No Content
202 Accepted
Response Code List – Failure

400 Bad Request
404 Not Found
405 Method Not Allowed
409 Conflict
422 Unprocessable Content
500 Internal Server Error
...

Chapter 8. Operations on Collection Resources

JSON Patch with `application/json-patch+json` MAY be used to perform multiple operations on a collection resource, such as creating or deleting multiple members of the collection in a single request.



The Internet media type for a JSON Patch document is `application/json-patch+json`. Alternatively, also JSON Patch Query defined in [TMF630-5] with `application/json-patch-query+json` can be used.

JSON Patch ("`application/json-patch+json`") documents are atomic, if one operation in the document fails, then no operation will be carried out as per [RFC6902]; this is also due to the HTTP PATCH method being atomic, as per [RFC5789].

The JSON PATCH operation is relative to the request's target collection resource. For each of the directives of the PATCH, the "path" element (a JSON Pointer relative to the target collection, possibly extended by a query expression, or a JSON Path) can be used to select a child-resource or a collection of child-resources of the request's target.



When a JSON Pointer used as `path` and it is of the form `/somepath/` (i.e. if it ends with a `/` character), this is interpreted as the element `""` (empty string) of `/somepath`, which usually will lead to an error, because it does not point to a valid child-resource, see [RFC6901] (section 5).

To address a non-existing element at the end of an array, `/-` must be appended to the pointer to the array.

8.1. Adding a non-existing element to a collection

To add a new entity to a collection, the `op` directive MUST be set to `add`, and the new entity will be created at the element identified by the `path`. To add the entity at the end of the collection array, the `path` MUST end with `/-`, see [RFC6902] (section 4.1).

For added entities, not all the attributes have to be specified in the request, but at least all the required attributes MUST be populated.

8.2. Updating an existing element of a collection

To update an existing entity, the `op` directive MUST be set to `replace`, and the `path` MUST point to the element to be updated. The value of the `value` element in the PATCH document MUST contain the new element, which replaces the existing one completely.

If the `path` identifies several entities, all of them will be replaced with the new value.

Not all the attributes of the new element have to be specified in the request, but at least all the required attributes **MUST** be populated.

8.3. Removing an existing element from a collection

To remove an existing entity, the **op** directive **MUST** be set to **remove**, and the **path** **MUST** point to the element to be removed.

If the **path** identifies several entities, all of them will be removed.

A **value** element **MUST NOT** be present in the PATCH directive for a **remove** operation.

For removed resources, the return message **MUST NOT** contain any resource representation.

8.4. Mixed operations

A Patch operation on a collection **MAY** contain a mix of **add**, **replace**, and **remove** operations, as long as the **path** elements are valid and do not conflict with each other.

When a PATCH request contains several directives, the order of the directives is significant. The operations are applied in the order they are specified in the request body.

8.5. Response Codes

A successful **PATCH** must return either **200 OK** HTTP code when processed synchronously, or a **202 Accepted**. **201 Created** or **204 No Content** **MUST NOT** be used, because a **PATCH** request may also include several types of directives.

A **PATCH** operation **MUST** return a **400 Bad Request** if the request body is not valid JSON, if the **path** elements are not valid JSON Pointers, or if the **op** directive is not one of the allowed values (**add**, **replace**, **remove**).

Otherwise, a **PATCH** operation **MUST** return the status code corresponding to the first failing operation, respecting the sequence of the directives contained in the request body, and possibly the sequence of elements identified by the **path**.

Response Code List – Success
200 OK 202 Accepted
Response Code List – Failure

400 Bad Request
404 Not Found
405 Method Not Allowed
409 Conflict
422 Unprocessable Content
500 Internal Server Error
...

Example:

Create multiple trouble tickets

REQUEST

```
PATCH /api/troubleTicket
Content-type: application/json-patch+json
```

```
[
  {
    "op": "add",
    "path": "/-",
    "value": {
      "@type": "TroubleTicket",
      "name": "Network Outage Report",
      "description": "Experiencing a network outage in my area. No internet or
phone service available.",
      "severity": "High",
      "ticketType": "NetworkIssues",
      "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
      "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
      "priority": "High",
      "attachment": [
        {
          "@type": "AttachmentRef",
          "id": "e18a..",
          "href": "{apiRoot}/documentManagement/v5/document/e18a..",
          "name": "Network Status Screenshot",
          "description": "Screenshot of network status",
          "url": "https://mycsp.com/documents/docloader?docnum=e18a.."
        }
      ]
    },
    "channel": {
```

```
"@type": "ChannelRef",
"id": "6a86..",
"href": "https://mycsp.com/channel/6a86..",
"name": "Web"
},
"externalIdentifier": [
{
"@type": "ExternalIdentifier",
"owner": "DocumentManagementSystem",
"externalIdentifierType": "Screenshot",
"id": "e18a..",
"href": "https://mycsp.com/dms/screenshot/e18a.."
}
],
"note": [
{
"@type": "Note",
"id": "f0f0..",
"author": "Ms. M. Johnson",
"date": "2023-09-04T08:59:27.255Z",
"text": "This is critical, please address it urgently."
}
],
"relatedEntity": [
{
"@type": "RelatedEntity",
"role": "Exchange",
"entity": {
"@type": "EntityRef",
"id": "e18a..",
"href": "{apiRoot}/resourceInventoryManagement/v5/resource/e18a..",
"name": "EXC-MK-1872",
"@referredType": "Resource"
}
}
],
"relatedParty": [
{
"role": "reporter",
"partyOrPartyRole": {
"id": "afca..",
"name": "Ms. M. Johnson",
"@type": "PartyRef",
"@referredType": "Individual"
}
},
```

```

        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    ]
  },
  {
    "op": "add",
    "path": "/-",
    "value": {
      "@type": "TroubleTicket",
      "name": "Network Exchange Issue Report",
      "description": "Experiencing issues with the network exchange in the
Bourmouth area. Interruption in internet and phone services.",
      "severity": "Critical",
      "ticketType": "NetworkIssues",
      "requestedResolutionDate": "2023-09-04T08:30:22.143Z",
      "expectedResolutionDate": "2023-09-04T08:30:22.143Z",
      "priority": "High",
      "attachment": [
        {
          "@type": "AttachmentRef",
          "id": "ac1c..",
          "href": "{apiRoot}/documentManagement/v5/document/ac1c..",
          "name": "Network Exchange Status Screenshot",
          "description": "Screenshot of network exchange status",
          "url": "https://mycsp.com/documents/docloader?docnum=ac1c.."
        }
      ],
      "channel": {
        "@type": "ChannelRef",
        "id": "f238..",
        "href": "https://mycsp.com/channel/f238..",
        "name": "BackOffice"
      },
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "owner": "DocumentManagementSystem",
          "externalIdentifierType": "Screenshot",
          "id": "ac1c..",
          "href": "https://mycsp.com/dms/screenshot/ac1c.."
        }
      ],
      "note": [
        {

```

```

        "@type": "Note",
        "id": "b38e..",
        "author": "Mr. J. Adams",
        "date": "2023-09-04T08:30:22.143Z",
        "text": "This is critical, all the addresses from this exchange are
affected."
    },
    ],
    "relatedEntity": [
        {
            "@type": "RelatedEntity",
            "role": "Exchange",
            "entity": {
                "@type": "EntityRef",
                "id": "f018..",
                "href": "{apiRoot}/resourceInventoryManagement/v5/resource/f018..",
                "name": "EXC-MK-2023",
                "@referredType": "Resource"
            }
        }
    ],
    "relatedParty": [
        {
            "role": "reporter",
            "partyOrPartyRole": {
                "id": "bd0e..",
                "name": "Mr. J. Adams",
                "@type": "PartyRef",
                "@referredType": "Individual"
            }
        }
    ]
}
]

```

RESPONSE

```

[
  {
    "@type": "TroubleTicket",
    "id": "a6a0..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/a6a0..",
    "name": "Network Outage Report",

```

```

    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Trouble Ticket created by the customer",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "e18a..",
        "href": "{apiRoot}/documentManagement/v5/document/e18a..",
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",
        "url": "https://mycsp.com/documents/docloader?docnum=e18a.."
      }
    ],
    "channel": {
      "@type": "ChannelRef",
      "id": "6a86..",
      "href": "https://mycsp.com/channel/6a86..",
      "name": "Web"
    },
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "owner": "DocumentManagementSystem",
        "externalIdentifierType": "Screenshot",
        "id": "e18a..",
        "href": "https://mycsp.com/dms/screenshot/e18a.."
      }
    ],
    "note": [
      {
        "@type": "Note",
        "id": "f0f0..",
        "author": "Ms. M. Johnson",
        "date": "2023-09-04T08:59:27.255Z",
        "text": "This is critical, please address it urgently."
      }
    ]
  }

```

```

    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntity",
        "role": "Exchange",
        "entity": {
          "@type": "EntityRef",
          "id": "e18a..",
          "href": "{apiRoot}/resourceInventoryManagement/v5/resource/e18a..",
          "name": "EXC-MK-1872",
          "@referredType": "Resource"
        }
      }
    ],
    "relatedParty": [
      {
        "role": "reporter",
        "partyOrPartyRole": {
          "id": "afca..",
          "name": "Ms. M. Johnson",
          "@type": "PartyRef",
          "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    ],
    "statusChangeHistory": [
      {
        "@type": "StatusChange",
        "statusChangeDate": "2023-09-04T08:59:27.255Z",
        "statusChangeReason": "Trouble Ticket created by the customer",
        "status": "acknowledged"
      }
    ]
  },
  {
    "@type": "TroubleTicket",
    "id": "7256..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/7256..",
    "name": "Network Exchange Issue Report",
    "description": "Experiencing issues with the network exchange in the
    Bourmouth area. Interruption in internet and phone services.",
    "severity": "Critical",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:30:22.143Z",

```

```

"requestedResolutionDate": "2023-09-04T08:30:22.143Z",
"expectedResolutionDate": "2023-09-04T08:30:22.143Z",
"lastUpdate": "2023-09-04T08:30:22.143Z",
"priority": "High",
"status": "acknowledged",
"statusChangeDate": "2023-09-04T08:30:22.143Z",
"statusChangeReason": "Trouble Ticket created by the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "ac1c..",
    "href": "{apiRoot}/documentManagement/v5/document/ac1c..",
    "name": "Network Exchange Status Screenshot",
    "description": "Screenshot of network exchange status",
    "url": "https://mycsp.com/documents/docloader?docnum=ac1c.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "f238..",
  "href": "https://mycsp.com/channel/f238..",
  "name": "BackOffice"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "ac1c..",
    "href": "https://mycsp.com/dms/screenshot/ac1c.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "b38e..",
    "author": "Mr. J. Adams",
    "date": "2023-09-04T08:30:22.143Z",
    "text": "This is critical, all the addresses from this exchange are
affected."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",

```

```

    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "f018..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/f018..",
      "name": "EXC-MK-2023",
      "@REFERREDType": "Resource"
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "bd0e..",
        "name": "Mr. J. Adams",
        "@type": "PartyRef",
        "@REFERREDType": "Individual"
      }
    }
  ],
  "statusChangeHistory": [
    {
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T08:30:22.143Z",
      "statusChangeReason": "Trouble Ticket created by the customer",
      "status": "acknowledged"
    }
  ]
}
]

```



In the above example, certain attributes are created by the API producer, the default ones **id**, **href** and maybe some other based on the API implementation logic (e.g., status, lastUpdate, statusChangeHistory).

The resources created are identified by a service-generated URI. The ID in the resource representation and the URI are related.

Chapter 9. Task Resource Pattern

This section describes the use of Task resources to expose complex operations that are not representable as standard CRUD entity based operations.

Examples:

- topupBalance
- cancelAlarm
- cancelProductOrder
- queryProductRecommendation

9.1. Modeling complex Operations with Task Resources

If proper REST design is limited during translation from action-based operational interfaces to REST interfaces, **it is allowable** to define a resource for the operation and **POST** to that operation to execute.

A Task resource name **MUST** be a **verb** representing the task to be executed, E.g. validate, check, heal, migrate, clear, close, transfer, etc.

Execution of a task is requested by creating (POST) of a Task Resource.

Task creation may result in the creation of a number of related resources or other tasks.

Task operation return codes:

Response Code List – Success
200 OK 201 Created 202 Accepted 204 No content
Response Code List – Failure
400 Bad Request 405 Method Not Allowed 409 Conflict 417 Expectation Failed 422 Unprocessable Content 500 Internal Server Error ...

For details when to use which of the success codes, see the following sections.

Tasks can be split into two major categories:

- short-running tasks, where the result of the task can be returned in the same call to the client
- long-running tasks, where the result cannot be computed immediately; then the task resource can be created and returned to the client in the same API call

When an implementation for a Task supports both alternatives, it MAY decide at runtime to execute the task as short-running or long-running task, taking into consideration its expectation of the duration of task execution, e.g. by investigating any request parameters or payload values.

In that case, the consumer must be able to distinguish and handle both variants of the task operation.

See the [Type-selection by Server](#) section below for more detailed description and an example.



Irrespective of the task category, all task operations can also be handled asynchronously (see [asynchronous operations pattern](#)).



If the server supports it, the client might request a switch of the task operation, e.g. from synchronous to asynchronous via the *Expect* header (see [Expect Header](#)).

If a Task Resource is created and persisted due to the task operation, the server MUST return an *id*, *href*, *@type* and the *Location* header for it.

If an *id* is returned, it MUST represent the Task Resource created for the task.

Iterators MAY be supported.

9.2. Short-running Tasks

Short-running tasks represent the tasks where the task result is returned to the client in the same API call. Usually, they are tasks that are quick to perform and don't require a long processing time (e.g. *cancelAlarm*).

Due to their nature, the short-running tasks usually don't require a lifecycle as the task will transition through the task lifecycle and reach the final status, and the final result will be presented (however, it is implementation specific if a short-running task should have a

state transition or not).

It is the implementation or even runtime decision if the Task Resource for a short-running task is persisted or not.

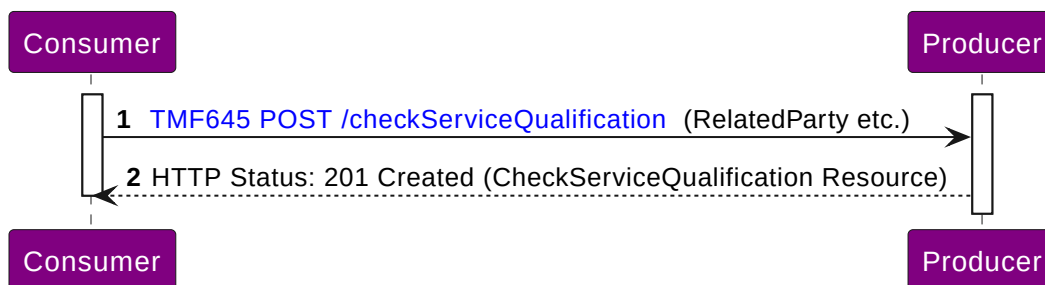
- If it is not persisted, the server returns the task's result data in the payload, with a **200 OK** status, or, if there is no result to be returned, with **204 No Content** status.
- If it is persisted, the server **MUST** return the created task resource, with **id**, **href**, **@type** and the **Location** header identifying the created task resource, with a **201 Created** status. The returned task resource **SHOULD** contain the operation's result data, if any. The task resource together with the results can also be retrieved later, as long as the implementation chooses to retain it.



Left to each API's design, a short-running task could be mandated to always create a task resource, to never create a task resource, or to leave it up to the implementation decision.

Task example with return code **201 Created**

In this example, the task has been completed successfully, and the full representation of the persisted task resource is returned.



REQUEST

```
POST {apiRoot}/serviceQualificationManagement/v5/checkServiceQualification
Content-Type: application/json
```

```
{
  "@type": "CheckServiceQualification",
  "description": "Broadband Fiber Service Qualification for Residential Area",
  "expectedQualificationDate": "2023-08-31T14:30:00.000Z",
  "externalIdentifier": [
    {
```

```

        "id": "a634..",
        "@type": "ExternalIdentifier"
    }
],
"provideAlternative": true,
"instantSyncQualification": false,
"provideUnavailabilityReason": true,
"relatedParty": [
    {
        "role": "Buyer contact",
        "@type": "RelatedPartyOrPartyRole",
        "partyOrPartyRole": {
            "@type": "PartyRef",
            "id": "2186..",
            "href": "{apiRoot}/partyManagement/v5/individual/2186..",
            "name": "Alice Johnson",
            "@referredType": "Individual"
        }
    }
],
"serviceQualificationItem": [
    {
        "id": "1",
        "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "service": {
            "place": [
                {
                    "@type": "RelatedPlaceRefOrValue",
                    "role": "Installation place",
                    "place": {
                        "@type": "PlaceRef",
                        "@referredType": "GeographicAddress",
                        "id": "8d80..",
                        "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
                        "name": "Johnson Residence"
                    }
                }
            ]
        },
        "serviceCharacteristic": [
            {
                "id": "540c..",
                "name": "Connection Speed",
                "valueType": "string",
                "value": "1 Gbps",
            }
        ]
    }
]

```

```

        "@type": "StringCharacteristic"
      }
    ],
    "serviceSpecification": {
      "id": "bc41..",
      "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
      "name": "Fiber Broadband",
      "@type": "ServiceSpecification"
    },
    "@type": "Service"
  },
  "@type": "ServiceQualificationItem"
}
]
}

```

RESPONSE

```

201 Created
Content-Type: application/json
Location:
{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069..

```

```

{
  "id": "b069..",
  "href":
"{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069..",
  "checkServiceQualificationDate": "2023-08-31T14:30:00.000Z",
  "description": "Broadband Fiber Service Qualification for Residential Area",
  "effectiveQualificationDate": "22023-08-31T14:30:00.000Z",
  "estimatedResponseDate": "2023-08-31T14:30:00.000Z",
  "expectedQualificationDate": "2023-08-31T14:30:00.000Z",
  "expirationDate": "2023-08-31T18:30:00.000Z",
  "externalIdentifier": [
    {
      "id": "a634..",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideUnavailabilityReason": true,

```

```
"instantSyncQualification": false,
"qualificationResult": "alternate",
"relatedParty": [
  {
    "role": "Buyer contact",
    "@type": "RelatedPartyOrPartyRole",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "2186..",
      "href": "{apiRoot}/partyManagement/v5/individual/2186..",
      "name": "Alice Johnson",
      "@referredType": "Individual"
    }
  }
],
"serviceQualificationItem": [
  {
    "id": "1",
    "expectedActivationDate": "2023-09-15T16:45:00.000Z",
    "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
    "expirationDate": "2023-09-15T16:45:00.000Z",
    "qualificationResult": "alternate",
    "state": "done",
    "alternateServiceProposal": [
      {
        "id": "1",
        "alternateServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "alternateService": {
          "serviceCharacteristic": [
            {
              "id": "2f47..",
              "name": "Connection Speed",
              "valueType": "string",
              "value": "512 Mb",
              "@type": "StringCharacteristic"
            }
          ],
          "serviceSpecification": {
            "id": "3b88..",
            "href":
              "{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/3b88..",
            "name": "Fiber Broadband - 512Mb",
            "@type": "ServiceSpecification"
          }
        },
        "@type": "Service"
      }
    ]
  }
]
```

```

    },
    "@type": "AlternateServiceProposal"
  }
],
"eligibilityUnavailabilityReason": [
  {
    "code": "26-001",
    "label": "Not enough bandwidth at this place",
    "type": "ServiceEligibilityUnavailabilityReason"
  }
],
"service": {
  "place": [
    {
      "@type": "RelatedPlaceRefOrValue",
      "role": "Installation place",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "id": "8d80..",
        "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
        "name": "Johnson Residence"
      }
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "540c..",
      "name": "Connection Speed",
      "valueType": "string",
      "value": "1 Gbps",
      "@type": "StringCharacteristic"
    }
  ],
  "serviceSpecification": {
    "id": "bc41..",
    "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
    "name": "Fiber Broadband",
    "@type": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "ServiceQualificationItem"

```

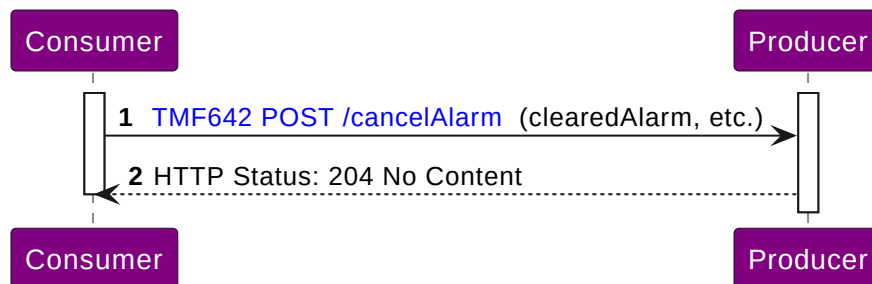
```

    }
  ],
  "state": "done",
  "@type": "CheckServiceQualification"
}

```

Task example with the return code **204 No Content**

In this example, the task has completed successfully but no resource representation is returned.



REQUEST

```

POST {apiRoot}/alarmManagement/v5/clearAlarm
Content-Type: application/json

```

```

{
  "clearSystemId": "ems-1",
  "clearUserId": "bob@example.net",
  "alarmClearedTime": "2023-08-31T14:30:00.000Z",
  "alarmPattern": [
    {
      "id": "7f06.."
    }
  ]
}

```

RESPONSE

204 No Content

9.3. Long-running Tasks

Long-running tasks represent the tasks where the time required to complete the task operation is long, and it is not feasible to keep the client connection open (e.g., *processAllDailyOrders*).

In this case, the server **MUST** create and persist the Task Resource and return it to the client in the same API call, but task execution will not be completed, and **MAY** even not have been started.

The Task Resource **MUST** have **id**, **href**, **@type** attributes and a **state** that indicates whether the task is in progress, and return it together with the **Location** header identifying the created task resource; as such, the supported success return code is **201 Created**. The returned task resource will probably not yet contain result data of the requested operation.

The task resource **SHOULD** be retained at least as long as the task is in progress, and it **MAY** be retained for a longer period of time, depending on the implementation.

To allow the consumers to check the progress of the tasks, the task resource **MUST** have a state. This will enable the API consumer to check the transitions of the task resource through the lifecycle to determine when the task has been completed.

To allow the consumers to check the current progress of a long-running task, the following patterns can be used:

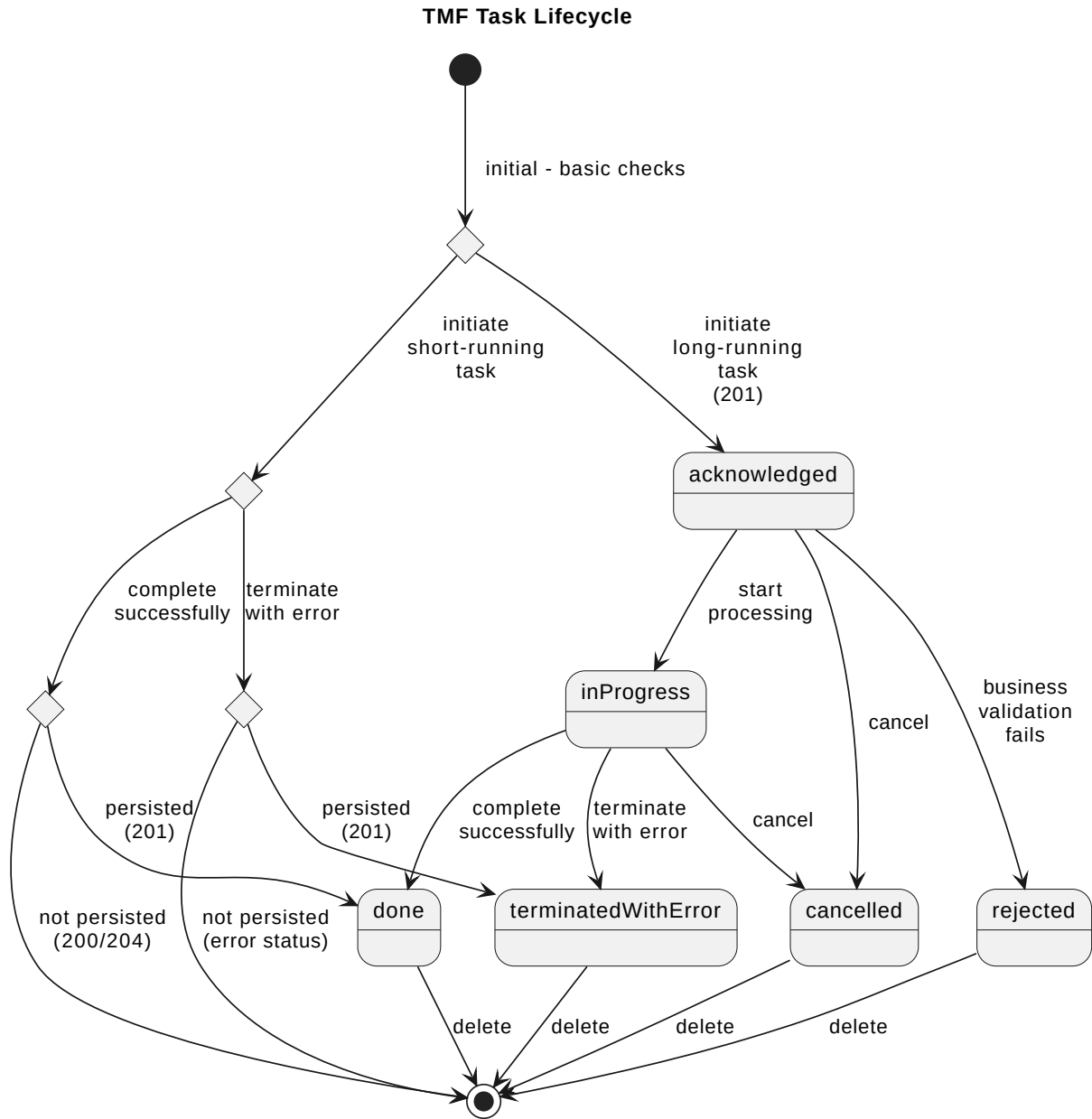
- Active Retrieval of the created task resource – via a **GET** operation for the task resources exposed by the API. For example, the task is **searchTimeSlot**, and the **GET** operation is **searchTimeSlot/{id}**:
POST {apiRoot}/appointmentManagement/v5/searchTimeSlot
returns the task resource with id, so that it can be retrieved later:
GET {apiRoot}/appointmentManagement/v5/searchTimeSlot/{id}
- Monitor Pattern – described [here](#) in the Design Guidelines – Part 1
- Publish/Subscribe Pattern – described [here](#) in the Design Guidelines – Part 1

9.4. Task Lifecycle

The following state-transition diagram depicts the lifecycle of a task resource.



Long-running tasks **MUST** implement this lifecycle, whereas for short-running tasks this is optional.



A task operation **MUST** return:

- a **200 OK** if the task completed successfully, the task resource is not persisted; the full representation of the task's results is returned
- a **201 Created** if the task resource was successfully created and has been persisted, and the full representation of the task resource is returned.
- a **202 Accepted** (not shown in the diagram) if the operation, i.e. the creation of the task resource, is performed asynchronously
- a **204 No Content** if the task completed successfully but no resource representation (neither of the task resource nor any task result) is being returned.

- the appropriate error status, if the task operation and no task resource was created.



When the Task Resource has been persisted, in particular for long-running tasks, a **GET** operation for the task resource identifier **MUST** be exposed by the API.



All the tasks **MUST** follow the task lifecycle depicted above.



The attribute **state** must be used for the lifecycle value of the task.



Long-running tasks can be implemented and monitored either through the task lifecycle approach, monitor, or the pub/sub notifications approach.

9.5. Task Cancellation

Canceling a long-running task can be a very complex activity that requires the implementation of multiple compensation steps to ensure data integrity.

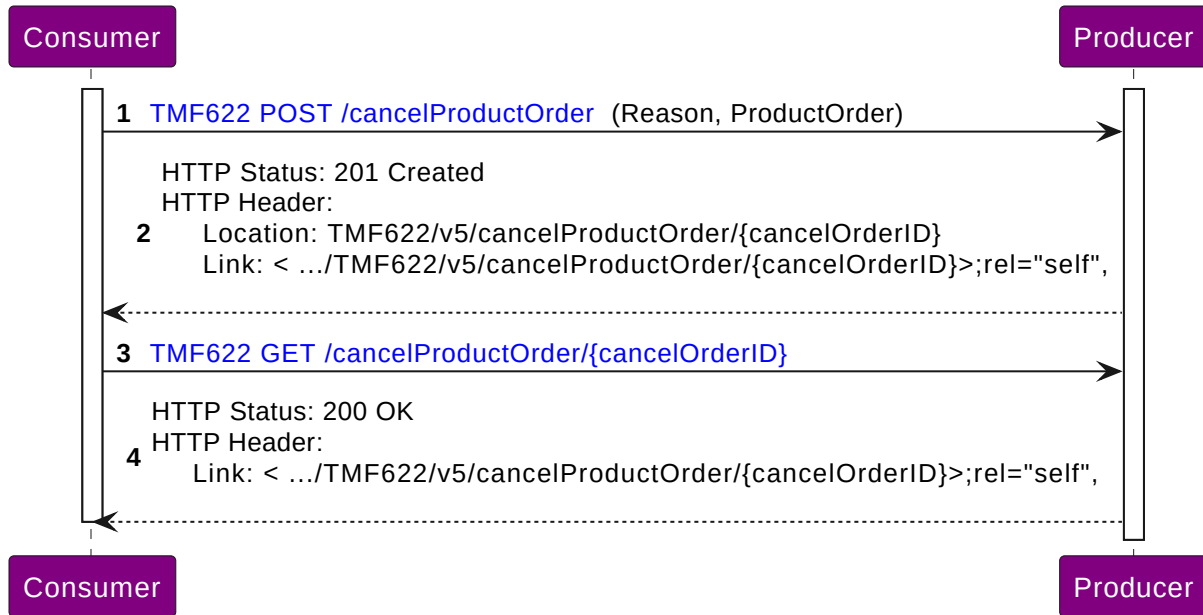
As such, the cancellation of a long-running task **SHOULD** be implemented as a separate task, which will provide a structured and standardized approach to tracking and managing the cancellation of activities.

As with any task resource, the cancellation task follows the same life cycle described above, and as such, it is not always guaranteed to succeed – it can be rejected, it can be completed with error, and any implementation must take into account the appropriate constraints that must be implemented on the cancellation activity.

Example: Task with return code **201 Created**:

In this example, a task resource is created representing the task, hence the response includes **201 Created** and the representation of the task resource is being returned.

Once the resource **id** has been returned, the client can poll the task progress using the **GET** operation or use the asynchronous mechanism via the [notifications pattern](#).



REQUEST

```
POST {apiRoot}/productOrderingManagement/v5/cancelProductOrder
Content-Type: application/json
```

```
{
  "cancellationReason": "Duplicate order",
  "requestedCancellationDate": "2023-03-14T11:56:27.098Z",
  "productOrder": {
    "id": "8558..",
    "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558..",
    "@REFERREDType": "ProductOrder"
  },
  "@type": "CancelProductOrder"
}
```

In this example, business validations have not yet completed, processing has not yet started:

RESPONSE

```
201 Created
Content-Type: application/json
Location: {apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..
Link:
```

```
<{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..>;rel="self"
```

```
{
  "id": "3aea..",
  "href": "{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..",
  "cancellationReason": "Duplicate order",
  "requestedCancellationDate": "2023-03-14T11:56:27.098Z",
  "productOrder": {
    "id": "8558..",
    "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558..",
    "@REFERREDType": "ProductOrder"
  },
  "state": "acknowledged",
  "@type": "CancelProductOrder"
}
```

The following describes the scenario when the client is checking the progress of the task, and the result is that the task has progressed to the state **inProgress**:

REQUEST

```
GET {apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..
Content-Type: application/json
```

RESPONSE

```
200 Ok
Content-Type: application/json
Location: {apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..
Link:
<{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..>;rel="self"
```

```
{
  "id": "3aea..",
  "href": "{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..",
  "cancellationReason": "Duplicate order",
  "requestedCancellationDate": "2023-03-14T11:56:27.098Z",
  "productOrder": {
    "id": "8558..",
    "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558..",

```

```

    "@referredType": "ProductOrder"
  },
  "state": "InProgress",
  "@type": "CancelProductOrder"
}

```

The following covers the case when the task has completed:

REQUEST

```

GET {apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..
Content-Type: application/json

```

RESPONSE

```

200 Ok
Content-Type: application/json
Location: {apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..
Link:
<{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..>;rel="self",

```

```

{
  "id": "3aea..",
  "href": "{apiRoot}/productOrderingManagement/v5/cancelProductOrder/3aea..",
  "cancellationReason": "Duplicate order",
  "requestedCancellationDate": "2023-03-14T11:56:27.098Z",
  "productOrder": {
    "id": "8558..",
    "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558..",
    "@referredType": "ProductOrder"
  },
  "state": "done",
  "@type": "CancelProductOrder"
}

```

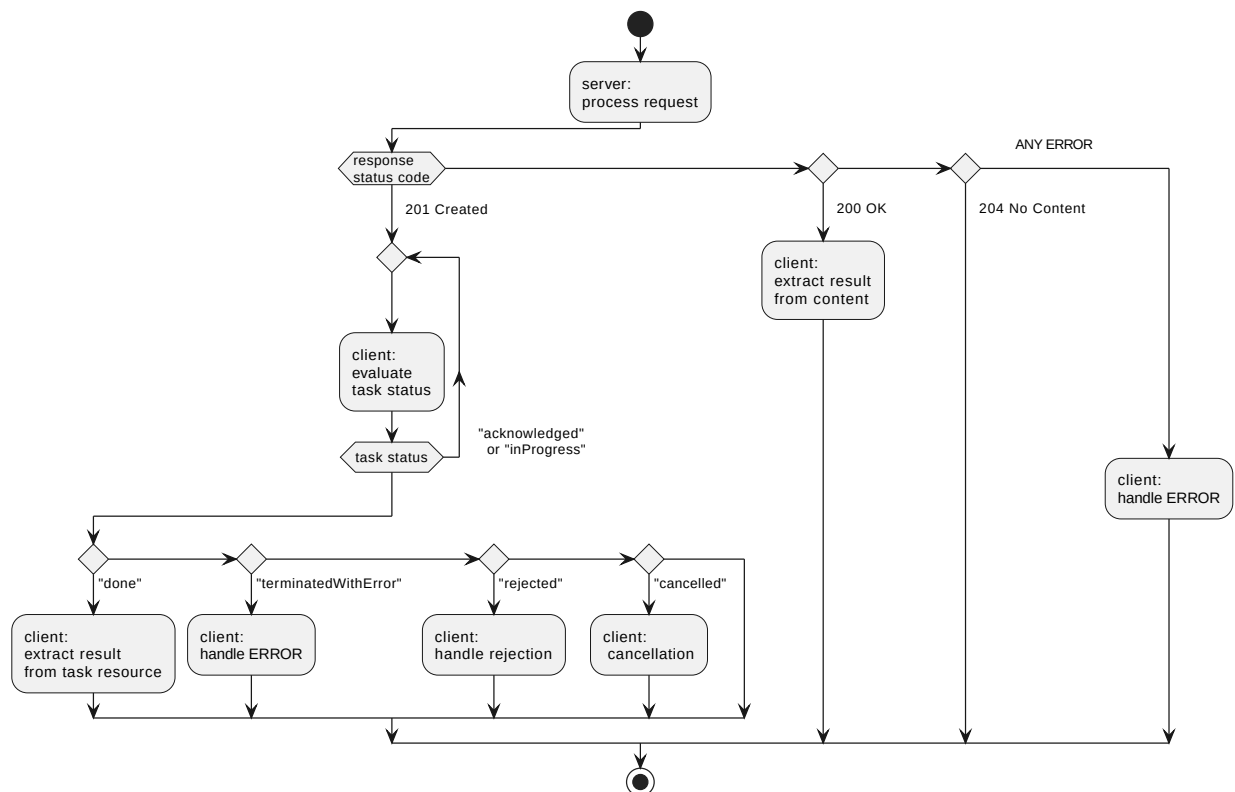
9.6. Type-selection by Server

When an implementation for a Task supports both short-running and long-running execution, it MAY decide at runtime to execute the task as short-running or long-running task, taking into consideration its expectation of the duration of task execution, e.g. by

investigating any request parameters or payload values.

In that case, the consumer must be able to distinguish and handle both variants of the task operation.

The following diagram depicts the decision-making process of the client when receiving the response to his POST request for the task resource:



The server processes the request, and decides how to handle it. For a long-running task, it will create and persist the Task Resource. For short-running tasks, he may also decide to create and persist the Task Resource, or not.

When the server created the Task Resource, it returns **201 Created**, and the Task Resource contains the actual status of the task execution, which can be **acknowledged**, **inProgress**, **terminatedWithError**, **rejected**, **cancelled** or **done**.

Depending on the method applied, the client either polls the Task Resource, uses the Monitor, or gets notified by events. It evaluates the state of the Task Resource. As long it is **acknowledged** or **inProgress**, it iterates. When any of the other the status values is reached, task execution has completed, and the client acts according to the result.

When in case of a short-running task the server did not create a Task Resource, the response contains the appropriate status code for the client to act upon, and in case of

success, indicates if the client should extract result data from the response payload, or not

9.7. Using Task with Sub-Resource based Response

There are scenarios when the Task Resource is created, and the result of the task operation contains a possibly large array of sub-resources. In such cases, there MAY be a need to iterate/paginate on the response part of the TASK (e.g. paginate on arrays of the response of the TASK).

In order to accomplish this the [sub-resource pattern](#) can be used.

The [sub-resource pattern](#), using hypermedia, allows any implementation to expose different task sub-resource arrays, enabling as such the capability to paginate/iterate over these array of resources in future without retrieving again the full resource representation.

Example

REQUEST

```
POST {apiRoot}/serviceQualificationManagement/v5/checkServiceQualification
Content-Type: application/json
```

```
{
  "@type": "CheckServiceQualification",
  "description": "Broadband Fiber Service Qualification for Residential Area",
  "expectedQualificationDate": "2023-08-31T14:30:00.000Z",
  "externalIdentifier": [
    {
      "id": "a634..",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "instantSyncQualification": false,
  "provideUnavailabilityReason": true,
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyOrPartyRole",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "2186..",
        "href": "{apiRoot}/partyManagement/v5/individual/2186..",

```

```

        "name": "Alice Johnson",
        "@referredType": "Individual"
    }
},
"serviceQualificationItem": [
    {
        "id": "1",
        "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "service": {
            "place": [
                {
                    "@type": "RelatedPlaceRefOrValue",
                    "role": "Installation place",
                    "place": {
                        "@type": "PlaceRef",
                        "@referredType": "GeographicAddress",
                        "id": "8d80..",
                        "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
                        "name": "Johnson Residence"
                    }
                }
            ],
            "serviceCharacteristic": [
                {
                    "id": "540c..",
                    "name": "Connection Speed",
                    "valueType": "string",
                    "value": "1 Gbps",
                    "@type": "StringCharacteristic"
                }
            ],
            "serviceSpecification": {
                "id": "bc41..",
                "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
                "name": "Fiber Broadband",
                "@type": "ServiceSpecification"
            },
            "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
    }
]

```

```
}
```

RESPONSE

201 Created
Content-Type: application/json

Location:
{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069..
Link:
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069..>;
rel="self"

<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem>; rel="collection"

```
{
  "id": "b069..",
  "href":
    "{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069..",
  "checkServiceQualificationDate": "2023-08-31T14:30:00.000Z",
  "description": "Broadband Fiber Service Qualification for Residential Area",
  "effectiveQualificationDate": "22023-08-31T14:30:00.000Z",
  "estimatedResponseDate": "2023-08-31T14:30:00.000Z",
  "expectedQualificationDate": "2023-08-31T14:30:00.000Z",
  "expirationDate": "2023-08-31T18:30:00.000Z",
  "externalIdentifier": [
    {
      "id": "a634..",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyOrPartyRole",
      "partyOrPartyRole": {
```

```

    "@type": "PartyRef",
    "id": "2186..",
    "href": "{apiRoot}/partyManagement/v5/individual/2186..",
    "name": "Alice Johnson",
    "@referredType": "Individual"
  }
},
"serviceQualificationItem": [
  {
    "id": "1",
    "expectedActivationDate": "2023-09-15T16:45:00.000Z",
    "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
    "expirationDate": "2023-09-15T16:45:00.000Z",
    "qualificationResult": "alternate",
    "state": "done",
    "alternateServiceProposal": [
      {
        "id": "1",
        "alternateServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "alternateService": {
          "serviceCharacteristic": [
            {
              "id": "2f47..",
              "name": "Connection Speed",
              "valueType": "string",
              "value": "512 Mb",
              "@type": "StringCharacteristic"
            }
          ],
          "serviceSpecification": {
            "id": "3b88..",
            "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/3b88..",
            "name": "Fiber Broadband - 512Mb",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "AlternateServiceProposal"
      }
    ],
    "eligibilityUnavailabilityReason": [
      {
        "code": "26-001",

```

```

        "label": "Not enough bandwidth at this place",
        "type": "ServiceEligibilityUnavailabilityReason"
    }
],
"service": {
    "place": [
        {
            "@type": "RelatedPlaceRefOrValue",
            "role": "Installation place",
            "place": {
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress",
                "id": "8d80..",
                "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
                "name": "Johnson Residence"
            }
        }
    ],
    "serviceCharacteristic": [
        {
            "id": "540c..",
            "name": "Connection Speed",
            "valueType": "string",
            "value": "1 Gbps",
            "@type": "StringCharacteristic"
        }
    ],
    "serviceSpecification": {
        "id": "bc41..",
        "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
        "name": "Fiber Broadband",
        "@type": "ServiceSpecification"
    },
    "@type": "Service"
},
"@type": "ServiceQualificationItem"
}
],
"state": "done",
"@type": "CheckServiceQualification"
}

```

The task resource, via hypermedia (*Link* header), will inform the consumers what subresources are available as a collection, enabling as such pagination capabilities over them.

As seen in the above example, the `checkServiceQualification` task resource is exposing, via *Link* header, the `serviceQualificationItem` collection, that allows the consumer to paginate/iterate over it.

Link:
`<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem>; rel="collection"`



Paginating over the task subresources MAY not be available until the task is completed.



As described in the [sub-resources pattern](#) or in the [Query Resources with attribute filtering](#) it is up to the implementation to decide which pagination mechanism will be used.

Example: Pagination over the subresource `serviceQualificationItem`

Consider the above example task response, with 10 items in the `serviceQualificationItem` array and that the consumer wants to iterate over pages of 2 elements from the second page.

REQUEST

```
GET
{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=2&limit=2
Accept: application/json
```

RESPONSE

```
200 OK
Content-Type: application/json
X-Result-Count:2
X-Total-Count:10
```

Link:

```
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=2&limit=2>; rel="self",
```

```
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=0&limit=2>; rel="first",
```

```
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=3&limit=2>; rel="next",
```

```
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=0&limit=2>; rel="prev",
```

```
<{apiRoot}/serviceQualificationManagement/v5/checkServiceQualification/b069../serviceQualificationItem?offset=8&limit=2>; rel="last"
```

```
[
  {
    "id": "3",
    "expectedActivationDate": "2023-09-15T16:45:00.000Z",
    "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
    "expirationDate": "2023-09-15T16:45:00.000Z",
    "qualificationResult": "alternate",
    "state": "done",
    "alternateServiceProposal": [
      {
        "id": "1",
        "alternateServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "alternateService": {
          "serviceCharacteristic": [
            {
              "id": "2f47..",
              "name": "Connection Speed",
              "valueType": "string",
              "value": "512 Mb",
              "@type": "StringCharacteristic"
            }
          ],
          "serviceSpecification": {
            "id": "3b88..",
            "href":
              "{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/3b88..",
            "name": "Fiber Broadband - 512Mb",
            "@type": "ServiceSpecification"
          }
        }
      }
    ]
  }
]
```

```

        },
        "@type": "Service"
    },
    "@type": "AlternateServiceProposal"
}
],
"eligibilityUnavailabilityReason": [
{
    "code": "26-001",
    "label": "Not enough bandwidth at this place",
    "type": "ServiceEligibilityUnavailabilityReason"
}
],
"service": {
    "place": [
        {
            "@type": "RelatedPlaceRefOrValue",
            "role": "Installation place",
            "place": {
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress",
                "id": "8d80..",
                "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
                "name": "Johnson Residence"
            }
        }
    ],
    "serviceCharacteristic": [
        {
            "id": "540c..",
            "name": "Connection Speed",
            "valueType": "string",
            "value": "1 Gbps",
            "@type": "StringCharacteristic"
        }
    ],
    "serviceSpecification": {
        "id": "bc41..",
        "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
        "name": "Fiber Broadband",
        "@type": "ServiceSpecification"
    },
    "@type": "Service"
}

```

```

    },
    "@type": "ServiceQualificationItem"
  },
  {
    "id": "4",
    "expectedActivationDate": "2023-09-15T16:45:00.000Z",
    "expectedServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
    "expirationDate": "2023-09-15T16:45:00.000Z",
    "qualificationResult": "alternate",
    "state": "done",
    "alternateServiceProposal": [
      {
        "id": "1",
        "alternateServiceAvailabilityDate": "2023-09-15T16:45:00.000Z",
        "alternateService": {
          "serviceCharacteristic": [
            {
              "id": "2f47..",
              "name": "Connection Speed",
              "valueType": "string",
              "value": "200 Mb",
              "@type": "StringCharacteristic"
            }
          ],
          "serviceSpecification": {
            "id": "3b88..",
            "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/3b88..",
            "name": "Fiber Broadband - 200Mb",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "AlternateServiceProposal"
      }
    ],
    "eligibilityUnavailabilityReason": [
      {
        "code": "26-001",
        "label": "Not enough bandwidth at this place",
        "type": "ServiceEligibilityUnavailabilityReason"
      }
    ],
    "service": {
      "place": [

```

```
{
  "@type": "RelatedPlaceRefOrValue",
  "role": "Installation place",
  "place": {
    "@type": "PlaceRef",
    "@referredType": "GeographicAddress",
    "id": "8d80..",
    "href":
"{apiRoot}/geographicAddressManagement/v5/geographicAddress/8d80..",
    "name": "Johnson Residence"
  }
},
"serviceCharacteristic": [
  {
    "id": "540c..",
    "name": "Connection Speed",
    "valueType": "string",
    "value": "1 Gbps",
    "@type": "StringCharacteristic"
  }
],
"serviceSpecification": {
  "id": "bc41..",
  "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/bc41..",
  "name": "Fiber Broadband",
  "@type": "ServiceSpecification"
},
"@type": "Service"
},
"@type": "ServiceQualificationItem"
}
```

Chapter 10. Asynchronous Operation Pattern

The following section describes the asynchronous operation pattern defining the mechanics and rules where an API operation might be treated in an asynchronous fashion (through the **202 - Accepted** return code).

The asynchronous operation pattern allows implementations to handle API operations asynchronously, whether they are standard create, update, delete, or task-based operations. This pattern is typically used when the operation cannot be completed synchronously, either because the operation itself cannot be completed or because the resources required for the operation cannot be created or computed in a timely manner.

An operation is considered synchronous in behaviour when the resource/data is returned to the client in the same interaction.

An operation is *asynchronous* in behaviour when the server implementation accepts the request without requiring the client to persist the connection until the processing is completed.



Overall, the asynchronous operation pattern can be applied to at least all the immutable operations.

In the asynchronous operation pattern, the API operation does not return a response immediately. Instead, the server will acknowledge the receipt of the request and provide mechanisms (e.g. *monitor* or *notifications*) that will allow the consumer to check the progress.

The API implementation then completes the operation in the background. Once the operation is complete, the implementation sends a notification to the caller or the caller can poll for the state operation.

Possible return code for an asynchronous operations is **202 Accepted** [RFC9110] (section 15.3.3)

By returning a **202 Accepted**, the server indicates that the request has been accepted, however the processing may not have started (in fact it may never be processed at all) or has not been completed yet.

See [RFC9110], section 15.3.3:



The 202 (Accepted) status code indicates that the request has been accepted for processing, but the processing has not been completed. The request might or might not eventually be acted upon, as it might be

disallowed when processing actually takes place. ... The 202 response is intentionally noncommittal.

...

The representation sent with this response ought to describe the request's current status and point to (or embed) a status monitor that can provide the user with an estimate of when the request will be fulfilled.

Currently, in the TM Forum Design Guidelines there are two patterns defined for the asynchronous behavior of the operations:

Publish/Subscriber Pattern

Described [here](#).

Monitor Pattern

Described [here](#).

The above patterns cover the asynchronous behavior in two different ways:

- based on a push notification (*pub/sub*).
- based on a poll mechanism (*monitor*)



If the asynchronous operations are being supported by the server implementation, then either the *monitor* or the *notification* pattern MUST be used.



If the *monitor* pattern is being used, then the server MUST create and return to the consumer the monitor resource ID, in order to allow the consumer to check the progress of the operation.

Depending on the server implementation, as a result of the request, at least one of the two options listed below MUST be returned:

- an **id** and **href** and **@type**

or

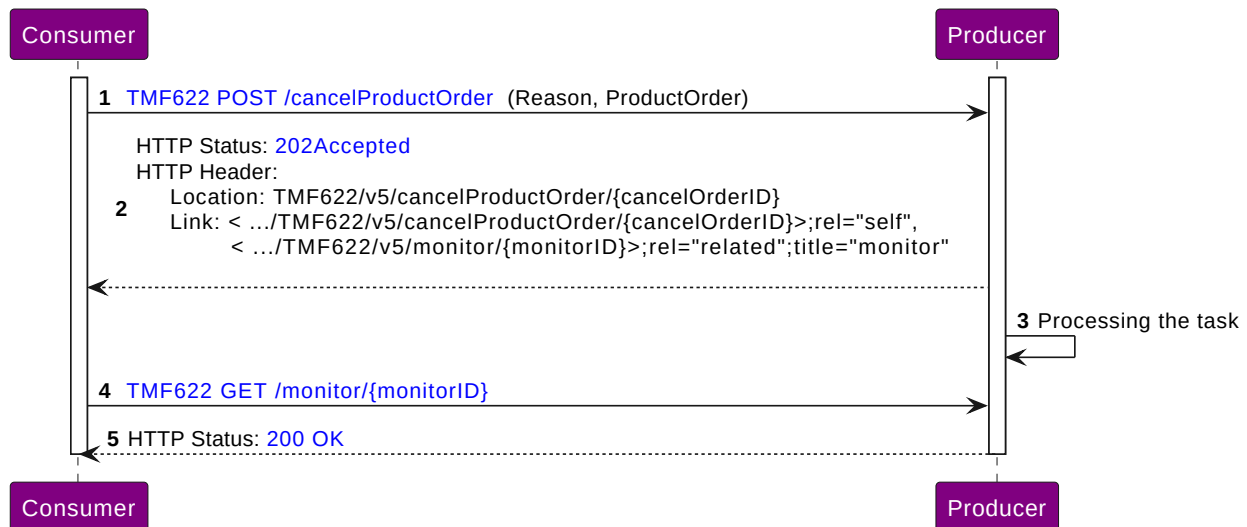
- a monitor resource



If the service is not able to provide an **id** and **href** to the consumer, then a monitor MUST be provided.

10.1. Asynchronous Operation & Monitor

A classic interaction with asynchronous response and monitor pattern will look like below:



In the above example we can notice the following:

- The client does not send any header to indicate to the server if it would like a synchronous answer or an asynchronous answer (no [Expect Header](#) used).
 - By listing the respective responses, the API definition specifies, if the server can return either **201 Created** or **202 Accepted**, or both patterns (synchronous and asynchronous).
 - It also specifies, via including the *Expect* header, if there is support for the client to decide.
- The server is:
 - returning a **202 Accepted** HTTP Status Code – representing the asynchronous behavior
 - returning a **Location** header to indicate to the consumer the path to the cancel order resource. In this case, the server is able to provide as part of the response the *cancelProductOrderID*.
 - The location path can look something like this:

Location: {apiRoot}/productOrderingManagement/v5/cancelProductOrder/5fbe..

(where **5fbe..** is the cancelProductOrder task identifier)



If the server is unable to provide a resource identifier, then it must provide the monitor resource.

- The link to the actual resource (**self**)

Link:

```
<{apiRoot}/productOrderingManagement/v5/cancelProductOrder/5fbe...>;rel=self,

```

- Due to the fact that the server is implementing a "monitor" pattern it will also return the relevant "hypermedia links"

- The link to the **monitor resource**, where the consumer can check regularly in order to check the status of the operation

Link:

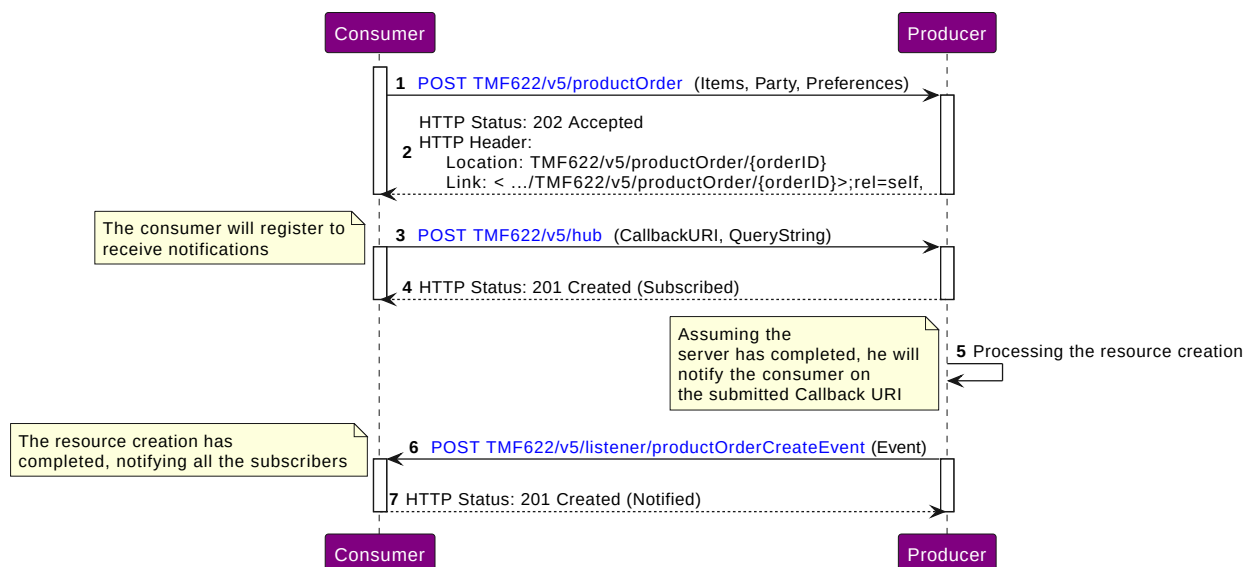
```
<{apiRoot}/productOrderingManagement/v5/monitor/dd55...>;rel=related;title=monitor,

```

(where the **dd55..** is the monitor resource identifier assigned for this resource)

10.2. Asynchronous Operations & Notifications

If the client does not want to use the "polling" mechanism, via the monitor pattern, but instead it wants to be notified via a callback when the resource creation is completed, then the client has to register to the "hub" resource with the callback URI, as depicted below:



As we can see in the above diagram, there are two API calls between the consumer (e.g. EngagementManagementUI) and the producer (e.g. OrderManagement):

- Step 1 & 2 → to create the Product Order resource
- Step 3 & 4 → to register to the `/hub` endpoint – in order to receive the notifications

The information required to register to the notifications is described in the Design Guidelines, Part 1, Chapter 10 (see [Notifications](#)):

```
{
  "callback": "https://api.clientdomain.com/listener/event",
  "query": "eventType=ProductOrderCreateEvent"
}
```

- "callback" – represents the callback URL where the notifications can be sent.
 - For example:
`{apiRoot}/productOrderingManagement/v5/listener/productOrderCreateEvent`
- "query" – represents an eventType or a condition that the resource must meet in order to trigger the notification.
 - For example: `eventType=ProductOrderCreateEvent` or `ID=5fbe..&state=inProgress`



None of the **callback URL** in the examples in this chapter are normative. The API client is free to choose any path (event the endpoint name) on his convenience.

The notification that is issued by the producer when the operation has completed is detailed in the Design Guidelines – Part 1 – see [Notification Patterns](#)

A sample on how the notification can look can be found below:

```
{
  "correlationId": "250b..",
  "description": "CancelProductOrderCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "d491..",
  "eventTime": "2021-09-28T07:43:59.059Z",
  "eventType": "CancelProductOrderCreateEvent",
  "priority": "1",
  "timeOccurred": "2021-09-28T07:43:59.059Z",
  "title": "CancelProductOrderCreateEvent",
}
```

```
"event": {
  "cancelProductOrder": {
    "id": "aeb3..",
    "href":
"https://host:port/productOrderingManagement/v5/cancelProductOrder/aeb3..",
    "cancellationReason": "Duplicate order",
    "creationDate": "2019-04-30T08:13:59.506Z",
    "requestedCancellationDate": "2021-08-30T09:14:46.145Z",
    "productOrder": {
      "id": "75fa..",
      "href":
"https://host:port/productOrderingManagement/v5/productOrder/75fa..",
      "@referredType": "ProductOrder"
    },
    "state": "acknowledged",
    "@type": "CancelProductOrder"
  },
  "relatedParty": [
    {
      "id": "8de5..",
      "role": "Issuer",
      "@type": "RelatedParty",
      "@referredType": "Organization"
    }
  ],
  "reportingSystem": {
    "id": "c7f0..",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "b710..",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CancelCancelProductOrderCreateEvent"
}
```

10.3. Expect Header

Using *Expect* header, the client can request a specific behaviour from the API producer to handle the request either as a *synchronous* or *asynchronous* one.

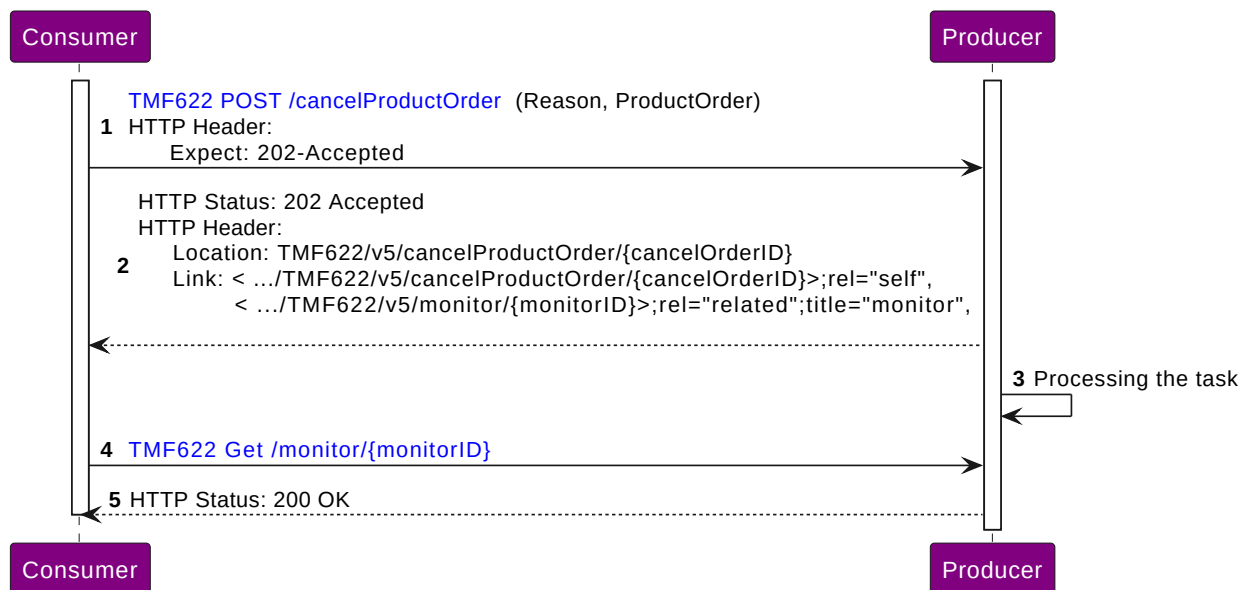
If the API Producer supports both mechanisms (synchronous and asynchronous) the API client can ask for its request to be handled in a particular way.

The Expect header behaviour is specified in [\[RFC9110\]](#) (section-10.1.1).

The *Expect* header is set during the API request:

- If the client requests an asynchronous behaviour, the Expect header MUST be set to **202 Accepted**.
- If the client requests a synchronous behaviour, the Expect header MUST be set to a corresponding return code as per operation (e.g., **201 Created**).

Example: Expect Header – 202-Accepted



REQUEST

```
POST {apiRoot}/productOrderingManagement/v5/cancelProductOrder
Content-Type: application/json
Expect: 202-Accepted
```

```
{
  "cancellationReason": "Duplicate order",
```

```

"requestedCancellationDate": "2022-03-14T11:56:27.098Z",
"productOrder": {
  "id": "8558..",
  "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558.."
}
}

```

RESPONSE

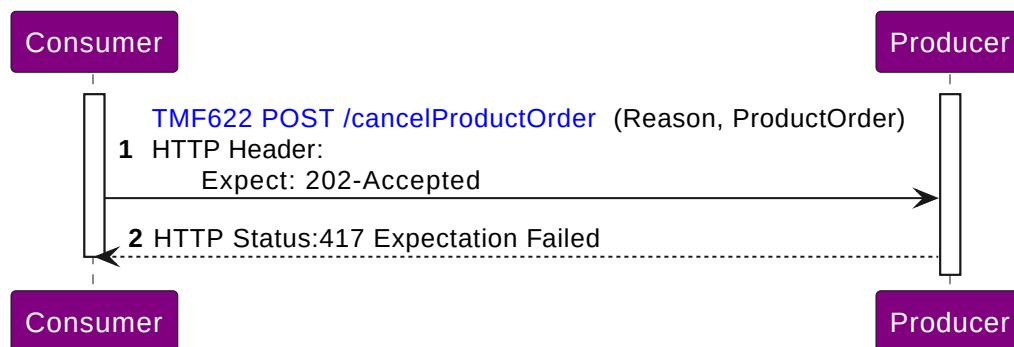
```

202 Accepted
Content-Type: application/json
Location: {apiRoot}/productOrderingManagement/v5/cancelProductOrder/c2a9..
Link:
<{apiRoot}/productOrderingManagement/v5/cancelProductOrder/c2a9..>;rel="self",

```

If the API producer receives an Expect header but can't fulfil the demand, it MAY respond with a **417 Expectation Failed** status code to indicate that the unexpected expectation cannot be met.

Example: Expect Header – Product Can't Comply



REQUEST

```

POST productOrderingManagement/v5/cancelProductOrder
Content-Type: application/json
Expect: 202-Accepted

```

```

{
  "cancellationReason": "Duplicate order",
  "requestedCancellationDate": "2022-03-14T11:56:27.098Z",
  "productOrder": {

```

```
"id": "8558..",  
  "href": "{apiRoot}/productOrderingManagement/v5/productOrder/8558.." }  
}
```

RESPONSE

```
---  
417 Expectation Failed  
---
```



If the API, by design, is exposing a resource parameter to ask for a synchronous response (e.g. *instantSyncRecommendation*) but in the expect header is passing a request for an asynchronous behaviour – as this is a contradictory request the server **MUST** return an appropriate error code (e.g., **409 - Conflict**).



Some existing APIs have implemented other methods to achieve similar purposes. These realizations **SHOULD** be migrated to the *Expect* header mechanism in future major versions of the TMF Open APIs.

10.4. Monitor Pattern

The monitor resource is specifically designed to track the execution (poll based mechanism) of long-running tasks. This enables clients to monitor the progress or outcome of these tasks without requiring immediate responses.

The monitor resource is a dedicated resource that provides information about the status of a long-running task. It allows clients to query the current state of the task, including whether it is still in progress, has completed successfully, or has encountered an error.

There are two primary patterns for tracking the status of long-running tasks:

- Monitor Pattern – The client actively queries (polls) a dedicated monitor resource to retrieve the current status of the task.
- Notification Pattern – The system proactively sends a notification to the client when the task completes or encounters an error.

If an implementation supports the monitor pattern, the following guidelines **MUST** be followed:

- A top-level resource collection named monitor MUST be defined in the API.
- Each long-running task that supports monitoring MUST have a corresponding monitor resource instance created by the server.
- A link to this monitor resource MUST be included in the response to the original request that initiated the long-running task.

The monitor resource MUST support the HTTP GET method to allow clients to query the task status.

Clients should use the monitor endpoint to repeatedly check for updates at appropriate intervals (polling).



Implementations may optionally implement and emit events such as `MonitorCreateEvent`, `MonitorStateChangeEvent` or `MonitorAttributeValueChange` allowing consumers to register for asynchronous notifications. By subscribing to these events, consumers can be asynchronously notified of monitor creation, state transitions, or attribute updates, thereby avoiding reliance on active polling.

To ensure consistency and prevent unintended modifications, the following operations MUST NOT be supported on the monitor resource:

- POST – Clients cannot manually create a monitor resource; it must be generated automatically by the server.
- PATCH – Clients cannot modify the monitor resource. The status and data should be updated solely by the system.
- DELETE – Clients cannot delete the monitor resource. Instead, the system should define an appropriate retention policy for automatically removing completed or expired monitor instances.

The monitor resource SHOULD persist for a reasonable duration after task completion to allow clients to retrieve the results.

The API MAY define a retention policy for monitor resources, specifying when completed monitors are automatically removed (e.g., after a fixed time period or when storage limits are reached).

REQUEST

```
POST|PATCH {apiRoot}/productOrderManagement/v5/monitor/
```

Accept: application/json

RESPONSE

405 Method Not Allowed

REQUEST

PATCH|DELETE {apiRoot}/productOrderManagement/v5/monitor/1bf9..
Accept: application/json

RESPONSE

405 Method Not Allowed

The following HTTP Status codes are used on the retrieving of the monitor resource:

HTTP Response Codes – Success
200 OK
HTTP Response Codes – Failure
400 Bad Request
404 Not Found
405 Method Not Allowed
500 Internal Server Error
...

10.4.1. Retrieving monitor

The monitor can be accessed via the Monitor collection or via the singleton instance.

Example:

The response to **POST /activation/service** operation cannot be sent synchronously as such a monitor resource hyperlink is given in the response

REQUEST

POST {apiRoot}/serviceActivationConfiguration/v5/service

Accept: application/json

```
{
  "category": "MobileService",
  "description": "Mobile Line ",
  "feature": [
    {
      "id": "1",
      "isBundle": false,
      "isEnabled": true,
      "name": "Voice",
      "featureCharacteristic": [
        {
          "id": "1",
          "name": "VoiceSettings",
          "valueType": "object",
          "value": {
            "voiceStandard": "enable",
            "voWifi": "disable",
            "roaming": "enable",
            "@type": "VoiceFeatureCharacteristic"
          },
          "@type": "ObjectCharacteristic"
        }
      ],
      "@type": "Feature"
    }
  ],
  "relatedParty": [
    {
      "role": "member",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "c7b6..",
        "href": "{apiRoot}/partyManagement/v5/party/c7b6..",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceCharacteristic": [
    {
```

```

        "id": "7251..",
        "name": "MSISDN",
        "valueType": "string",
        "value": "415-275-7439",
        "@type": "StringCharacteristic"
    }
],
"serviceSpecification": {
    "id": "1f7c..",
    "href": "{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/1f7c..",
    "name": "c_Mobile",
    "@type": "ServiceSpecificationRef"
},
"state": "active",
"supportingService": [
    {
        "serviceSpecification": {
            "id": "c678..",
            "href":
"{apiRoot}/serviceCatalogManagement/v5/serviceSpecification/c678..",
            "name": "c_MobileSupport",
            "@type": "ServiceSpecificationRef"
        },
        "serviceCharacteristic": [
            {
                "id": "501e..",
                "name": "IMSI",
                "valueType": "string",
                "value": "228 01 21 76510739 ",
                "@type": "StringCharacteristic"
            }
        ],
        "state": "active",
        "@type": "Service"
    },
    {
        "id": "794b0..",
        "href": "{apiRoot}/serviceInventoryManagement/v5/service/794b0..",
        "@referredType": "Service",
        "@type": "ServiceRef"
    }
],
"@type": "Service"
}

```

RESPONSE

```
202 Accepted
Content-Type: application/json
Location: {apiRoot}/serviceActivationConfiguration/v5/service/51b7..
Link:
<{apiRoot}/serviceActivationConfiguration/v5/monitor/c641..>;rel=related;title=mo
nitor,
    <{apiRoot}/serviceActivationConfiguration/v5/service/51b7..>;rel=self
```

In the above example, the monitor resource is created and the link to the monitor resource is provided in the response, allowing the consumer to check regularly the status of the operation.

```
Link:
<{apiRoot}/serviceActivationConfiguration/v5/monitor/c641..>;rel=related;title=mo
nitor
```

(the **c641..** is the monitor resource identifier assigned for this resource)

The monitor resources SHOULD support at least the following states: inProgress, inError, completed.



Structure of the monitor can be extended using the extension patterns described in [\[TMF630-2\]](#).

Example: List all the monitors

REQUEST

```
GET {apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor
Accept: application/json
```

RESPONSE

```
[
  {
    "@type": "Monitor",
    "id": "4947..",
    "href": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/4947..",
```

```
"state": "Completed",
"request": {
  "method": "POST",
  "to": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application",
  "body": {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "name": "Evolve Quest",
    "commercialName": "Evolue Quest",
    "description": "Evolve Quest mobile application",
    "category": "games",
    "operationalState": "enable",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_quest_channelPartner15",
      "redirectUrl": [
        "https://evolvequest.com/redirect"
      ],
      "jwksUri": "https://evolvequest.com/jwk"
    },
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "31"
    }
  }
},
"response": {
  "statusCode": "201",
  "body": {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "101",
    "href":
"{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/b62b..",
    "name": "Evolve Quest",
    "commercialName": "Evolue Quest",
    "description": "Evolve Quest mobile application",
    "category": "games",
    "operationalState": "enable",
    "approvalStatus": "approved",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_quest_channelPartner15",
      "redirectUrl": [
        "https://evolvequest.com/redirect"
      ]
    }
  }
}
```

```

    ],
    "jwksUri": "https://evolvequest.com/jwk"
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "31"
  },
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15"
  }
}
},
"sourceHref":
"{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/b62b.."
},
{
  "@type": "Monitor",
  "id": "785d..",
  "href": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/785d..",
  "state": "Completed",
  "request": {
    "method": "POST",
    "to": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "name": "Evolve Quest",
      "commercialName": "Evolle Quest",
      "description": "Evolve Quest mobile application",
      "category": "games",
      "operationalState": "enable",
      "digitalIdentity": {
        "@type": "ApiDigitalIdentity",
        "clientId": "evolve_quest_channelPartner15",
        "redirectUrl": [
          "https://evolvequest.com/redirect"
        ],
        "jwksUri": "https://evolvequest.com/jwk"
      },
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "31"
      }
    }
  }
}

```

```

    },
    "response": {
      "statusCode": "201",
      "body": {
        "@type": "Application",
        "@baseType": "LogicalResource",
        "id": "101",
        "href":
"{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/11d7..",
        "name": "Evolve Quest",
        "commercialName": "Evolle Quest",
        "description": "Evolve Quest mobile application",
        "category": "games",
        "operationalState": "enable",
        "approvalStatus": "approved",
        "digitalIdentity": {
          "@type": "ApiDigitalIdentity",
          "clientId": "evolve_quest_channelPartner15",
          "redirectUrl": [
            "https://evolvequest.com/redirect"
          ],
          "jwksUri": "https://evolvequest.com/jwk"
        },
        "applicationOwner": {
          "@type": "PartyRoleRef",
          "id": "31"
        },
        "channelPartner": {
          "@type": "PartyRoleRef",
          "id": "15"
        }
      }
    },
    "sourceHref":
"{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/11d7.."
  }
]

```



Implementations must enforce security checks to ensure that only monitors associated with the authenticated user's identity and access permissions are returned in response to a GET request. This means verifying the user's authentication credentials and authorization scope to prevent unauthorized access.

Example: Retrieve a monitor resource:

REQUEST

```
GET {apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/4947..  
Accept: application/json
```

RESPONSE

200 OK

```
{  
  "@type": "Monitor",  
  "id": "4947..",  
  "href": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/4947..",  
  "state": "Completed",  
  "request": {  
    "method": "POST",  
    "to": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application",  
    "body": {  
      "@type": "Application",  
      "@baseType": "LogicalResource",  
      "name": "Evolve Quest",  
      "commercialName": "Evolve Quest",  
      "description": "Evolve Quest mobile application",  
      "category": "games",  
      "operationalState": "enable",  
      "digitalIdentity": {  
        "@type": "ApiDigitalIdentity",  
        "clientId": "evolve_quest_channelPartner15",  
        "redirectUrl": [  
          "https://evolvequest.com/redirect"  
        ],  
        "jwksUri": "https://evolvequest.com/jwk"  
      },  
      "applicationOwner": {  
        "@type": "PartyRoleRef",  
        "id": "31"  
      }  
    }  
  },  
  "response": {
```

```

"statusCode": "201",
"body": {
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "101",
  "href": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/b62b..",
  "name": "Evolve Quest",
  "commercialName": "Evolle Quest",
  "description": "Evolve Quest mobile application",
  "category": "games",
  "operationalState": "enable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15",
    "redirectUrl": [
      "https://evolvequest.com/redirect"
    ],
    "jwksUri": "https://evolvequest.com/jwk"
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "31"
  },
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15"
  }
}
},
"sourceHref":
"{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/b62b.."
}

```

The monitor resource schema does provide information about the request and response of the operation, including the status code and the body of the response. In the scenario where the task has encountered an error, the monitor resource **MUST** provide the error details under the **response.body** attribute following the TMForum Error schema.

Example: Error encountered on the monitor resource

REQUEST

```
GET {apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/cefd..
```

Accept: application/json

RESPONSE

```
{
  "@type": "Monitor",
  "id": "987654",
  "href": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/monitor/cefd..",
  "state": "Completed",
  "request": {
    "method": "POST",
    "to": "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "name": "Evolve Quest",
      "commercialName": "EvoQuest",
      "description": "Evolve Quest mobile application",
      "category": "games",
      "operationalState": "enable",
      "digitalIdentity": {
        "@type": "ApiDigitalIdentity",
        "clientId": "evolve_quest_channelPartner15",
        "redirectUrl": [
          "https://evolvequest.com/redirect"
        ],
        "jwksUri": "https://evolvequest.com/jwk"
      },
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "31"
      }
    }
  },
  "response": {
    "statusCode": "422",
    "body": {
      "code": "456-dd-df45",
      "reason": "Missing product configuration attribute",
      "message": "Product characteristic information are not consistent",
      "timestamp": "2025-02-14T12:14:23.473Z",
      "status": "200-10",
      "@type": "Error"
    }
  }
}
```

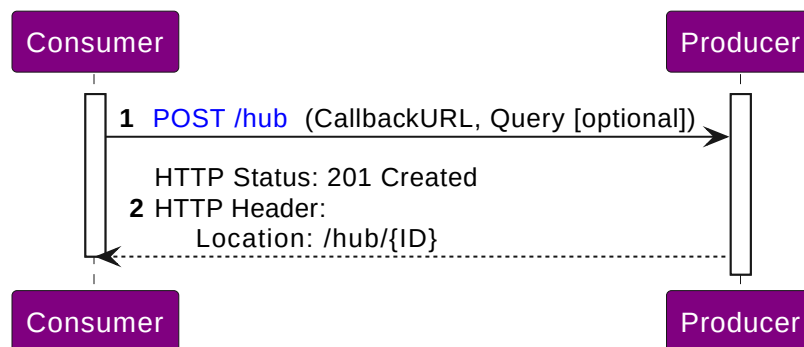
```
},  
  "sourceHref":  
  "{apiRoot}/openGatewayOnboardingAndOrdering/v5/application/cefd.."
```

Chapter 11. Notification Patterns

The following section describes the publish/subscribe pattern supported by REST based APIs supporting eventing.

A hub can be created at the API level or at each API resource level within an API.

A subscription to the Hub endpoint can be created by the consumers in order to receive notifications.



11.1. Create Subscription & register Listener

A client can register to the APIs **Hub** endpoints in order to receive notifications to a particular client listener endpoint when certain conditions have been met.

The client registers a listener, in the form of the callback URL, by creating a Hub subscription – a resource unique to the listener.



None of the **callback URL** in the examples in this chapter are normative. The API client is free to choose any path (event the endpoint name) on his convenience.

The HUB subscription resource is attached or bound to the API and its attribute specify the listener event callback address and the callback conditions.

The hub subscription is created via a **POST** operation call to the api **/hub** end-point, for example:

Example:

To create a subscription for the TMF621 – Trouble Ticket API

REQUEST

```
POST {apiRoot}/troubleTicket/v5/troubleTicket/hub
Content-Type: application/json
```

```
{
  "callback": "http://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent"
}
```

When creating the Hub subscription via the **POST** call, the communication endpoint address that the service instance must use to deliver notifications (by default on all supported events) is being set (e.g., <http://api.clientdomain.com/tmf-api/troubleTicket/v5/listener/troubleTicketCreateEvent>)



The TMForum APIs provide a set of listeners for each event type defined in the API, which can be used as callbacks. However, additional listeners (callback endpoints) can be defined according to specific implementation details. For example, a callback in the format of **"callback": "http://api.clientdomain.com/troubleTicketListener"** can be used to capture all events related to the Trouble Ticket API.



A query expression may be supplied with the subscription registration.

The query expression may be used to filter specific event types and/or any content of the event. Each TMForum Open API does specify the event types supported by the API (e.g., **TroubleTicketAttributeValueChangeEvent**, **TroubleTicketCreateEvent**, etc.) which can be used in the query expression to filter the notifications to be sent to the listener endpoint.

The query expression structure used for notification filtering supports the same one as used for queries i.e. GET. If no query expression is supplied when the subscription is created then all the notifications will be sent to the listener.

Subsequent **POST** calls may be rejected by the service if it does not support multiple listeners. In this case, the previous subscription must be deleted before the endpoint can be created again via a DELETE **/api/hub/{id}** call.

Possible return codes for the successful subscription creation MUST be: – **201 Created** defined in [\[RFC9110\]](#) (section 15.3.2)

If the request to create a subscription fails a corresponding HTTP status code can be returned in accordance (e.g., client error 4xx or server error 5xx). Default error codes

supplied in the API definitions include 400, 404, 405, 409 and 500.

Response Code List – Success
201 Created
Response Code List – Failure
400 Bad Request
404 Not Found
405 Method Not Allowed
409 Conflict
422 Unprocessable Content
500 Internal Server Error
...

Example: Create a Subscription:

Create a hub subscription to receive events from the TMF621 Trouble Ticket API on the callback URL <https://api.clientdomain.com/tmf-api/troubleTicket/v5/listener/troubleTicketCreateEvent>



The resource model for the listeners described in each API under "notification listeners (client side)" are just examples and additional listeners can be created according to the implementation specific details.

REQUEST

```
POST {apiRoot}/troubleTicket/v5/hub
Content-Type: application/json
```

```
{
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent"
}
```

RESPONSE

```
201 Created
Content-Type: application/json
Location: https://tmf-api/troubleTicket/v5/hub/490b..
```

```
{
  "id": "490b..",
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent"
}
```

Example: Subscribe to specific notifications using the query mechanism:

In the following example, the client is subscribing to the `TroubleTicketStatusChangeEvent` event notifications by using the query mechanism (`"query": "eventType=TroubleTicketStatusChangeEvent"`).

REQUEST

```
POST {apiRoot}/troubleTicket/v5/hub
Content-Type: application/json
```

```
{
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketStatusChangeEvent",
  "query": "eventType=TroubleTicketStatusChangeEvent"
}
```

RESPONSE

```
201 Created
Content-Type: application/json
Location: https://tmf-api/troubleTicket/v5/hub/ba78..
```

```
{
  "id": "ba78..",
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketStatusChangeEvent",
  "query": "eventType=TroubleTicketStatusChangeEvent"
}
```

Example: Subscribe to specific notifications using the query mechanism:

In the following example, the *filter* operator defined in [\[TMF630-6\]](#) is being used as part of

the query mechanism.

REQUEST

```
POST {apiRoot}/troubleTicket/v5/hub
Content-Type: application/json
```

```
{
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent",
  "query": "filter=event.troubleTicket.attachment[?(@.size==300)]"
}
```

RESPONSE

```
201 Created
Content-Type: application/json
Location: https://tmf-api/troubleTicket/v5/hub/d488..
```

```
{
  "id": "d488..",
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent",
  "query": "filter=event.troubleTicket.attachment[?(@.size==300)]"
}
```

Example: Subscribe to specific notifications using the query mechanism (fields):

In the following example, the *fields* operator is being used part of the query mechanism to restrict the payload of the notification according to the rules specified.

REQUEST

```
POST {apiRoot}/troubleTicket/v5/hub
Content-Type: application/json
```

```
{
  "callback": "https://api.clientdomain.com/tmf-
```

```
api/troubleTicket/v5/listener/troubleTicketCreateEvent",
  "query":
    "event.troubleTicket.status=rejected&fields=event.troubleTicket.id,event.TroubleTicket.href,event.troubleTicket.statusChangeReason"
}
```

RESPONSE

```
201 Created
Content-Type: application/json
Location: https://tmf-api/troubleTicket/v5/hub/d488..
```

```
{
  "id": "d488..",
  "callback": "https://api.clientdomain.com/tmf-api/troubleTicket/v5/listener/troubleTicketCreateEvent",
  "query":
    "event.troubleTicket.status=rejected&fields=event.troubleTicket.id,event.TroubleTicket.href, event.troubleTicket.statusChangeReason"
}
```

Example: Subscribe to specific notifications (filter):

The consumer has the option, via the hub registration flow, to specify additional conditions to filter the events he will receive.

REQUEST

```
POST {apiRoot}/troubleTicket/v5/hub
Accept: application/json
```

```
{
  "callback": "https://api.clientdomain.com/listener/event",

  "query": "eventType=TroubleTicketStatusChangeEvent&event.troubleTicket.severity=Urgent&fields=event.troubleTicket.id,event.troubleTicket.name,event.troubleTicket.severity"
}
```

RESPONSE

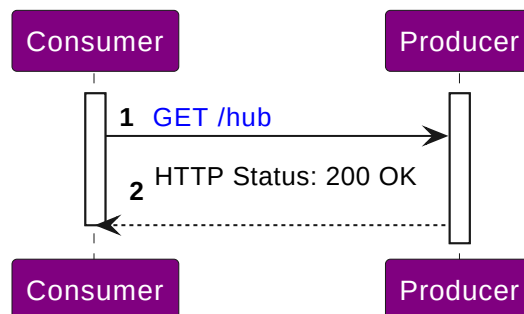
```
201 Created
Content-Type: application/json
Location: https://tmf-api/troubleTicket/v5/hub/3f2a..
```

```
{
  "id": "82b5..",
  "callback": "https://api.clientdomain.com/listener/event",
  "query":
    "eventType=TroubleTicketStatusChangeEvent&event.troubleTicket.severity=Urgent&fields=event.troubleTicket.id,event.troubleTicket.name,event.troubleTicket.severity"
}
```

11.2. Retrieve all Subscriptions

To retrieve all subscription details from the HUB, a GET must be used on the hub endpoint.

```
GET /hub
```



The query operation follow the same rules as the query on the collections.



Implementations must enforce security checks to ensure that only subscriptions associated with the authenticated user's identity and access permissions are returned in response to a GET request. This means verifying the user's authentication credentials and authorization scope to prevent unauthorized access. For example, in B2B scenarios, each partner should only be able to retrieve their own subscriptions—Partner A must not have visibility into Partner B's subscriptions.

Example: Retrieve all subscriptions:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/hub
Accept: application/json
```

RESPONSE

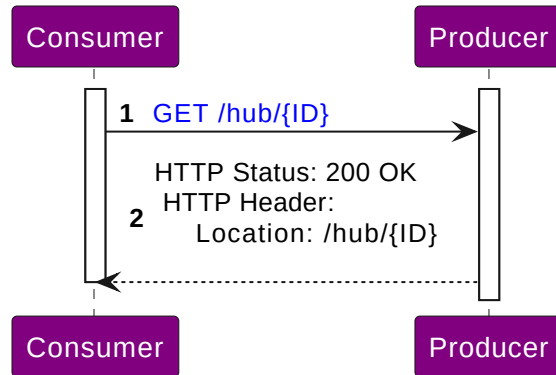
```
200 OK
Content-Type: application/json
```

```
[
  {
    "id": "82b5..",
    "callback": "https://api.clientdomain.com/listener/event",
    "query":
      "eventType=TroubleTicketStatusChangeEvent&event.troubleTicket.severity=Urgent&fields=event.troubleTicket.id,event.troubleTicket.name,event.troubleTicket.severity"
  },
  {
    "id": "c67e..",
    "callback": "https://api.clientdomain.com/listener/event",
    "query": "eventType=TroubleTicketAttributeValueChangeEvent"
  }
]
```

11.3. Retrieve Subscription

To retrieve a subscription details from the HUB, the GET must be used accompanied by the subscription identifier.

```
GET hub/{id}
```



Possible return codes for the successful subscription retrieval MUST be:

200 OK, see [\[RFC9110\]](#) (section 15.3.1).

If the request to retrieve a subscription fails a corresponding HTTP status code can be returned in accordance (e.g., client error 4xx or server error 5xxx). Default error codes supplied in the API definitions include 400, 404, 405 and 500, but the list can be extended as required by the implementation specifics.

Response Code List – Success
200 OK
Response Code List – Failure
400 Bad Request
404 Not Found
405 Method Not Allowed
500 Internal Server Error
...

Example: Retrieve a Subscription:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/hub/490b..
Accept: application/json
```

RESPONSE

```
200 OK
Content-Type: application/json
```

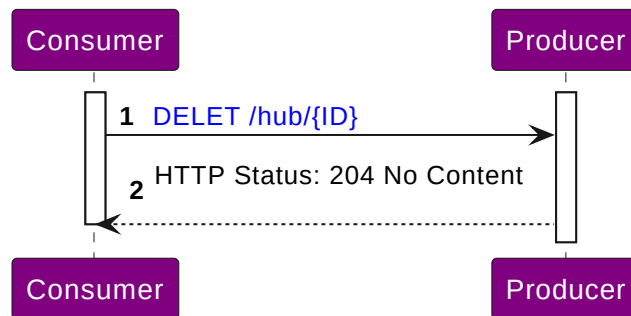
```
{
  "id": "490b..",
  "callback": "https://api.clientdomain.com/tmf-
api/troubleTicket/v5/listener/troubleTicketCreateEvent",
  "query":
"event.troubleTicket.status=rejected&fields=event.troubleTicket.id,event.TroubleT
icket.href, event.troubleTicket.statusChangeReason",
  "@type": "Hub"
}
```

11.4. Delete Subscription

To unregister a listener the HUB resource corresponding to the listener must be deleted.

```
DELETE hub/{id}
```

This removes the subscription that was created.



On successful deletion one of the two following return codes MUST be used:

- HTTP status **204 NO CONTENT** with no response body.

or

- HTTP status **202 Accepted**

Response Code List – Success
204 No Content
202 Accepted
Response Code List – Failure

404 Not Found
500 Internal Server Error
...

Example: Delete an existing subscription

REQUEST

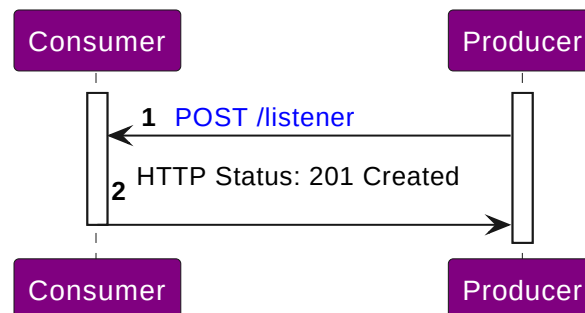
```
DELETE {apiRoot}/troubleTicket/v5/hub/ba78..
```

RESPONSE

204 No Content

11.5. Publishing Events

In the publish/subscribe pattern, once the subscription has been created, the producer/server will start to publish and send out the events that are matching the specified criteria (if any) by posting the event to the listener address specified in the subscription.



The structure of the event is defined as follows:

```
{
  "eventId": "The identifier of the notification",
  "eventTime": "Time of the event notification.",
  "eventType": "The type of the notification.",
  "correlationId": "The correlation id for this event.",
  "domain": "The domain of the event.",
  "title": "The title of the event.",
  "description": "An explanation of the event.",
  "priority": "Event priority",
}
```

```

"timeOccurred": "The time the event occurred.",
"fieldPath": "The path identifying the object field concerned by this
notification",
"event": {
  "resource": {
    ... // RESOURCE SAMPLE
  }
},
"@type": "Event",
"@baseType": "When sub-classing, this defines the super-class",
"@schemaLocation": "A URI to a JSON-Schema file that defines additional
attributes and relationships"
}

```



More information on the event types supported by an API and possible extensions of the event type's schema can be found in the API's specification and user guide.

On successful sending the event a **201 Created** or **204 No Content** return code **MUST** be used.

If the request to send the notification fails a corresponding HTTP status code can be returned in accordance (e.g., client error 4xx or server error 5xx). Default error codes supplied in the API definitions include 400, 401, 403, 404, 405 and 500, but the list can be extended as required by the implementation specifics.

Response Code List – Success
201 Created
204 No Content
Response Code List – Failure
400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
405 Method Not Allowed
422 Unprocessable Content
500 Internal Server Error
...

Example: Sending a TroubleTicket event:

REQUEST

```
POST {apiRoot}/troubleTicket/v5/listener/TroubleTicketStatusChangeEvent
Accept: application/json
```

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "TroubleTicketStatusChangeEvent",
  "event": {
    "troubleTicket": {
      "@type": "TroubleTicket",
      "id": "f6ac..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",
      "status": "acknowledged",
      "statusChangeDate": "2023-09-04T12:32:26.203Z",
      "statusChangeReason": "Need more information from the customer",
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "owner": "DocumentManagementSystem",
          "externalIdentifierType": "Screenshot",
          "id": "4aaf..",
          "href": "https://mycsp.com/dms/screenshot/4aaf.."
        }
      ],
      "note": [
        {
          "@type": "Note",
          "id": "6050..",
          "author": "Peter Helper",
          "date": "2023-09-04T12:32:26.203Z",
          "text": "Received missing information from the customer."
        }
      ]
    },
    "statusChangeHistory": [
      {
        "@type": "StatusChange",
        "statusChangeDate": "2023-09-04T08:59:27.255Z",
        "statusChangeReason": "Need more information from the customer",
        "status": "acknowledged"
      },
      {

```

```
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T12:32:26.203Z",
      "statusChangeReason": "Received information from the customer",
      "status": "InProgress"
    }

  ]
},
"@baseType": "Event",
"@type": "TroubleTicketCreateEvent"
}
```

RESPONSE

204 No Content

11.6. Event Types

Across the TM Forum Open API, a set of standard events are published with the API specifications.

This includes the default events reflecting changes performed against the API's resources:

- {resource}CreateEvent
- {resource}AttributeValueChangeEvent
- {resource}StatusChangeEvent / {resource}StateChangeEvent
- {resource}DeleteEvent
- {resource}VersionCreateEvent (for versioned resources only, see [\[TMF630-4\]](#))
- {resource}VersionCurrentEvent (for versioned resources only, see [\[TMF630-4\]](#))



The naming convention for the events follows the notation:
{resourceName}{EventName}

Example: event names:

- TroubleTicketCreateEvent
- TroubleTicketAttributeValueChangeEvent

In addition to the standard events described above, each API can define additional events

specific to the API.

Example:

Looking at the TroubleTicket API we can identify the following events being specified by the API specification:

Type	Event Name
Default	TroubleTicketCreateEvent
Default	TroubleTicketAttributeValueChangeEvent
Default	TroubleTicketStatusChangeEvent
Default	TroubleTicketDeleteEvent
Specific	TroubleTicketResolvedEvent
Specific	TroubleTicketInformationRequiredEvent



TroubleTicket is not versioned, so that there is no "TroubleTicketVersionCreateEvent".

All the *default* events follow the same *Event* schema.

All the *specific* events MUST follow the same *Event* schema, but they will be described in the associated API-specific documentation.

The structure of the *Event* schema has been extended for and has the following structure:

```
{
  "@type":          "When sub-classing, this defines the sub-class Extensible
name",
  "@baseType":      "When sub-classing, this defines the super-class",
  "@schemaLocation": "A URI to a JSON-Schema file that defines additional
attributes and relationships",
  "id":             "Unique Identifier",
  "correlationId":  "The correlation id for this event",
  "domain":         "The domain of the event.",
  "title":          "The title of the event.",
  "description":    "An explanation of the event.",
  "priority":       "Event priority",
  "timeOccurred":   "The time the event occurred.",
  "source":         "Source Entity described via EntityRef",
  "reportingSystem": "Reporting System described by EntityRef",
  "relatedParty":   "Related Entity reference. A related party defines party
```

```

linked to a specific entity",
  "analyticCharacteristic": "Analytic characteristics",
  "eventId":                "The identifier of the notification",
  "eventTime":              "Time of the event notification.",
  "eventType":              "The type of the notification.",
  "event": {
    "resource": {
      ... // Resource Payload }
    }
  }
}

```



The **resource** in the above structure is a placeholder for the actual lower camel case name of the resource, so in Trouble Ticket case it would be **troubleTicket**.

The *Event* schema is also *extensible* so it can be extended via sub-classing.



In specific scenarios, multiple events of different types might be emitted for a single resource.

For example, a change of status performed via PATCH operation might result in two different event types being emitted:

AttributeValueChangeEvent and *StatusChangeEvent*

11.6.1. CreateEvent

The *CreateEvent* will be issued when a new *resource* has been created.



It **MUST** include the full contents of the resource (including all sub-resources) that has been created.

```

{
  "correlationId": "ae70..",
  "description": "TroubleTicketCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "bb45..",
  "eventTime": "2022-08-25T12:18:12.171Z",
  "eventType": "TroubleTicketCreateEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:18:06.252Z",
  "title": "TroubleTicketCreateEvent",
  "event": {

```

```

"troubleTicket": {
  "@type": "TroubleTicket",
  "id": "f6ac..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or
phone service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T08:59:27.255Z",
  "lastUpdate": "2023-09-04T08:59:27.255Z",
  "priority": "High",
  "status": "acknowledged",
  "statusChangeDate": "2023-09-04T08:59:27.255Z",
  "statusChangeReason": "Need more information from the customer",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "4aaf..",
      "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
      "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
    }
  ],
  "channel": {
    "@type": "ChannelRef",
    "id": "8be1..",
    "href": "https://mycsp.com/channel/8be1..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "4aaf..",
      "href": "https://mycsp.com/dms/screenshot/4aaf.."
    }
  ],
  "note": [
    {

```

```

        "@type": "Note",
        "id": "3153..",
        "author": "Mr. N. Bene",
        "date": "2023-09-04T08:59:27.255Z",
        "text": "This is critical, please address it urgently."
    },
    ],
    "relatedEntity": [
        {
            "@type": "RelatedEntity",
            "role": "Exchange",
            "entity": {
                "@type": "EntityRef",
                "id": "4aaf..",
                "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
                "name": "EXC-MK-1872",
                "@referredType": "Resource"
            }
        }
    ],
    "relatedParty": [
        {
            "role": "reporter",
            "partyOrPartyRole": {
                "id": "620d..",
                "name": "Mr. N. Bene",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
            "@type": "RelatedPartyRefOrPartyRoleRef"
        }
    ],
    "statusChangeHistory": [
        {
            "@type": "StatusChange",
            "statusChangeDate": "2023-09-04T08:59:27.255Z",
            "statusChangeReason": "Need more information from the customer",
            "status": "acknowledged"
        }
    ]
}
},
"@baseType": "Event",
"@type": "TroubleTicketCreateEvent"

```

}

11.6.2. AttributeValueChangeEvent

The *AttributeValueChangeEvent* will be issued when there is a change in one or more of the attributes for a *resource*. Its primary purpose is to notify the listeners of a change in the resource(s) so they don't need to contact the source system (*Event-Carried State Transfer*).

When a *resource* is updated via the API-specific operations (e.g., Patch), the *AttributeValueChangeEvent* will be emitted, capturing the changes that occurred against the resource.

The *AttributeValueChangeEvent* will contain the details of the changed resource data based on the semantics of the **JSON MERGE**. For more details regarding the JSON Merge Patch semantics see [\[RFC7396\]](#).

For example – TroubleTicket event where the ticket priority and severity were changed:

```
{
  "correlationId": "26ef..",
  "description": "TroubleTicketAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "bb45..",
  "eventTime": "2022-08-25T12:18:12.171Z",
  "eventType": "TroubleTicketAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:18:06.252Z",
  "title": "TroubleTicketAttributeValueChangeEvent",
  "event": {
    "troubleTicket": {
      "id": "2a0a..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/2a0a..",
      "priority": "Low",
      "severity": "Low",
      "@type": "TroubleTicket"
    }
  },
  "@baseType": "Event",
  "@type": "TroubleTicketAttributeValueChangeEvent"
}
```



The *AttributeValueChangeEvent* resource must always include the resource's addressable information (e.g., ID).



As the *AttributeValueChangeEvent* is following the JSON Merge Patch, this means that the consumer must maintain its own local interpretation of the data (partial or full).

- As per JSON Merge semantics, merge patches cannot easily manipulate arrays. If you want to modify any of its elements, remove or add an element to the array, then the entire array must be included in the merge patch document, even if the changed parts are minimal. As a general rule, the new array completely replaces the old one.
- As the default behaviour for the event is to contain only the **delta** (changed values), the consumer must process all the events, in order to maintain the data consistency.
- Alternative solution is for the producer to always send the complete resource definition, however this option must be carefully considered as sending the full payload will considerably affect the network traffic, processing, etc.

11.6.3. StatusChangeEvent or StateChangeEvent

The *StatusChangeEvent* resp. *StateChangeEvent* will be issued when the resource lifecycle *status/state* has changed.



It MUST include the status of the resource.

```
{
  "correlationId": "4599..",
  "description": "TroubleTicketStatusChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "bb45..",
  "eventTime": "2022-08-25T12:18:12.171Z",
  "eventType": "TroubleTicketStatusChangeEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:18:06.252Z",
  "title": "TroubleTicketStatusChangeEvent",
  "event": {
    "troubleTicket": {
      "id": "2a0a..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/2a0a..",
      "status": "pending",
      "statusChangeDate": "2019-05-31T07:34:45.968Z",
      "statusChangeReason": "Need more information from the customer",

```

```

    "@type": "TroubleTicket"
  },
  "@baseType": "Event",
  "@type": "TroubleTicketStatusChangeEvent"
}

```

11.6.4. DeleteEvent

The *DeleteEvent* will be issued when a resource has been deleted.



The payload of the resource MUST include only the resource's addressable information (e.g., *id*, *href*).

For example:

```

{
  "correlationId": "45eb..",
  "description": "TroubleTicketDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "bb45..",
  "eventTime": "2022-08-25T12:18:12.171Z",
  "eventType": "TroubleTicketDeleteEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:18:06.252Z",
  "title": "TroubleTicketDeleteEvent",
  "event": {
    "troubleTicket": {
      "id": "2a0a..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/2a0a..",
      "@type": "TroubleTicket"
    }
  },
  "@baseType": "Event",
  "@type": "TroubleTicketDeleteEvent"
}

```

11.6.5. VersionCreateEvent

The *VersionCreateEvent* will be issued when a version of a versioned resource (see part 4 of the Guideline) has been created.



As for the CreateEvent, the VersionCreateEvent MUST include the full contents of the new version metadata that has been created. The actual new version of the entity is created as a clone of the versioned entity and thus is not included in the event payload.

For example:

```
{
  "correlationId": "45eb..",
  "description": "ProductOfferingVersionCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "074c..",
  "eventTime": "2022-08-25T12:18:12.171Z",
  "eventType": "ProductOfferingVersionCreateEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:18:06.252Z",
  "title": "ProductOfferingVersionCreateEvent",
  "event": {
    "version": {
      "id": "45712365868452352",
      "href":
"tmfapi/productCatalogManagement/v5/productOffering/P06755662309056452/version/45
712365868452352",
      "@type": "Version",
      "creationDate": "2023-12-15T16:42:23-04:00",
      "version": "3.0.0",
      "description": "Major enhancement to Virtual Storage, allowing RAID and
secure backups",
      "isCurrent": false
    }
  },
  "@baseType": "Event",
  "@type": "ProductOfferingVersionCreateEvent"
}
```

Chapter 12. History Pattern

The following section describes the history pattern defining the mechanics and rules where systems exposing the TM Forum APIs provide support for API resource revision tracking.



The here described *revisions* of API resources must be distinguished from resource *versions* defined in Part 4 of the Guidelines.



The history pattern is not mandatory for all TM Forum APIs, and so far are not described in API Specifications. It is up to the decision of implementations to add support of the history pattern.

The history pattern allows implementations to expose the history and revision information on previous revisions of the resource(s) managed by a particular TM Forum API (note: the pattern is applied per API).

There are different reasons why there might be a need to capture the history of a resource, legal purposes, audit trailing or enabling business intelligence systems or advanced analytics to optimize, correlate and predict the next course of action based on the historical decision points and behaviour.

With revision control, every change performed against the resource is tracked, allowing consumers to see the entire history of the resource and what was changed, when and by whom.

When enabled through a TM Forum API, the history will be exposed through a separate endpoint: `{apiRoot}/{resourceName}/{resourceID}/{history}`

For example:

```
{apiRoot}/troubleTicket/v5/troubleTicket/df7e../history
```

where

```
{apiRoot} - {apiRoot}/troubleTicket/v5  
{resourceName} - troubleTicket  
{resourceID} - df7e..  
{history} - /history
```

A particular revision from the *history* collection will be available under the endpoint: `{apiRoot}/{resourceName}/{resourceID}/{history}/{revisionID}`

For example:

```
{apiRoot}/troubleTicket/v5/troubleTicket/df7e../history/965
```

where

```
{apiRoot} - {apiRoot}/troubleTicket/v5/  
{resourceName} - troubleTicket  
{resourceID} - df7e..  
{history} - /history  
{revision} - /965
```

12.1. Revisioning of Resources

When a resource is created (e.g. `POST {apiRoot}/troubleTicket/v5/troubleTicket`), it is put under *revision control/history*, and it becomes a "*revisioned resource*".

Every time the resource is updated (or when it is deleted), another revision in the *history* collection for that resource will be created.

The *history* resource is a collection that contains all the revisions (e.g. *history entries*) of the particular revisioned resource.

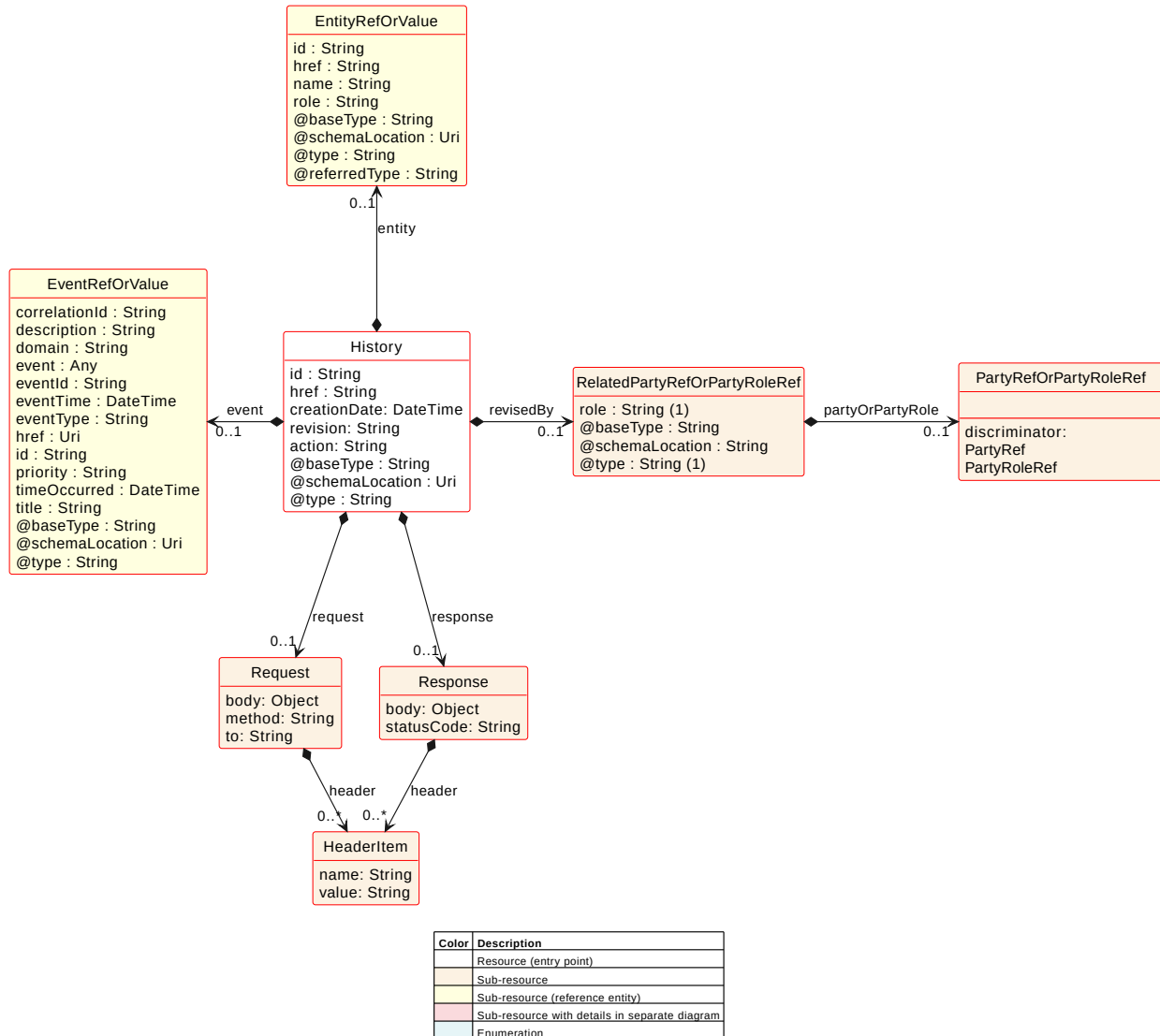


The *history* resources are considered protected resources from modifications as they are considered a snapshot of the resource; as such, they are **immutable**.

The consumer of the API will only be allowed to retrieve a previous revision from the *history*, but it won't be able to perform any modifications on the revision.

12.2. History Resource

The '*history*' resource has the following definition:



It provides the following information:

- Who changed the resource?
 - This is provided by the **RelatedPartyRefOrPartyRoleRef** entity. Through the authentication and authorization mechanism, the server is responsible for populating the entity with the relevant information.
 - *For example*, if the user 'John' is updating the troubleTicket with ID=bdOa., it is the server's responsibility when a new ticket revision is created in the history collection to populate the **RelatedPartyRefOrPartyRoleRef** information with John details.
- When was it changed?
 - This is provided by the **creationDate**. Every a new revision is created under **history**, the timestamp is captured against the **creationDate** attribute

- What is the current content of the resource
 - The new revision of the resource is captured as part of the **EntityRefOrValue** entity
- What is the current revision of the resource?
 - Captured against the **revision** attribute. The implementation will decide what the revisioning scheme is; nevertheless, the revision number must be unique
- What action was performed against the resource?
 - The initial action against the resource will be captured as part of the **action** attribute with possible values enum: created, updated, deleted.
- What was the original HTTP Request and its corresponding headers that resulted in a new snapshot/revision to be created?
 - The original HTTP request payload and headers will be recorded as part of the **Request & HeaderItem** entities
- What was the HTTP response and the corresponding headers?
 - The response payload and corresponding headers will be captured as part of the **Response & HeaderItem** entities.
- What was the event that caused a new revision of the entity to be created ?
 - In the case the source of the change is via an event the details of the event are captured in the 'EventRefOrValue' entity. In this case the Request/Response entities won't be populated.

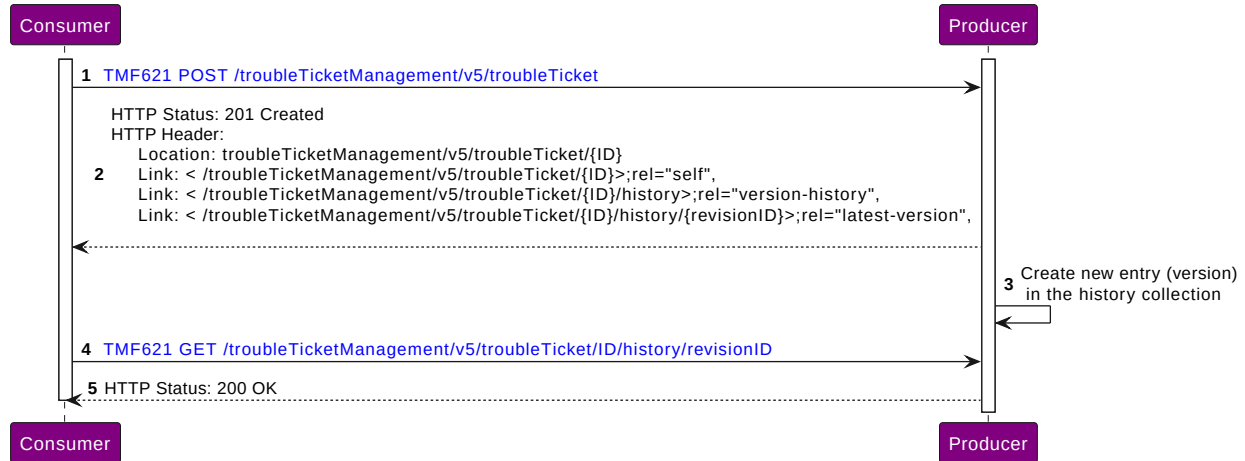
12.3. History Operations

To support the *history* the implementation must expose the '{apiRoot}/{resourceName}/{resourceID}/{history}' endpoint.

A new snapshot (revision) will be created in the **history** collection when a resource is created for the first time.



The **only** operation allowed on the history resource is "GET".



For example, creation of a new troubleTicket:

REQUEST

```
POST {apiRoot}/troubleTicket/v5/troubleTicket
Accept: application/json
```

```
{
  "@type": "TroubleTicket",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone
service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T08:59:27.255Z",
  "lastUpdate": "2023-09-04T08:59:27.255Z",
  "priority": "High",
  "status": "acknowledged",
  "statusChangeDate": "2023-09-04T08:59:27.255Z",
  "statusChangeReason": "Need more information from the customer",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "id": "4aaf..",
      "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
      "name": "Network Status Screenshot",
      "description": "Screenshot of network status",
    }
  ]
}
```

```
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  },
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "620d..",
```

```

        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ]
}

```

RESPONSE

201 Created
Content-Type: application/json

Location: {apiRoot}/troubleTicket/v5/troubleTicket/de8b..
Link: <{apiRoot}/troubleTicket/v5/troubleTicket/de8b..>; rel="self",
 <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>; rel="version-history",
 <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/df4d..>;
 rel="latest-version",

```

{
  "@type": "TroubleTicket",
  "id": "de8b..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
  "name": "Network Outage Report",
  "description": "Experiencing a network outage in my area. No internet or phone service available.",
  "severity": "High",
  "ticketType": "NetworkIssues",
  "creationDate": "2023-09-04T08:59:27.255Z",
  "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
  "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
  "resolutionDate": "2023-09-04T08:59:27.255Z",
  "lastUpdate": "2023-09-04T08:59:27.255Z",
  "priority": "High",
  "status": "acknowledged",
  "statusChangeDate": "2023-09-04T08:59:27.255Z",
  "statusChangeReason": "Need more information from the customer",
  "attachment": [
    {

```

```
    "@type": "AttachmentRef",
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  },
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
```

```
"relatedParty": [  
  {  
    "role": "reporter",  
    "partyOrPartyRole": {  
      "id": "620d..",  
      "name": "Mr. N. Bene",  
      "@type": "PartyRef",  
      "@referredType": "Individual"  
    },  
    "@type": "RelatedPartyRefOrPartyRoleRef"  
  },  
]  
}
```

12.3.1. Retrieve all revisions from History

In order to retrieve all revisions from the **history** collection a GET request must be made.

The **history** as all the other collection levels MUST follow the same pagination rules (e.g., offset, cursor-based) and support the same **selectors** (e.g., filter, fields) as all the others.

Example: Retrieve all revisions from the history:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/de8b../history  
Accept: application/json
```

RESPONSE

```
200 OK  
Content-Type: application/json  
Link:  
<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?offset=0&limit=10>;  
rel="self",  
  
<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?offset=0&limit=10>;  
rel="last"
```

```
[  
  {
```

```

    "id": "c349..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "revision": "1",
    "action": "create",
    "revisedBy": {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    "entity": {
      "@type": "TroubleTicket",
      "id": "de8b..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
      "name": "Network Outage Report",
      "description": "Experiencing a network outage in my area. No internet or
phone service available.",
      "severity": "High",
      "ticketType": "NetworkIssues",
      "creationDate": "2023-09-04T08:59:27.255Z",
      "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
      "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
      "resolutionDate": "2023-09-04T08:59:27.255Z",
      "lastUpdate": "2023-09-04T08:59:27.255Z",
      "priority": "High",
      "status": "acknowledged",
      "statusChangeDate": "2023-09-04T08:59:27.255Z",
      "statusChangeReason": "Need more information from the customer",
      "attachment": [
        {
          "@type": "AttachmentRef",
          "id": "4aaf..",
          "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
          "name": "Network Status Screenshot",
          "description": "Screenshot of network status",
          "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
        }
      ],
      "channel": {
        "@type": "ChannelRef",

```

```
    "id": "8be1..",
    "href": "https://mycsp.com/channel/8be1..",
    "name": "Web"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "DocumentManagementSystem",
      "externalIdentifierType": "Screenshot",
      "id": "4aaf..",
      "href": "https://mycsp.com/dms/screenshot/4aaf.."
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "3153..",
      "author": "Mr. N. Bene",
      "date": "2023-09-04T08:59:27.255Z",
      "text": "This is critical, please address it urgently."
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntity",
      "role": "Exchange",
      "entity": {
        "@type": "EntityRef",
        "id": "4aaf..",
        "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
        "name": "EXC-MK-1872",
        "@referredType": "Resource"
      }
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef"
```

```

    }
  ]
},
"request": {
  "method": "POST",
  "body": "...",
  "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
  "statusCode": "201",
  "body": "...",
  "header": [
    {
      "name": "Link",
      "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
    }
  ]
}
},
{
  "id": "b714..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/b714..",
  "creationDate": "2023-06-03T07:35:56.465Z",
  "revision": "2",
  "action": "update",
  "revisedBy": {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "620d..",
      "name": "Mr. N. Bene",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  "entity": {
    "@type": "TroubleTicket",
    "id": "de8b..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",

```

```
"creationDate": "2023-09-04T08:59:27.255Z",
"requestedResolutionDate": "2023-09-04T08:59:27.255Z",
"expectedResolutionDate": "2023-09-04T08:59:27.255Z",
"resolutionDate": "2023-09-04T08:59:27.255Z",
"lastUpdate": "2023-09-04T08:59:27.255Z",
"priority": "High",
"status": "acknowledged",
"statusChangeDate": "2023-09-04T08:59:27.255Z",
"statusChangeReason": "Need more information from the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  },
  {
    "@type": "Note",
    "id": "14b9..",
```

```

        "author": "Mr. N. Bene",
        "date": "2023-06-03T07:35:56.465Z",
        "text": "As an update the customer is also reporting iptv issues."
    }
],
"relatedEntity": [
    {
        "@type": "RelatedEntity",
        "role": "Exchange",
        "entity": {
            "@type": "EntityRef",
            "id": "4aaf..",
            "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
            "name": "EXC-MK-1872",
            "@referredType": "Resource"
        }
    }
],
"relatedParty": [
    {
        "role": "reporter",
        "partyOrPartyRole": {
            "id": "620d..",
            "name": "Mr. N. Bene",
            "@type": "PartyRef",
            "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    }
]
},
"request": {
    "method": "PATCH",
    "body": "...",
    "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
    "statusCode": "200",
    "body": "...",
    "header": [
        {
            "name": "Link",
            "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
        }
    ]
}

```

```
]
  }
}
]
```

Example: Retrieve all revisions from the history using the offset based pagination:

REQUEST

```
GET {apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?offset=0&limit=2
Accept: application/json
```

RESPONSE

```
200 OK
Content-Type: application/json
Link:
<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?offset=0&limit=2>;
rel="self",

<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?offset=0&limit=2>;
rel="last"
```

```
[
  {
    "id": "c349..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "revision": "1",
    "action": "create",
    "revisedBy": {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    "entity": {
```

```
"@type": "TroubleTicket",
"id": "de8b..",
"href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
"name": "Network Outage Report",
"description": "Experiencing a network outage in my area. No internet or
phone service available.",
"severity": "High",
"ticketType": "NetworkIssues",
"creationDate": "2023-09-04T08:59:27.255Z",
"requestedResolutionDate": "2023-09-04T08:59:27.255Z",
"expectedResolutionDate": "2023-09-04T08:59:27.255Z",
"resolutionDate": "2023-09-04T08:59:27.255Z",
"lastUpdate": "2023-09-04T08:59:27.255Z",
"priority": "High",
"status": "acknowledged",
"statusChangeDate": "2023-09-04T08:59:27.255Z",
"statusChangeReason": "Need more information from the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
```

```

        "id": "3153..",
        "author": "Mr. N. Bene",
        "date": "2023-09-04T08:59:27.255Z",
        "text": "This is critical, please address it urgently."
    },
    ],
    "relatedEntity": [
        {
            "@type": "RelatedEntity",
            "role": "Exchange",
            "entity": {
                "@type": "EntityRef",
                "id": "4aaf..",
                "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
                "name": "EXC-MK-1872",
                "@referredType": "Resource"
            }
        }
    ],
    "relatedParty": [
        {
            "role": "reporter",
            "partyOrPartyRole": {
                "id": "620d..",
                "name": "Mr. N. Bene",
                "@type": "PartyRef",
                "@referredType": "Individual"
            },
            "@type": "RelatedPartyRefOrPartyRoleRef"
        }
    ],
    },
    "request": {
        "method": "POST",
        "body": "...",
        "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
    },
    "response": {
        "statusCode": "201",
        "body": "...",
        "header": [
            {
                "name": "Link",
                "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",

```

```

    }
  ]
}
},
{
  "id": "b714..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/b714..",
  "creationDate": "2023-06-03T07:35:56.465Z",
  "revision": "2",
  "action": "update",
  "revisedBy": {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "620d..",
      "name": "Mr. N. Bene",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  "entity": {
    "@type": "TroubleTicket",
    "id": "de8b..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "resolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "4aaf..",
        "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",

```

```

        "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
    },
    "channel": {
        "@type": "ChannelRef",
        "id": "8be1..",
        "href": "https://mycsp.com/channel/8be1..",
        "name": "Web"
    },
    "externalIdentifier": [
        {
            "@type": "ExternalIdentifier",
            "owner": "DocumentManagementSystem",
            "externalIdentifierType": "Screenshot",
            "id": "4aaf..",
            "href": "https://mycsp.com/dms/screenshot/4aaf.."
        }
    ],
    "note": [
        {
            "@type": "Note",
            "id": "3153..",
            "author": "Mr. N. Bene",
            "date": "2023-09-04T08:59:27.255Z",
            "text": "This is critical, please address it urgently."
        },
        {
            "@type": "Note",
            "id": "14b9..",
            "author": "Mr. N. Bene",
            "date": "2023-06-03T07:35:56.465Z",
            "text": "As an update the customer is also reporting iptv issues."
        }
    ],
    "relatedEntity": [
        {
            "@type": "RelatedEntity",
            "role": "Exchange",
            "entity": {
                "@type": "EntityRef",
                "id": "4aaf..",
                "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
                "name": "EXC-MK-1872",
                "@referredType": "Resource"
            }
        }
    ]

```

```

    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ]
},
"request": {
  "method": "PATCH",
  "body": "...",
  "to": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b.."
},
"response": {
  "statusCode": "200",
  "body": "...",
  "header": [
    {
      "name": "Link",
      "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
    }
  ]
}
}
]

```

12.3.2. Retrieve Revision from History matching Criteria

Example: Retrieve the revision with number 1:

REQUEST

```

GET {apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?revision=1
Accept: application/json

```

RESPONSE

200 OK
Content-Type: application/json

```
[
  {
    "id": "c349..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "revision": "1",
    "action": "create",
    "revisedBy": {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    "entity": {
      "@type": "TroubleTicket",
      "id": "de8b..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
      "name": "Network Outage Report",
      "description": "Experiencing a network outage in my area. No internet or
phone service available.",
      "severity": "High",
      "ticketType": "NetworkIssues",
      "creationDate": "2023-09-04T08:59:27.255Z",
      "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
      "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
      "resolutionDate": "2023-09-04T08:59:27.255Z",
      "lastUpdate": "2023-09-04T08:59:27.255Z",
      "priority": "High",
      "status": "acknowledged",
      "statusChangeDate": "2023-09-04T08:59:27.255Z",
      "statusChangeReason": "Need more information from the customer",
      "attachment": [
        {
          "@type": "AttachmentRef",
```

```
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  },
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
],
"relatedParty": [
```

```

    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ]
},
"request": {
  "method": "POST",
  "body": "...",
  "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
  "statusCode": "201",
  "body": "...",
  "header": [
    {
      "name": "Link",
      "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
    }
  ]
}
}
]

```

Example: retrieve the revision which has a note from John Smith:

REQUEST

```

GET
{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history?filter=entity.note[?(@.author=="Jack Smith")]
Accept: application/json

```

RESPONSE

200 OK

Content-Type: application/json

```
[
  {
    "id": "c349..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "revision": "3",
    "action": "create",
    "revisedBy": {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    "entity": {
      "@type": "TroubleTicket",
      "id": "de8b..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
      "name": "Network Outage Report",
      "description": "Experiencing a network outage in my area. No internet or
phone service available.",
      "severity": "High",
      "ticketType": "NetworkIssues",
      "creationDate": "2023-09-04T08:59:27.255Z",
      "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
      "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
      "resolutionDate": "2023-09-04T08:59:27.255Z",
      "lastUpdate": "2023-09-04T08:59:27.255Z",
      "priority": "High",
      "status": "acknowledged",
      "statusChangeDate": "2023-09-04T08:59:27.255Z",
      "statusChangeReason": "Need more information from the customer",
      "attachment": [
        {
          "@type": "AttachmentRef",
          "id": "4aaf..",
          "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
```

```
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",
        "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
    },
    ],
    "channel": {
        "@type": "ChannelRef",
        "id": "8be1..",
        "href": "https://mycsp.com/channel/8be1..",
        "name": "Web"
    },
    ],
    "externalIdentifier": [
        {
            "@type": "ExternalIdentifier",
            "owner": "DocumentManagementSystem",
            "externalIdentifierType": "Screenshot",
            "id": "4aaf..",
            "href": "https://mycsp.com/dms/screenshot/4aaf.."
        }
    ],
    ],
    "note": [
        {
            "@type": "Note",
            "id": "3153..",
            "author": "Jack Smith",
            "date": "2023-09-04T08:59:27.255Z",
            "text": "This is critical, please address it urgently."
        }
    ],
    ],
    "relatedEntity": [
        {
            "@type": "RelatedEntity",
            "role": "Exchange",
            "entity": {
                "@type": "EntityRef",
                "id": "4aaf..",
                "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
                "name": "EXC-MK-1872",
                "@referredType": "Resource"
            }
        }
    ],
    ],
    "relatedParty": [
        {
            "role": "reporter",
```

```

    "partyOrPartyRole": {
      "id": "620d..",
      "name": "Jack Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  ]
},
"request": {
  "method": "POST",
  "body": "...",
  "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
  "statusCode": "201",
  "body": "...",
  "header": [
    {
      "name": "Link",
      "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
    }
  ]
}
}
]

```

12.3.3. Retrieve a specific Revision based on ID

A specific revision can be retrieved from the **history** collection using the revision id.

Example: Retrieve the revision with **ID=c349..**:

REQUEST

```

GET tmf-api/troubleTicket/v5/troubleTicket/de8b../history/c349..
Accept: application/json

```

RESPONSE

```
200 OK
Content-Type: application/json
Link: <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..;
rel="self",
      <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>; rel="version-
history",
      <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/b714..>;
rel="latest-version",
```

```
{
  "id": "c349..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
  "creationDate": "2023-05-31T07:34:45.968Z",
  "revision": "1",
  "action": "create",
  "revisedBy": {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "620d..",
      "name": "Mr. N. Bene",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  "entity": {
    "@type": "TroubleTicket",
    "id": "de8b..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "resolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
```

```
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
      "@type": "EntityRef",
      "id": "4aaf..",
      "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
      "name": "EXC-MK-1872",
      "@referredType": "Resource"
    }
  }
]
```

```

    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Jack Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ]
},
"request": {
  "method": "POST",
  "body": "...",
  "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
  "statusCode": "201",
  "body": "...",
  "header": [
    {
      "name": "Link",
      "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\\\"version-history\\\", \"
    }
  ]
}
}

```

12.3.4. Retrieve the Revision that was valid at a given Date/Time

The revision that was the current one at a given date/time can be retrieved from the **history** collection as follows:

Example: Retrieve the revision valid on June 2nd, 2023, at 10 pm UTC:

REQUEST

```
GET tmf-api/troubleTicket/v5/troubleTicket/de8b../history?creationDate.lte=2023-06-02T22:00:00.000Z&sort=-creationDate&limit=1
```

Accept: application/json

RESPONSE

```
200 OK
Content-Type: application/json
Link: <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..;
rel="self",
      <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>; rel="version-
history",
      <{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/b714..>;
rel="latest-version",
```

```
[
  {
    "id": "c349..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history/c349..",
    "creationDate": "2023-05-31T07:34:45.968Z",
    "revision": "1",
    "action": "create",
    "revisedBy": {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "620d..",
        "name": "Mr. N. Bene",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    "entity": {
      "@type": "TroubleTicket",
      "id": "de8b..",
      "href": "{apiRoot}/troubleTicket/v5/troubleTicket/de8b..",
      "name": "Network Outage Report",
      "description": "Experiencing a network outage in my area. No internet or
phone service available.",
      "severity": "High",
      "ticketType": "NetworkIssues",
      "creationDate": "2023-09-04T08:59:27.255Z",
      "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
      "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
```

```
"resolutionDate": "2023-09-04T08:59:27.255Z",
"lastUpdate": "2023-09-04T08:59:27.255Z",
"priority": "High",
"status": "acknowledged",
"statusChangeDate": "2023-09-04T08:59:27.255Z",
"statusChangeReason": "Need more information from the customer",
"attachment": [
  {
    "@type": "AttachmentRef",
    "id": "4aaf..",
    "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
    "name": "Network Status Screenshot",
    "description": "Screenshot of network status",
    "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
  }
],
"channel": {
  "@type": "ChannelRef",
  "id": "8be1..",
  "href": "https://mycsp.com/channel/8be1..",
  "name": "Web"
},
"externalIdentifier": [
  {
    "@type": "ExternalIdentifier",
    "owner": "DocumentManagementSystem",
    "externalIdentifierType": "Screenshot",
    "id": "4aaf..",
    "href": "https://mycsp.com/dms/screenshot/4aaf.."
  }
],
"note": [
  {
    "@type": "Note",
    "id": "3153..",
    "author": "Mr. N. Bene",
    "date": "2023-09-04T08:59:27.255Z",
    "text": "This is critical, please address it urgently."
  }
],
"relatedEntity": [
  {
    "@type": "RelatedEntity",
    "role": "Exchange",
    "entity": {
```

```

        "@type": "EntityRef",
        "id": "4aaf..",
        "href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
        "name": "EXC-MK-1872",
        "@REFERREDType": "Resource"
    }
}
],
"relatedParty": [
{
    "role": "reporter",
    "partyOrPartyRole": {
        "id": "620d..",
        "name": "Jack Smith",
        "@type": "PartyRef",
        "@REFERREDType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
}
]
},
"request": {
    "method": "POST",
    "body": "...",
    "to": "{apiRoot}/troubleTicket/v5/troubleTicket/"
},
"response": {
    "statusCode": "201",
    "body": "...",
    "header": [
        {
            "name": "Link",
            "value": "<{apiRoot}/troubleTicket/v5/troubleTicket/de8b../history>;
rel=\"version-history\"",
        }
    ]
}
}
]

```

12.4. History Links via the Link Header

The **history** pattern is following the [\[RFC5829\]](#).

The following Link Relations are supported:

Link Relation	Description
<code>version-history</code>	When included on a revisioned resource, this link points to a resource containing the revision history for this resource.
<code>latest-version</code>	When included on a revisioned resource, this link points to a resource containing the latest (i.e., current) revision.
<code>predecessor-version</code>	When included on a revisioned resource, this link points to a resource containing the predecessor revision in the revision history.
<code>successor-version</code>	When included on a revisioned resource, this link points to a resource containing the successor revision in the revision history.



The use of "version" in the above link relations originates from [\[RFC5829\]](#) and is not replaced by a more appropriate term "revision". Do not confuse with the term "Version" used for Resource Versioning in [\[TMF630-4\]](#).

Chapter 13. Miscellaneous

13.1. API Versioning



In the following, "API Version" refers to the TMF API specification version as indicated in the TMF API OAS "version" attribute (for instance "version: 5.0.0"); it does NOT refer of the version of the API implementation.

The API's full version ID has major, minor and patch numbers, e.g. **v5.0.0** and is specified inside the TM Forum OpenAPI OAS:

```
{apiFullVersion} = {majorApiVersion}.{minorApiVersion}.{patchApiVersion}
```



For more details see [Semantic Versioning](#).

The versioning is applied uniformly to all the entities under a versioned API. That is, it is assumed that entities under the management scope of the API are all aligned.

13.1.1. Major Version

The major version number (**{majorApiVersion}**) is incremented for an incompatible change.

For major modifications of the API, not backward compatible and forcing client implementations to be changed, the version number must be updated.

13.1.2. Minor Version

The minor version number (**{minorApiVersion}**) is incremented for a compatible change.

For minor modifications of the API specification, major version numbering SHOULD not be updated, provided the following backward compatibility rules are respected:

- New elements in a data type must be optional (minOccurs=0)
- Changes in the cardinality of an attribute in a data type must be from mandatory to optional or from lower to greater
- New attributes defined in an element must be optional (absence of use="required").
- If new enumerated values are included, the former ones and its meaning must be kept.
- If new operations are added, the existing operations must be kept
- New parameters added to existing operations must be optional and existing parameters must be kept

13.1.3. Patch Version

The patch version number (`{patchApiVersion}`) is incremented for any backward compatible bug fixes.

13.1.4. Version in API URL

The URLs of TMF REST APIs MUST state the API's major version number, prefixed by "v", following the API Name.

The format for the API ID with version number is defined as:

`{serverRoot}/{apiName}/{apiVersion}`

where:

- `{serverRoot}` is implementation specific (e.g.: <https://api.service.company.com>)
- `{apiName}` is the name of the API
- `{apiVersion}` is the major version of the API, prefixed by "v" (e.g. "v5")

Example:

<https://api.domain.com/tmf-api/troubleTicket/v5>

Here,

- `{serverRoot}` is <https://api.domain.com/tmf-api>
- `{apiName}` is `troubleTicket`
- `{apiVersion}` is `v5`

The *major* version number (without the minor, patch number) MUST be present in the URL.

However, if there are changes in the minor versions and there is a demand for a dual or multi API versions run scenario, where the consumers can make an explicit choice of the API version, the *minor* version can be added in the URL.

For example, imagine a `v5` Party API that returned basic Party information, that was deployed and working. Now a second implementation is available, that returns a lot more Party information, and the producer wants to offer an optional migration.

This new implementation could be hosted alongside as `/v5.1/`, so that consumers can choose to continue on `/v5/` or route to `/v5.1/` – or (more forcefully) an intermediate proxy might redirect all `/v5/` requests to `/v5.1/`, or route them randomly or rule-controlled to the different versions, e.g. for A/B testing.

For example to address an Individual resource (specialized party representing a natural person):

<https://api.domain.com/tmf-api/party/v5/individual>

and

<https://api.domain.com/tmf-api/party/v5.1/individual>

13.2. National Language Support

13.2.1. Introduction

In some localities, software systems need to manage data in multiple national languages. For example:

- Canada – IT systems must support English and French, by law
- Belgium – four languages (Dutch, French, German, English)
- Switzerland – four languages (German, French, Italian, Romansh)
- South Africa – many languages (English, isiZulu, isiXhosa, Afrikaans, others)

Even in markets with a single national language, software systems need to support that language correctly.

Support for languages includes:

- Display of data in the language requested by (or implied by) the end user
- Data entry in the requested language
- Filtering/searching of data according to the data entered in the requested language
- Sorting of lists of data according to the sorting rules (collating sequence) of the requested language
- Attachments in the requested language

13.2.2. General

A deployed software system that implements Open APIs **SHOULD** have a list of languages that it can support. If more than one language is supported there **SHOULD** be some default language for the system deployment.

If a specific language is needed by an API consumer, requested language **SHOULD** be communicated to the software system using the HTTP **Accept-Language** header. If the requested language is not supported by the software system, it will ignore the request and

use the default language defined for the deployment.

This guideline section deals with the following use cases, discussed in more detail in the subsequent subsections.

- Single free-text value (language-independent)
- Single value for persistence and business logic, but which should be input and displayed in the preferred language of the end-user
- Value per language at design time, single language at runtime
- Value per language for display and persistence

13.2.3. Single free-text value

The attribute has a single textual value, for example:

- TroubleTicket.description – would be entered by the end customer, CSR, etc. as free text.
- Note.text – would be entered by whoever is handling the entity that contains the list of notes.

The software system SHOULD store and retrieve value of the text field in the API resource as-is, without modification. For example:

```
{
  "description": "My phone cannot dial abroad"
}
```

```
{
  "note": [
    {
      "text": "Be careful, there is a savage dog in the front garden"
    }
  ]
}
```

Filtering on the attribute value (if supported by the API) SHOULD be direct, without modification. Sorting on the attribute (if supported by the API) SHOULD be according to the collating sequence of the default language, since there is no control on the language of the value entered for the attribute.

13.2.4. Single value for persistence and business logic, but language-dependent display

The attribute has a single value (may be textual or numeric) that is stored and may also be used in business logic. But it is entered and displayed in the requested language. For example:

- TroubleTicket.status (in fact most entity states or statuses)
- OrderItem.action

The API implementation SHOULD accept and return a language-independent value for the storage and retrieval of the attribute. It is the responsibility of the presentation (or experience) layer to translate between the language-independent value and the displayable language-dependent value, for both input and output. For example:

```
{
  "status":"acknowledged"
}
```

might be translated in the presentation layer as "acquittement" in French or "acknowledged" in English.

```
{
  "productOrderItem": [
    {
      "action":"add"
    }
  ]
}
```

might be translated in the presentation layer as "Ajout" in French or "Add" in English.

Similarly, filtering SHOULD be done on the language-independent value, and it is this value is what should be specified in the filter parameter. So:

- On a search screen the presentation layer will display a list of language-dependent values allowing the user to choose one (or more, if multiple choice is supported by the API)
- Before invoking the GET operation the presentation layer will translate the chosen value(s) to the language-independent representation.

Regarding sorting:

- When a list of valid values is displayed (to populate the field or query, such as pulldown or radio button control), it is the responsibility of the presentation layer to decide on the sequence of display of the values.
- When a list entities is returned (from a GET query on a managed entity, or an internal list of sub-entities), and sort is requested and supported on this attribute, sorting SHOULD be done on the language-independent value, according to the collating sequence of the default language.

13.2.5. Value per language at authoring time, single language at runtime

The attribute has a separate value for each language, authored in advance. This applies to entities in the various catalog APIs, and other APIs that have <entity>specification. The expectation is that:

- At authoring time the values can be saved, retrieved, filtered in multiple languages
- At runtime the values are read-only, and are treated according to the requested language

For example:

- ProductOffering.name
- ServiceSpecification.name

Full guidance for this type of attribute will be given in a future version of the design guidelines. But the general direction is to have a separate flexible array of attribute names, languages, and attribute value per language.

13.2.6. Value per language for display and persistence

The attribute has a separate value for each language, without authoring in advance. This applies in localities where there is a requirement to maintain multiple languages for field values. For example:

- Customer.name
- Party.name

Full guidance for this type of attribute will be given in a future version of the design guidelines. But the general direction is to have a separate flexible array of attribute names, languages, and attribute value per language.

13.3. Resource Scheme Customisation – removing non-mandatory Attributes

The TMF Open APIs are designed to cater for a large number of use cases, and in doing so, the resource schemes associated with the APIs can be rather verbose.

Each implementation, depending on the use cases implemented, can update the resource schemes to remove the attributes that are not mandatory and, by doing so, minimize the resource payloads.

However, it **MUST** not remove any mandatory attribute (e.g., **href**) or API parameter (e.g., **filter**, **fields**) with a dedicated semantic as specified by the Design Guidelines or any mandatory behaviour patterns.

Chapter 14. Event Management

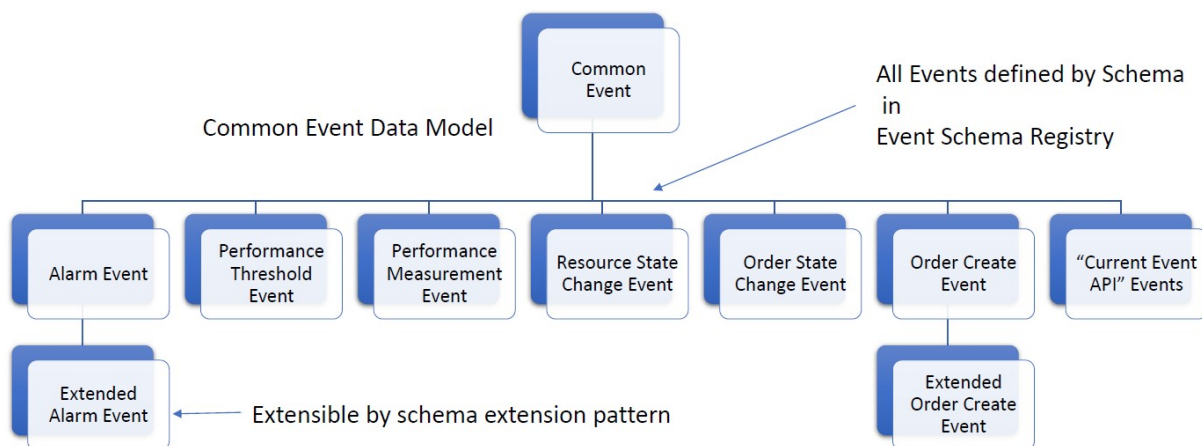
14.1. Decoupling

Event Management supports decoupling directly. It decouples TMF APIs from each other and powers decoupling on domain and API level.

14.2. Event API Relationship to Notification Events by a generalized Event Model

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that are described below. Polymorphism in collections occurs when entities inherit from base entities, for instance the “Alarm Event” or the “Order Create Event” or any kind of event payload can be described by the event entity. This means the common event data model can be extended by any kind of event payload because of the polymorphic data structure definition. By that polymorphic pattern the common event data model describes the event taxonomy of any event type.

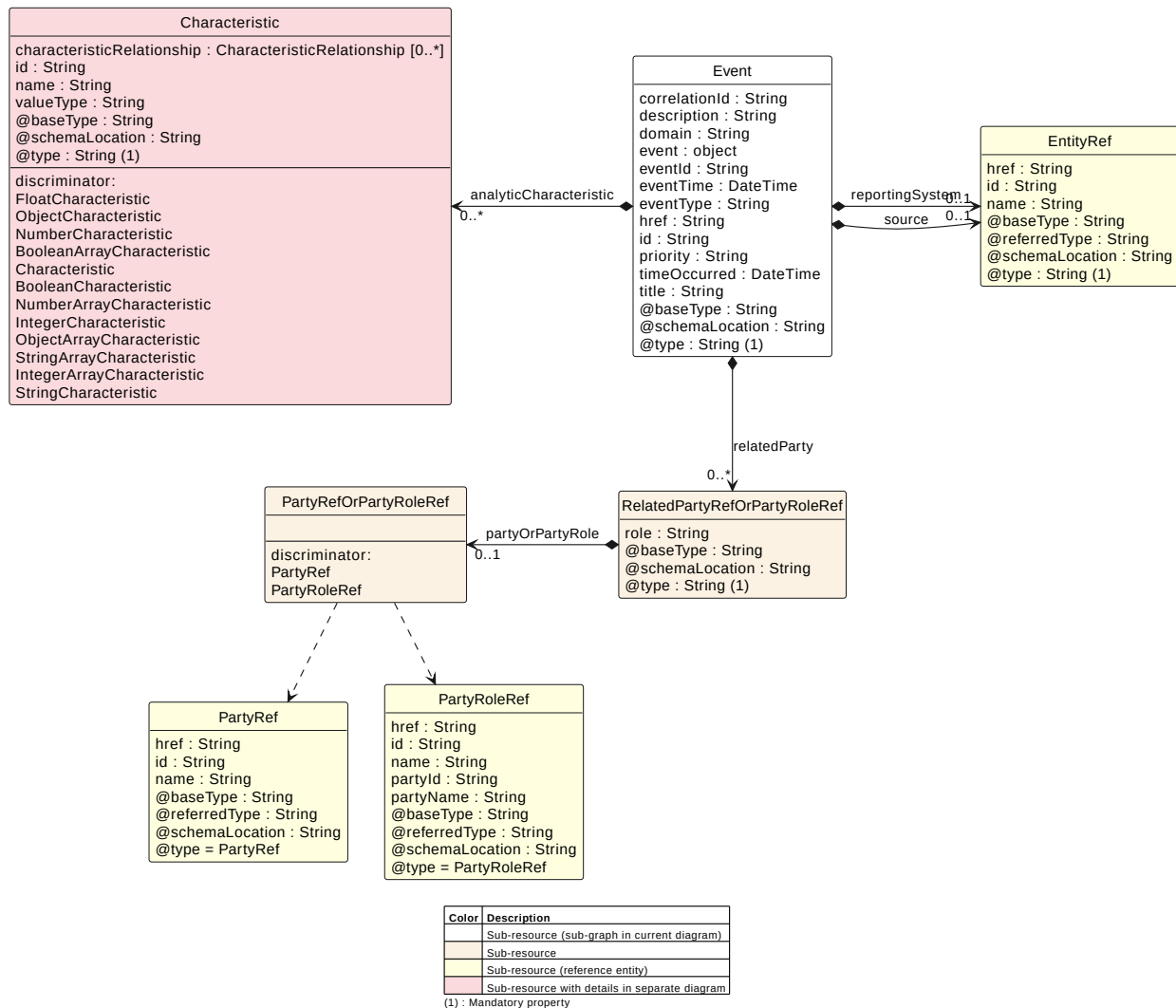
Event Taxonomy



Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines Part 2 document.

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a list of Event instances some may be instances of Alarm Events where other could be instances of OrderCreation Events. The `@type` gives this information. All resources and sub-resources of this API have an `@type` attributes that must be provided.

14.3. Event Resource Model



The “event” attribute element contains the payload of the sourced entity resource (e.g. Alarm, Resource, Order..).

The “eventType” attribute contains the triggered event type (e.g. AlarmCreateEvent, ResourceAttributeValueChangeEvent,..) thrown by the notification interface of the TMF Open API.

Additional attributes (e.g. “domain”, “reportingSystem”, “source”, “relatedParty”,...) describe the complete sourcing of the Event.

Especially by the **@type** and **@schemaLocation** of the “event” (eventPayload) the Event taxonomy is described by TMF Data Model (schemas) and can be used for further processing (e.g. Analytics).

As example the Event `TroubleTicketCreateEvent` is given below:

AlarmNotification

```
{
  "correlationId": "a347..",
  "description": "TroubleTicketCreateEvent illustration",
  "domain": "Assurance",
  "eventId": "6ae2..",
  "eventTime": "2023-09-04T08:59:27.255Z",
  "eventType": "TroubleTicketCreateEvent",
  "priority": "1",
  "timeOccurred": "2023-09-04T08:59:27.255Z",
  "title": "TroubleTicketCreateEvent",
  "event": {
    "@type": "TroubleTicket",
    "id": "f6ac..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicket/f6ac..",
    "name": "Network Outage Report",
    "description": "Experiencing a network outage in my area. No internet or
phone service available.",
    "severity": "High",
    "ticketType": "NetworkIssues",
    "creationDate": "2023-09-04T08:59:27.255Z",
    "requestedResolutionDate": "2023-09-04T08:59:27.255Z",
    "expectedResolutionDate": "2023-09-04T08:59:27.255Z",
    "resolutionDate": "2023-09-04T08:59:27.255Z",
    "lastUpdate": "2023-09-04T08:59:27.255Z",
    "priority": "High",
    "status": "acknowledged",
    "statusChangeDate": "2023-09-04T08:59:27.255Z",
    "statusChangeReason": "Need more information from the customer",
    "attachment": [
      {
        "@type": "AttachmentRef",
        "id": "4aaf..",
        "href": "{apiRoot}/documentManagement/v5/document/4aaf..",
        "name": "Network Status Screenshot",
        "description": "Screenshot of network status",
        "url": "https://mycsp.com/documents/docloader?docnum=4aaf.."
      }
    ]
  },
  "channel": {
    "@type": "ChannelRef",
```

```
"id": "8be1..",
"href": "https://mycsp.com/channel/8be1..",
"name": "Web"
},
"externalIdentifier": [
{
"@type": "ExternalIdentifier",
"owner": "DocumentManagementSystem",
"externalIdentifierType": "Screenshot",
"id": "4aaf..",
"href": "https://mycsp.com/dms/screenshot/4aaf.."
}
],
"note": [
{
"@type": "Note",
"id": "3153..",
"author": "Mr. N. Bene",
"date": "2023-09-04T08:59:27.255Z",
"text": "This is critical, please address it urgently."
}
],
"relatedEntity": [
{
"@type": "RelatedEntity",
"role": "Exchange",
"entity": {
"@type": "EntityRef",
"id": "4aaf..",
"href": "{apiRoot}/resourceInventoryManagement/v5/resource/4aaf..",
"name": "EXC-MK-1872",
"@referredType": "Resource"
}
}
],
"relatedParty": [
{
"role": "reporter",
"partyOrPartyRole": {
"id": "620d..",
"name": "Mr. N. Bene",
"@type": "PartyRef",
"@referredType": "Individual"
}
},
"@type": "RelatedPartyRefOrPartyRoleRef"
```

```
    }
  ],
  "statusChangeHistory": [
    {
      "@type": "StatusChange",
      "statusChangeDate": "2023-09-04T08:59:27.255Z",
      "statusChangeReason": "Need more information from the customer",
      "status": "acknowledged"
    }
  ],
  "troubleTicketSpecification": {
    "@type": "TroubleTicketSpecificationRef",
    "id": "4aaf..",
    "href": "{apiRoot}/troubleTicket/v5/troubleTicketSpecification/4aaf..",
    "name": "NetworkOutageTroubleTicket",
    "version": "1.2"
  },
  "troubleTicketCharacteristic": [
    {
      "id": "4aaf..",
      "name": "OutageAddress",
      "valueType": "string",
      "value": "912143 London Road, BOURNEMOUTH, BH83 2EN",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "4aaf..",
          "relationshipType": "dependency"
        }
      ],
      "@type": "StringCharacteristic"
    }
  ],
  "reportingSystem": {
    "id": "4b5d..",
    "name": "APP-745",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "ab18..",
    "name": "APP-317",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  }
}
```

```

    },
    "relatedParty": [
      {
        "role": "reporter",
        "partyOrPartyRole": {
          "id": "efa1..",
          "name": "Mr. N. Bene",
          "@type": "PartyRef",
          "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    ],
    "@baseType": "Event",
    "@type": "TroubleTicketCreateEvent"
  }

```

14.4. Topic/Event Resource Graph

Event Management supports the Event streaming concept by the Topic resource. The Topic resource is the target container for the event resource. The topic/event resource can be used for storing different events into different topics separated into domains (e.g. Party, Resource, Product,...). A topic/event resource graph can be set up for further processing and analytic functions.

The different topics can also be used for different access control on the topic/event resource because of the different endpoints of the Event API by topics.

Example: Creation of a "Resource" topic for Alarm events:

REQUEST

```
POST //eventhub/eventapi/topic
```

```

{
  "contentQuery": "AlarmValue",
  "headerQuery": "AlarmType",
  "name": "Resource",
  "@baseType": "topic",
  "@schemaLocation": "https://schemaregistry/Topic.schema.json",
  "@type": "topic"
}

```

```
}
```

RESPONSE

```
201 Created
Content-Type: application/json
Location: https://eventhub/eventapi/topic/13
```

```
{
  "id": "13",
  "href": "https://eventhub/eventapi/topic/13",
  "contentQuery": "AlarmValue",
  "headerQuery": "AlarmType",
  "name": "Resource",
  "@baseType": "topic",
  "@schemaLocation": "https://schemaregistry/Topic.schema.json",
  "@type": "topic"
}
```

14.5. Event Management as API for Consumers

Event Management has, like all other APIs, a consumer API. That consumer API allows Consumers to have access to the different topic/event resources into the graph. Different topics might provide content for different Consumers.

e.g. the “Party/TroubleTicket” Topic contains TroubleTicket events for the Consumers service support.

14.6. Event Management in a multi-cloud Scenario

Event Management supports the sourcing and storing of any API notification. It allows storing and consuming events (data) as streams.

For example storing the complete PhysicalResource data replication (copy) as stream of events into topic “Resource” (channel/domain):

```
POST //eventhub/topic/{topicId}/event (e.g. topicId = 111,
topic.name = Resource)
```

The corresponding Resource Inventory API should support the data replication by a

resource data export operation according to the event notification interface. (s. TMF639_Inventory_Resource specification)

The Event stream can be consumed by list of events selected by the offset keyword.

List of events as a stream (example):

```
GET
/eventhub/topic/111/event?eventType=PhysicalResourceDataReplication&offset=10000&limit=1000
```

Or by eventTime

```
GET
/eventhub/topic/111/event?eventType=PhysicalResourceDataReplication&eventTime>=20-02-2020:12:00:00&limit=1000
```

It allows the usage of the Event API into a multi cloud scenario, where every cloud might contain an Event Management component.

Because of the unified and polymorph designed Event API a distribution of events in any cloud container is possible and can support decoupling multi cloud APIs by event (data) distribution and replication.

Event management supports data replication into the multi cloud environment by event streaming.

References

□ [IANA-LINKS]

Navigation Links

iana (Internet Assigned Numbers Authority)

<https://www.iana.org/assignments/link-relations/link-relations.xhtml>

□ [IANA-STATUS]

Hypertext Transfer Protocol (HTTP) Status Code Registry

iana (Internet Assigned Numbers Authority)

<http://www.iana.org/assignments/http-status-codes/http-status-codes.xml>

□ [ISO639]

Codes for the Representation of Names of Languages

ISO (the International Organization for Standardization)

<https://www.iso.org/iso-639-language-code>

□ [ISO3166] + The International Standard for country codes and codes for their subdivisions;

ISO (the International Organization for Standardization)

<https://www.iso.org/iso-3166-country-codes.html>

□ [ISO8601]

Date and time format

ISO (the International Organization for Standardization)

<https://www.iso.org/iso-8601-date-and-time-format.html>

□ [Mongo-ID]

Generating Globally Unique Identifiers for Use with MongoDB

MongoDB, Inc.

<https://www.mongodb.com/blog/post/generating-globally-unique-identifiers-for-use-with-mongodb>

□ [OAS3]

OpenAPI Specification, Version 3

OpenAPI Initiative, The Linux Foundation

Version 3.0.0: <https://spec.openapis.org/oas/v3.0.0>

Version 3.0.1: <https://spec.openapis.org/oas/v3.0.1>

Version 3.0.2: <https://spec.openapis.org/oas/v3.0.2>

Version 3.0.3: <https://spec.openapis.org/oas/v3.0.3>

Version 3.0.4: <https://spec.openapis.org/oas/v3.0.4>

Version 3.1.0: <https://spec.openapis.org/oas/v3.1.0>

Version 3.1.1: <https://spec.openapis.org/oas/v3.1.1>

□ [OWASP-ReDoS]

Regular expression Denial of Service – ReDoS

OWASP (The Open Worldwide Application Security Project)

https://owasp.org/www-community/attacks/Regular_expression_Denial_of_Service_-_ReDoS

□ [OWASP-API4:2023-RL]

API4:2023 Unrestricted Resource Consumption

OWASP (The Open Worldwide Application Security Project)

<https://owasp.org/API-Security/editions/2023/en/Oxa4-unrestricted-resource-consumption>

□ [PoLP]

Principle of least privilege

WIKIPEDIA, The Free Encyclopedia https://en.wikipedia.org/wiki/Principle_of_least_privilege (retrieved June 27, 2025)

□ [RFC3339]

Date and Time on the Internet: Timestamps

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc3339.html>

□ [RFC3986]

Uniform Resource Identifier (URI): Generic Syntax

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc3986.html>

□ [RFC4122]

A Universally Unique IDentifier (UUID) URN Namespace

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc4122.html>

□ [RFC5234]

Augmented BNF for Syntax Specifications: ABNF

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc5234.html>

□ [RFC5322]

Internet Message Format

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc5322.html>

□ [RFC5789]

PATCH Method for HTTP

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc5789.html>

□ [RFC5829]

Link Relation Types for Simple Version Navigation between Web Resources

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc5829.html>

□ [RFC6901]

JavaScript Object Notation (JSON) Pointer

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc6901.html>

□ [RFC6902]

JavaScript Object Notation (JSON) Patch

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc6902.html>

□ [RFC7396]

JSON Merge Patch

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc7396.html>

□ [RFC8174]

Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc8174.html>

□ [RFC8259]

The JavaScript Object Notation (JSON) Data Interchange Format

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc8259.html>

□ [RFC8288]

Web Linking

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc8288.html>

□ [RFC9110]

HTTP Semantics

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc9110.html>

□ [RFC9111]

HTTP Caching

IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc9111.html>

□ [RFC9112]

HTTP/1.1 IETF (Internet Engineering Task Force)

<https://www.ietf.org/rfc/rfc9112.html>

- [RFC9535]
JSONPath – Query Expressions for JSON
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc9535.html>
- [RMM]
The Richardson Maturity Model
"Justice Will Take Us Millions Of Intricate Moves, Act Three: The Maturity Heuristic"
Leonard Richardson
<http://www.crummy.com/writing/speaking/2008-QCon/act3.html>
- [SemVer]
Semantic Versioning 2.0
Tom Preston-Werner
<https://semver.org>
- [SID]
TMForum Information Framework
- [TMF630-1]
TMForum
REST API Design Guidelines
Part 1: Practical guidelines for RESTful APIs naming, CRUD, filtering, tasks, history, notifications
- [TMF630-2]
TMForum
REST API Design Guidelines
Part 2: Advanced guidelines for RESTful APIs polymorphism, extension patterns, depth and expand directive, entity RefOrValue
- [TMF630-3]
TMForum
REST API Design Guidelines
Part 3: Guidelines for extending TMF Open API's with hypermedia support
- [TMF630-4]
TMForum
REST API Design Guidelines
Part 4: Advanced guidelines for import and export tasks and for entity versioning
- [TMF630-5]
TMForum
REST API Design Guidelines
Part 5: JSON patch extensions to manage arrays
- [TMF630-6]

TMForum
REST API Design Guidelines
Part 6: JSON path extensions

□ [TMF630-7]

TMForum
REST API Design Guidelines
Part 7: JSON Schemas guideline

Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired, however a Document History must always be included.

Document History

Version History

This section records the changes between this and the previous document version as it is edited by the team concerned.



this is an incremental number which does not have to match the release number and used for change control purposes only.

Version Number	Date Modified	Modified by	Description of changes
1.0.0	9-Oct-2013	Pierre Gauthier	First release of document
1.1.0	November 2014	Pierre Gauthier	Updates for Frameworx 14.5
1.1.1	March 2015	Alicja Kawecki	Updated cover, footer and Notice to reflect TM Forum Approved status
3.0.0	20 Mar 2018	Nicoleta Stoica	Updates for Frameworx 17.5
4.0.0	23 April 2020	Several contributors / Pierre Gauthier	add security, event, fix typos, authentication diagram and some RFC references
4.0.2	13 January 2021	Several contributors / Pierre Gauthier	Fix authentication diagram and some RFC references
4.2.0	06 April 2021	Several contributors / Pierre Gauthier	Introduce cursor based pagination, plus minor precisions

Version Number	Date Modified	Modified by	Description of changes
5.0.0	04 September 2025	Several contributors / Florin Tene / Lutz Bettge	Upgrade to V5, introduce sub-resources, clarifications on task operations, introduce asynchronous pattern clarifications, expect header behaviour, lost update prevention, range-based pagination, clarifications on event types, history pattern, plus other changes.

Acknowledgments

Contributors to Prior Versions

Previous versions of this document were prepared by the following members of the TM Forum API team (in alphabetical order):

Alexis de Peufeilhoux	Deutsche Telekom	Contributor
Anoop Gupta	Amdocs	Contributor
Christophe Michel	Amdocs	Contributor
Fengjing Tan	Huawei	Contributor
Hongxia Hao	Huawei	Contributor,
Jacob Avraham	Amdocs	Contributor
Johanne Mayer	MayerConsult	Contributor
Jonathan Goldberg	Amdocs	Contributor
Kamal Maghsoudlou	Ericsson	Contributor
Ludovic Robert	Orange	Contributor
Luis Velarde	Telefonica	Contributor
Nicoleta Stoica	Vodafone	Contributor
Pierre Gauthier	TM Forum	Chief Architect, Team Leader
Sophie Bouleau	Orange	Contributor
Thomas Braun	Deutsche Telekom	Contributor

Contributors to Current Version

The following members of the TM Forum API team contributed to this version of the document (in alphabetical order):

Florin Tene	Cityfibre	Contributor
Jonathan Goldberg	TMForum	Contributor
Kamal Maghsoudlou	Ericsson	Contributor
Lutz Bettge	Deutsche Telekom	Contributor, Editor
Olivier Arnaud	Orange	Contributor